

COMMUNICATION SYSTEMS

15.1 INTRODUCTION

1. What is meant by the term communication ? Give a historical view of long distance communication.

Introduction. The word communication refers to the faithful transfer of information or message from one point to another point in an intelligible form. In ancient times, long distance messages were carried by messengers on foot, horse, cart etc. People also used coded signalling methods through burning fires, beating drums or waving flags etc. In these methods, human involvement was necessary at every stage of communication. In modern communication systems, the information is first converted into electrical signals and then sent

electronically. In our everyday life, we use many of these systems such as telephones, TV and radio communication, satellite communication, etc. The communication, through these systems is highly speedy, more reliable and much more effective over longer distances.

The work of the earlier scientists of 19th and 20th centuries like J.C. Bose, F.B. Morse, G. Marconi and Alexander Bell has proved to be the backbone of the modern communication. Today the growth of communication sector is much faster than any other industry.

The major events in the history of long distance communication are arranged chronically in Table 15.1.

Table 15.1 Some major milestones in the history of long distance communication

Year	Event	Remark
Around 1565 A.D.	The reporting of the delivery of a child by queen using drum beats from a distant place to king Akbar	It is believed that minister Birbal experimented with the arrangement to decide the number of drummers posted between the place where the queen stayed and the place where the king stayed.
1835	Invention of telegraph by Samuel F.B. Morse and Sir Charles Wheatstone	It resulted in tremendous growth of messages through post offices and reduced physical travel of messengers considerably.
1876	Telephone invented by Alexander Graham Bell and Antonio Meucci	Perhaps the most widely used means of communication in the history of mankind.
1895	Jagadis Chandra Bose and Guglielmo Marconi demonstrated wireless telegraphy	Their experiments marked the beginning of wire-less radio communication. In 1899, Guglielmo Marconi succeeded in transmitting radiowaves across the British Channel and next across the Atlantic Ocean in 1901.

Year	Event	Remark
1936	Television broadcast (John Logi Baird)	First television broadcast by BBC.
1955	First radio FAX transmitted across continent (Alexander Bain)	The idea of FAX transmission was patented by Alexander Bain in 1843.
1968	ARPANET – the first internet came into existence (J.C.R. Licklider)	ARPANET was a project undertaken by the U.S. defence department. It allowed file transfer from one computer to another connected to the network.
1975	Fibre optics developed at Bell Laboratories	Fibre optical systems are superior and more economical compared to traditional communication systems.
1989-91	Tim Berners-Lee invented the World Wide Web	WWW may be regarded as the mammoth encyclopedia of knowledge accessible to everyone round the clock throughout the year.

2. Distinguish between the terms communication, information, message and signal.

Communication. It is the process by which information is transferred from one point (source) to the other point (destination).

Information. It is basically the news which one wishes to convey.

Message. It is the physical manifestation of the information produced by the source. It may appear as a sequence of discrete symbols, time-varying quantity etc.

Signal. It is the electrical analog of the message at the source. The device which converts a physical quantity (message) into electrical signal is called a **transducer**. Clearly, the electrical signal contains the information. A **signal** may be defined as the single-valued function of time (that conveys the information) and which, at every instant of time, takes a unique value.

15.2 COMMUNICATION SYSTEM

3. What is a communication system? Describe briefly the major constituents of a communication system.

Communication system. The set up used to transmit information from one point to another is called a communication system. The essential parts of

a communication system are transmitter, communication channel and receiver, as shown in the block diagram of Fig. 15.1.

(i) **Transmitter.** It is a set-up that transmits the message to the receiving end through a communication channel. Its main function is to convert the message signal produced by the information source into a form suitable for transmission through the channel and to transmit it. If the information source gives a voice signal, a transducer converts it into an electrical signal before feeding it to the transmitter.

(ii) **Communication channel or transmission medium.** It is the medium or the physical path that connects a transmitter to a receiver. It carries the modulated wave from the transmitter to the receiver. It can be a transmission line, an optical fibre or free space. When the information signal propagates along the channel, some unwanted signals called noise, may interfere with it and thus the receiver receives a corrupted version of the transmitted signal.

(iii) **Receiver.** It is a set-up that receives the transmitted signals from the transmission medium and converts those signals back to their original form. This process of recovering the original signal is called **demodulation** or **detector**, which is reverse of the modulation process used at the transmitter.

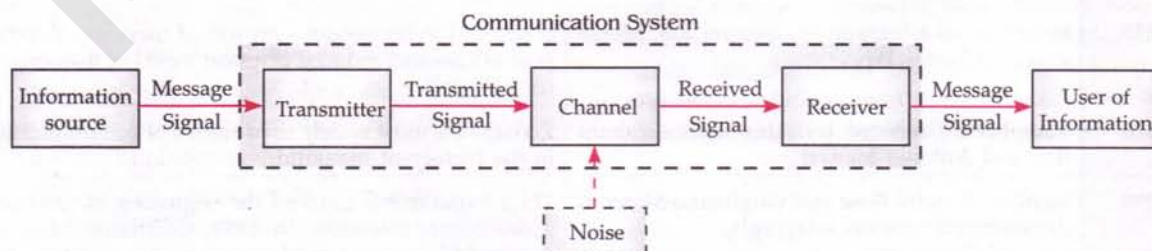


Fig. 15.1 Block diagram of an electronic communication system.

4. What are the two basic modes of communication ?

Two basic modes of communication. These are as follows :

1. Point-to-point communication. In this mode, communication occurs over a link between a single transmitter and receiver. For example, telephony.

2. Broadcast mode. In this mode, a large number of receivers are linked to a single transmitter. For example, radio and television.

15.3 BASIC TERMINOLOGY USED IN ELECTRONIC COMMUNICATION SYSTEMS

5. Define the various terms commonly used in describing various electronic communication systems.

Basic terminology used in electronic communication systems. Before we understand the basics of any communication system, it will be useful to have the knowledge of following terms :

1. Transducer. Any device which converts energy from one form to another is called a transducer. We may define an electrical transducer as a device which converts variations in physical quantity such as pressure, displacement, force, temperature etc., into corresponding variations in the electrical signal at its output. For example, a microphone, converts a sound signal into an electrical signal.

2. Signal. It is the electrical analog of the information produced by the source. It may be defined as a single-valued function of time (that conveys the information) and which, at every instant of time, takes a unique value. Signals can be analog or digital, as discussed in Article 15.5.

3. Noise. The unwanted electrical signals which get interfered with the information signal during its propagation through the transmission medium constitute noise. Consequently, a distorted or corrupted version of the transmitted signal reaches the receiver.

4. Transmitter. It is a device which processes a message signal into a form suitable for transmission and then transmits it to the receiving end through a transmission medium.

5. Receiver. It is a device which recovers the original message signal from the signal received at the output of the communication channel.

6. Attenuation. The loss of strength of a signal during its propagation through the transmission medium is called attenuation.

7. Amplification. It is the process of increasing the amplitude and hence the strength of an electrical signal by using an electric circuit (consisting of at least one transistor) called the amplifier. Amplification is required to compensate for the loss of strength of a signal during its

transmission. The additional energy is obtained from a dc source. The process of amplification is done at a location between the source and the destination where the strength of signal becomes weaker than the required level.

8. Range. It is defined as the largest distance between the source and the destination upto which a signal can be received with sufficient strength.

9. Bandwidth. It refers to the frequency range over which an equipment operates or the range over which the frequencies in a signal vary.

10. Baseband. It refers to the band of frequencies of the original signal (a signal not changed by modulation) as is produced by the source of information.

11. Modulation. Low frequency waves cannot be transmitted to long distances. So low frequency information signal is superimposed on a high frequency wave, which acts as a carrier of the information. Modulation is the process in which some characteristic, usually a amplitude, frequency or phase angle of a high frequency carrier wave is varied in accordance with the instantaneous value of the low frequency information signal, called the modulating signal.

12. Demodulation. The process of recovering the original information signal from the modulated wave at the receiver end is called demodulation or detection. It is the reverse process of the modulation process that occurs at the transmitting end.

13. Repeater. A signal while passing through the transmission medium may get attenuated due to the various energy losses along its path. So a signal booster or amplifying repeater is placed at suitable distance along its path. A repeater is a combination of a transmitter, an amplifier and a receiver which picks up a signal from the transmitter, amplifies and retransmits it to the receiver sometimes with a change of carrier frequency.

Multiple repeaters are placed at suitable distances along the transmission path. This extends the range of a communication system, as shown in Fig. 15.2. A communication satellite is essentially a repeater station in space.

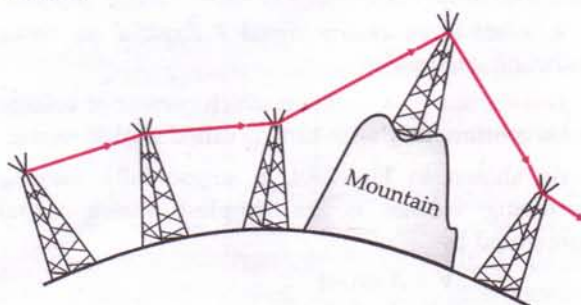


Fig. 15.2 Multiple usage of amplifier repeater extends the range of communication.

15.4 TRANSMITTER

6. What is a transmitter? What is its main function? Write the names of its basic components.

Transmitter. A transmitter is a set up that transmits the message to the receiving end through a communication channel. Its main function is to modify the message signal in a form suitable for transmission over the channel and to transmit it.

The basic components of a transmitter are

- (i) Message signal generator (ii) Modulator
- (iii) Antenna.

7. With the help of a schematic diagram, briefly describe an arrangement for transmission and reception of a message signal over a distance of several thousand kilometers.

A schematic arrangement for transmitting and receiving a message signal. Fig. 15.3 shows a block

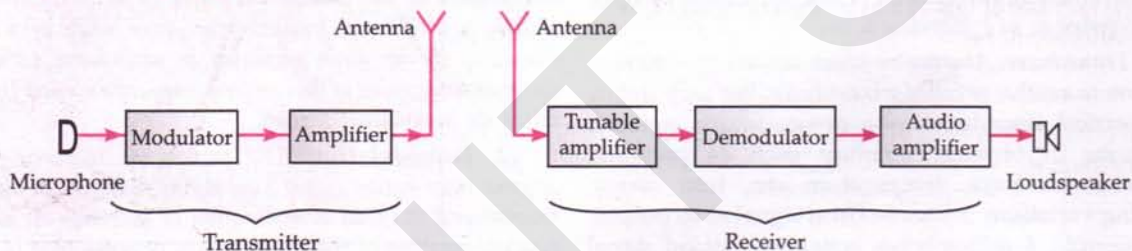


Fig. 15.3 A schematic arrangement for transmission and reception of a message signal.

15.5 MESSAGE SIGNALS

8. What are message signals? Mention their two types.

Message signals. In communication, a signal implies a time-varying electrical signal obtained by transforming original signal with the help of a suitable transducer. It is a single-valued function of time that conveys the information.

The message signals are of two types :

- (i) Analog signal (ii) Digital signal.

9. What is an analog signal? Explain the terms bandwidth and baseband.

Analog signal. A signal in which current or voltage varies continuously with time is called analog signal.

As shown in Fig. 15.4, a sinusoidally varying alternating voltage is the simplest analog signal represented by

$$V = A \sin \omega t$$

Here A is the amplitude and ω is the angular frequency of the analog signal.

diagram for an arrangement for transmitting a signal when transmitter and receiver are several thousand kilometers apart.

At the **transmitting end**, a microphone converts the sound signal into a time-varying electrical signal called the message signal. With the help of a modulator, the message signal is translated into radio frequency range. The power of the modulated wave is boosted by a suitable amplifier and then the wave is radiated into space from an antenna.

At the **receiving end**, another antenna picks up the waves from different transmitting stations. A tunable amplifier can selectively tune in and boost up the radio frequency wave from a particular station. The original signal is recovered by using a demodulator. After being amplified, the electrical signal is converted into sound signal by using a loudspeaker.

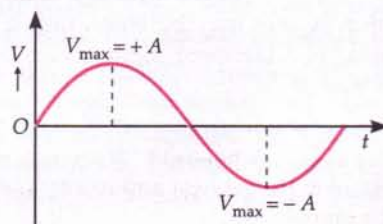


Fig. 15.4 An analog signal.

A vibrating tuning fork produces such a sound signal of single frequency. Every speech or music is a complex signal formed by the superposition of several sinusoidal signals of different amplitudes and frequencies.

10. What is a digital or discrete signal? What is meant by the terms bits and bytes?

Digital signal. A signal in which current or voltage can take only two discrete values is called a digital signal. A digital signal can take only two values 1 and 0 which are labelled as *high* and *low* values. In the square wave

shown in Fig. 15.5, a signal of 0 V represents binary 0 and a signal of 5 V represents binary 1. Thus digital signals are in the form of pulses of equal level.

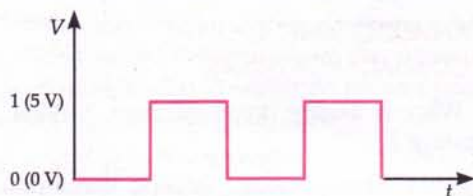


Fig. 15.5 Digital signal.

Digital signals can be used as effective commands. Digital signal 0 refers to an open circuit while 1 refers to a close circuit. 0 is also referred as *no* or *space* while 1 is referred as *yes* or *mark*.

Digits 0 and 1 are called *bits*. A group of bits is called a *binary word* or a *byte*. A byte made of 2 bits can give four code combinations: 00, 01, 10 and 11. In general, if x is the number of bits in a byte, then number of code combinations will be

$$N = 2^x$$

In a commonly used 8 bit byte, the number of code combinations is

$$N = 2^8 = 256$$

15.6 BANDWIDTH OF SIGNALS

11. What do you mean by bandwidth of a signal? What is its importance? Give examples of some bandwidths.

Bandwidth of signals. The message signal in any communication system may be voice, music, picture, computer data, etc. Different signals have different ranges of frequencies.

The range over which frequencies in an information signal vary is called **bandwidth**. It is equal to the difference between the highest and lowest frequencies present in the signal.

One of the most important limitations on the performance of a communication system is the bandwidth of the information signal. The type of the communication channel needed for a given signal depends on the band of frequencies to be transmitted.

For example, speech signals contain frequencies between 300 Hz to 3100 Hz. Such signals require a bandwidth of 2800 Hz (3100 Hz – 300 Hz) for telephonic transmission.

Audio signals have frequencies between 20 Hz to 20 kHz. So the transmission of a good music requires a bandwidth of about 20 kHz.

Video signals required for transmission of picture have a bandwidth of about 4.2 MHz.

A TV signal, which contains both voice and picture, is allocated a bandwidth of 6 MHz for transmission.

12. How can a rectangular wave be generated from the superposition of sinusoidal waves?

Approximation of a rectangular wave in terms of a fundamental sine wave and its harmonics. We have learnt that a sinusoidal wave is an analog signal while a square/rectangular wave is a digital signal. As shown in Fig. 15.6, we can decompose a rectangular wave into a superposition of sinusoidal waves of frequencies f_0 , $2f_0$, $3f_0$, $4f_0$, ..., nf_0 , where n is an integer extending to infinity and $f_0 = 1/T_0$. In the same figure, we have shown the fundamental (f_0), fundamental + second harmonic ($f_0 + 2f_0$), fundamental + second harmonic + third harmonic ($f_0 + 2f_0 + 3f_0$).

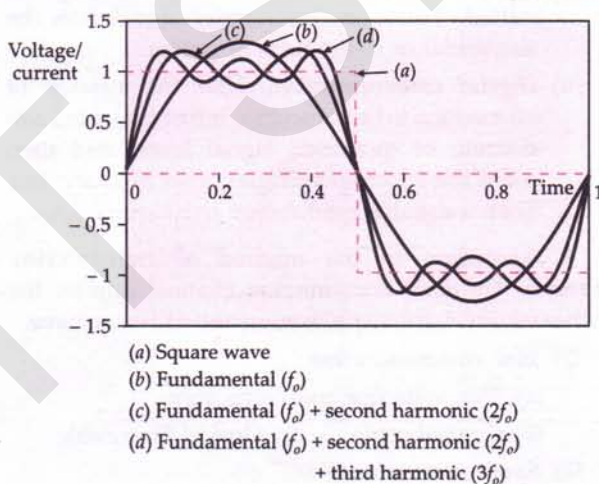


Fig. 15.6 Approximation of a square wave in terms of a fundamental sine wave and its harmonics.

Clearly, to reproduce the rectangular wave shape more exactly, one needs to superimpose all the harmonics f_0 , $2f_0$, $3f_0$, $4f_0$, This implies an infinite bandwidth. Practically, the contributions from the higher harmonics are negligible, which limits the bandwidth. Consequently, the received wave is the distorted version of the transmitted wave. However, if the bandwidth is large enough to contain a few harmonics, the information is not lost and the rectangular signal is exactly reproduced. This is because higher the harmonics, the less is its contribution to the waveform.

15.7 TYPES OF COMMUNICATION SYSTEMS

13. Mention the various categories into which the various communication systems are classified.

Types of communication systems. We classify different communication systems according to the

nature of information or mode of transmission or type of transmission channel used as follows :

1. According to the nature of information source.

- (i) *Speech transmission* as in a radio.
- (ii) *Picture as well as speech transmission* as in a television.
- (iii) *Facsimile transmission (FAX)*. It is used for exact reproduction of still pictures, documents, maps, etc. at a distant place.
- (iv) *Data transmission* as in computers.

2. According to the mode of transmission.

- (i) *Analog communication*. Here the signal which modulates the carrier signal for transmission varies in a continuous manner and is the analog or representative of the information to be transmitted. However, the carrier signal may be sinusoidal or in the form of pulses.
- (ii) *Digital communication*. Here the message or information to be transmitted is first converted into discrete or quantised signal levels and then coded into a corresponding sequence of binary data. Then a suitable modulation technique is used.

3. According to the method of transmission channel. The term transmission channel implies the path over which the signal is transmitted. So we have

(i) Line communication

- (a) Two wire line communication
- (b) Coaxial cable (c) Optical fibre cable

(ii) Space communication.

4. According to the type of modulation. Generally, the message or information or speech signals have low frequencies and cannot be directly transmitted to long distances. These signals are loaded or superimposed on a high frequency wave which acts as the carrier of information. The process is called *modulation*. The carrier wave may be a continuous wave or a pulse.

For *sinusoidal continuous carrier waves*, the types of modulation are

- (i) Amplitude modulation (AM)
- (ii) Frequency modulation (FM)
- (iii) Phase modulation (PM)

For *pulsed carrier waves*, the types of modulation are :

- (i) Pulse amplitude modulation (PAM)
- (ii) Pulse time modulation (PTM)
 - (a) Pulse position modulation (PPM)
 - (b) Pulse width modulation (PWM) or Pulse duration modulation (PDM)
- (iii) Pulse code modulation (PCM)

PCM method is preferred for *digital communication* while other methods are suitable for analog communication.

15.8 ANALOG AND DIGITAL COMMUNICATIONS

14. What is analog communication ? What is its disadvantage ?

Analog Communication. Analog communication makes use of analog electronic circuit and analog signal in such a way that the output voltage varies continuously in accordance with the input voltage. For example, telegraphy, telex, TV network, etc., are examples of analog communication systems. The analog operation is less reliable due to its many valued output. So it is now rarely used in modern communication systems.

15. What is digital communication ? Enumerate some of its advantages.

Digital communication. Digital communication makes use of an electronic circuit that can handle only digital signals. Modern communication systems like FAX, E-mail, etc. are purely based on digital electronic circuits because of the following advantages :

1. As compared to many valued analog operation, the two-valued digital operation is more reliable because here all the signals are easily identified as low and high.
2. As the information to be transmitted is already in pulse form, so its transmission needs simple technique.
3. A large number of digital signals can be sent through a single channel only.

15.9 MODULATION

16. What is modulation ? What is the need of modulation in communication systems ?

Modulation. A high frequency carrier wave is used to carry the audio signal (message) to large distances. *Modulation is the process by which some characteristic, usually amplitude, frequency or phase angle of a high frequency carrier wave is varied in accordance with the instantaneous value of the low frequency audio signal, called the modulating signal.*

Need of modulation in communication systems.

Message signals are also called **baseband signals**. In fact, the term baseband refers to the band of frequencies of the original signal, as produced by the source of information. Audio signals have a bandwidth of about 20 kHz. Such low frequency signals cannot be

transmitted directly to long distances and hence need to be modulated due to the following reasons :

1. Practical antenna length. To transmit a signal effectively, the height of the antenna should be comparable to the wavelength of the signal (at least $\lambda/4$ in length) so that the antenna properly senses the time variations of the signal. Now

$$\begin{aligned}\text{Wavelength} &= \frac{\text{Wave velocity}}{\text{Frequency}} \\ &= \frac{3 \times 10^8}{\text{Frequency (Hz)}} \text{ metre}\end{aligned}$$

So to transmit a signal of frequency 20 kHz, we need an antenna of height $= \frac{3 \times 10^8}{20 \times 10^3} \text{ m} = 15 \text{ km}$.

Antenna of such a height cannot be constructed or operated. On the other hand, if a carrier wave of 1 MHz is used, required antenna height comes down to just 300 m.

There is thus a need of translating the information contained in our low frequency baseband signal into a high radio frequency signal before transmission.

2. Effective power radiated by an antenna. For linear antenna of length l , it is seen that

$$\text{Power radiated} \propto \left(\frac{l}{\lambda}\right)^2$$

Thus for the same antenna length, the power radiated by short wavelength or high frequency signals would be large. If the audio signals (of longer wavelength) are directly radiated into space, they die out after covering some distance due to the low power radiated by the antenna. To transmit them over large distances, they are superimposed on high frequency carrier waves.

3. Frequency multiplexing. In order to transmit different audio signals (which fall in the same spectral range) simultaneously through one antenna, each signal is translated to a low frequency range so that it is easily recoverable and distinguishable from other signals at the receiving station.

4. Narrow banding. If a signal in the audio range (20 Hz to 20 kHz) is transmitted directly, then the ratio of highest to lowest frequency $= 20 \text{ kHz} / 20 \text{ Hz} = 1000$. If a carrier wave of frequency 100 kHz is used, then the ratio of highest to lowest frequency

$$= (100 \text{ kHz} + 20 \text{ kHz}) / (100 \text{ kHz} - 20 \text{ kHz}) \approx 1.2$$

Hence by using modulation, wide band signal can be converted into a narrow band signal.

The above discussion shows that there is a need for translating the original low frequency message (or

information) signal into a high frequency wave before transmission such that the transmitted signal continues to possess the information contained in the original signal.

17. What are carrier waves ? Mention the two types of carrier waves generally used.

Carrier waves. An alternating current of a frequency, at least twice that of the highest frequency component in the information signal, is used to provide a carrier wave. Since this high frequency signal is the signal which is actually transmitted and carries the information, so it is known as a carrier wave.

As shown in Fig. 15.7, the following two types of carrier waves are commonly used :

1. Continuous sinusoidal waves, and
2. Pulse shaped signals.

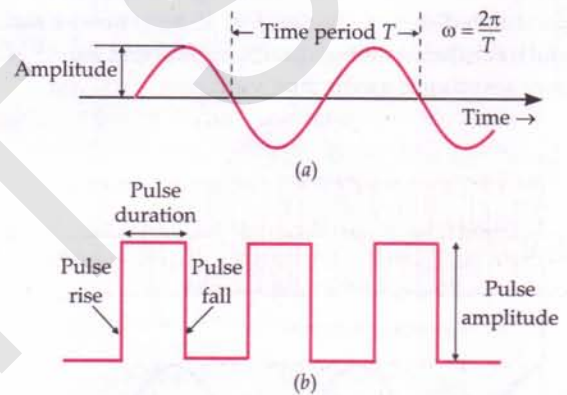


Fig. 15.7 Sinusoidal and pulse shaped carrier waves

15.10 TYPES OF CONTINUOUS WAVE MODULATION

18. Mention the different types of modulation techniques when the carrier waves are sinusoidal continuous waves.

Three types of modulation of continuous carrier waves. A sinusoidal carrier wave can be represented as

$$c(t) = A_c \cos(\omega_c t + \phi_0)$$

where $c(t)$ is the signal strength (voltage or current), A_c is the amplitude, $(\omega_c = 2\pi f_c)$ is the angular frequency and ϕ_0 is the initial phase of the carrier wave. The quantity $\phi = \omega_c t + \phi_0$ is called phase of the wave. The above equation shows that a carrier wave can be modulated in two ways :

1. By varying the amplitude of the carrier wave about mean value linearly with modulating signal, keeping the angular frequency ω_c fixed. This type of modulation is called **amplitude modulation (AM)**.

2. By varying phase angle θ of the carrier wave in accordance with the modulating signal, keeping the amplitude of the carrier wave fixed. This type of modulation is called **angular modulation**. The angular modulation has two forms :

- (a) Frequency modulation (FM)
- (b) Phase modulation (PM)

19. What is amplitude modulation ? Draw an A.M. wave. Give its important features.

Amplitude modulation. It is the process in which the amplitude of the high frequency carrier wave changes in accordance with the instantaneous value of modulating signal. Here a voltage proportional to the amplitude of the modulating signal gets added to the carrier amplitude.

Waveform of A.M. voltage. Fig. 15.8(a) shows an unmodulated carrier voltage, Fig. 15.8(b) shows a sinusoidal modulating voltage (audio signal) and Fig. 15.8(c) shows amplitude modulated voltage.

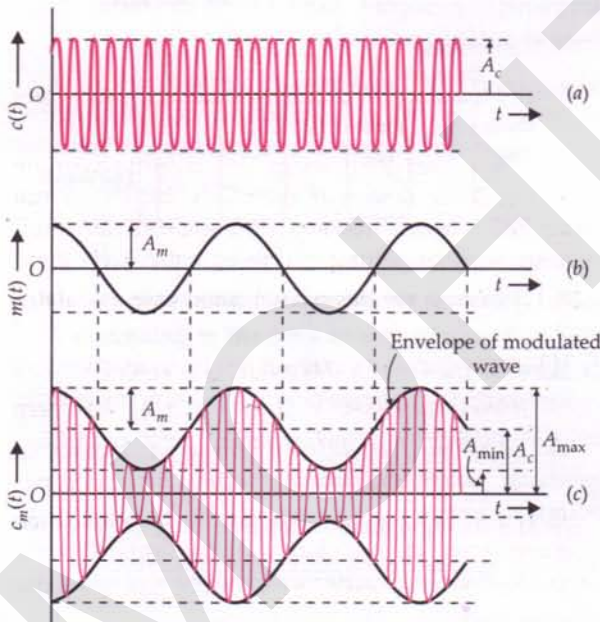


Fig. 15.8 Waveform of amplitude modulated carrier voltage.

Clearly, the amplitude of the carrier wave changes in accordance with the modulating wave. During the positive half cycle of the modulating signal, the amplitude of carrier wave increases. During the negative half cycle of the modulating signal, the amplitude of carrier wave decreases. Thus the top and bottom envelopes of the modulated wave are just the replica of the modulating signal.

Important features of the amplitude modulated wave :

1. Amplitude of the carrier wave changes according to the modulating signal.
2. Amplitude of the carrier wave changes at the frequency ' f_m ' of the modulating signal.
3. Frequency of the A.M. wave is equal to the unmodulated carrier frequency ' f_c '.

20. Define modulation factor. Express it in terms of maximum and minimum voltages of A.M. wave.

Modulation factor or depth of modulation. It represents the extent to which the amplitude of the carrier wave is changed by the modulating signal. It is defined as the ratio of the change in the amplitude of the carrier wave to the amplitude of the original carrier wave. It is also known as degree of modulation or modulation index. If A_m and A_c are the amplitudes of modulating and carrier signals respectively, then the modulation factor is given by

$$m = \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} = \frac{A_m}{A_c}$$

Modulation factor in terms of $A_{\max}$ and $A_{\min}$. Let $A_{\max}$ and $A_{\min}$ be the maximum and minimum voltages of A.M. wave.

From Fig. 15.8(c), it is clear that

$$A_m = \frac{A_{\max} - A_{\min}}{2}$$

and

$$\begin{aligned} A_c &= A_{\max} - A_m \\ &= A_{\max} - \frac{A_{\max} - A_{\min}}{2} \\ &= \frac{A_{\max} + A_{\min}}{2} \end{aligned}$$

Hence the modulation factor is

$$\begin{aligned} \mu &= \frac{A_m}{A_c} = \frac{(A_{\max} - A_{\min})/2}{(A_{\max} + A_{\min})/2} \\ &= \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \end{aligned}$$

Modulation factor m generally lies between 0 and 1.

The **percentage modulation** is given by

or
$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \times 100\%$$

21. Derive an expression for amplitude modulated wave.

Expression for amplitude modulated wave.

Let modulating signal, $m(t) = A_m \sin \omega_m t$
and carrier signal, $c(t) = A_c \sin \omega_c t$

where, $m(t)$ = instantaneous voltage of modulating wave

A_m = amplitude of modulating wave

$$\omega_m = 2\pi f_m$$

= angular frequency of modulating wave

$c(t)$ = instantaneous voltage of carrier wave

A_c = amplitude of carrier wave

$$\omega_c = 2\pi f_c$$

= angular frequency of carrier wave

The amplitude of the carrier wave varies at the frequency ω_m of the modulating wave. So amplitude of the modulated wave is given by

$$\begin{aligned} A &= A_c + m(t) = A_c + A_m \sin \omega_m t \\ &= A_c + \mu A_c \sin \omega_m t = A_c (1 + \mu \sin \omega_m t) \end{aligned}$$

Here $\mu = \frac{A_m}{A_c}$ is the modulation index.

The instantaneous voltage of the A.M. is given by

$$\begin{aligned} c_m(t) &= A \sin \omega_c t = A_c (1 + \mu \sin \omega_m t) \times \sin \omega_c t \\ &= A_c \sin \omega_c t + \frac{\mu A_c}{2} \times 2 \sin \omega_c t \cdot \sin \omega_m t \\ &= A_c \sin \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c - \omega_m) t \\ &\quad - \cos(\omega_c + \omega_m) t] \end{aligned}$$

$$\text{or } c_m(t) = A_c \sin \omega_c t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_m) t + \frac{\mu A_c}{2} \cos(\omega_c - \omega_m) t \quad \dots(1)$$

22. What is meant by sideband frequencies in a carrier wave? Justify that bandwidth of an A.M. wave is $2f_m$, where f_m is the modulating frequency.

Sideband frequencies and bandwidth of A.M. wave. Equation (1) shows that a sinusoidally modulated wave is the sum of three different waves:

1. One wave is the original carrier wave of amplitude A_c and frequency $f_c = \omega_c / 2\pi$.
2. The second wave has amplitude $\mu A_c / 2$ and frequency $(f_c + f_m) [= (\omega_c + \omega_m) / 2\pi]$.
3. The third wave has amplitude $\mu A_c / 2$ and frequency $(f_c - f_m) [= (\omega_c - \omega_m) / 2\pi]$.

Clearly, the process of modulation does not change the original carrier frequency but produces two new frequencies $(f_c + f_m)$ and $(f_c - f_m)$ which are called **sideband frequencies**.

The sum of the carrier frequency and the modulating frequency $(f_c + f_m)$ is called **upper side band frequency (USB)**. Thus

$$\text{USB} = f_c + f_m$$

The difference of the carrier frequency and the modulating frequency $(f_c - f_m)$ is called **lower side band frequency (LSB)**. Thus

$$\text{LSB} = f_c - f_m$$

As the carrier frequency is many times greater than the modulating frequency, so the sideband frequencies lie close to the carrier frequency, as shown in Fig. 15.9.

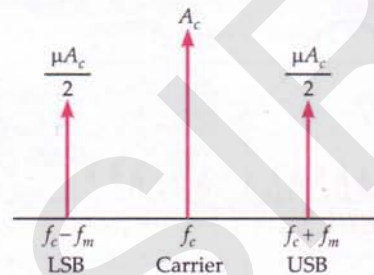


Fig. 15.9 Frequency spectrum of sinusoidally A.M. wave.

The difference of the highest and the lowest frequencies present in the A.M. wave is called its **bandwidth**.

$$\therefore \text{Bandwidth} = (f_c + f_m) - (f_c - f_m) = 2f_m$$

Hence the bandwidth of an amplitude modulated wave is twice the frequency of the modulating signal.

If the broadcast frequencies (of carrier waves) are so sufficiently spaced that their side bands do not overlap, then we can broadcast different stations simultaneously without getting interfered with each other.

23. (a) Explain various types of amplitude modulated waveforms with the help of diagrams.

Or

Sketch the waveform of a carrier that has been amplitude modulated to each of the following levels:

- (i) 0% (ii) 50% (iii) 100% (iv) 150%.

(b) Hence explain the significance of modulation factor.

(a) **Various types of amplitude modulated waves.** We know that

$$\begin{aligned} \mu &= \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} \\ &= \frac{A_m}{A_c} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \end{aligned}$$

- (i) When $\mu = 0\%$, i.e., the modulating signal $A_m = 0$ and so the amplitude of the carrier wave remains unchanged, as shown in Fig. 15.10(b).
- (ii) When $\mu = 50\% = 0.50$, $A_{\max} = 3 A_{\min}$. The modulated carrier wave is as shown in Fig. 15.10(c). The carrier is said to be **under modulated**.

- (iii) When $\mu = 100\% = 1$, $A_{\min} = 0$. The modulated carrier wave is as shown in Fig. 15.10(d). The carrier is said to be *fully modulated*.
- (iv) When $\mu = 150\% = 1.5$, $A_{\max}$ becomes 2.5 times the carrier amplitude. The carrier wave is said to be *overmodulated*. But the negative half cycle of the modulating signal gets clipped, as shown in Fig. 15.10(e).

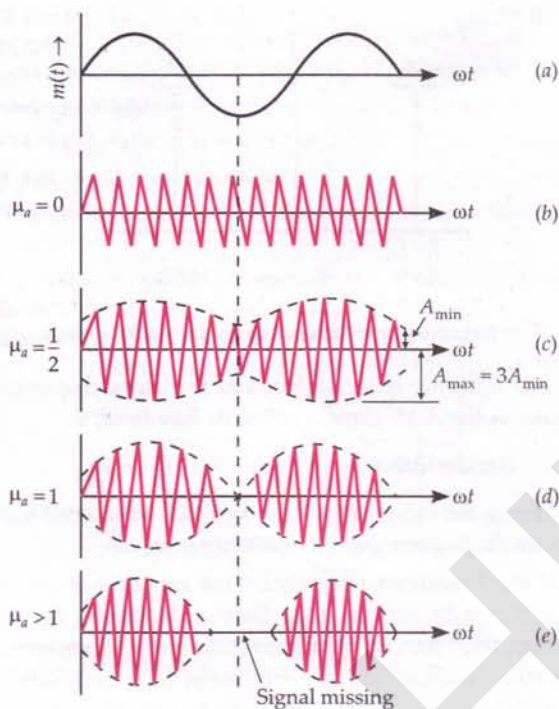


Fig. 15.10

(b) **Importance of modulation factor.** The modulation factor determines the strength and quality of the transmitted signal. If m is small, the amount of variation in the carrier amplitude is small. As a result, the transmitted audio signal will not be strong enough. The greater the degree of modulation, the stronger and clearer will be the audio signal. However, if the carrier is overmodulated ($\mu > 1$), distortion will occur during reception. Modulated wave is missing during negative peak of the modulating (audio) signal and the envelope is discontinuous. So modulation index must not be allowed to exceed unity.

24. Discuss the advantages and disadvantages of amplitude modulation.

Advantages of amplitude modulation :

1. It is an easier method for transmitting and receiving voice signals.
2. It requires simple and cheaper transmitters and receivers.

3. Its transmission requires low carrier frequencies of 0.5 – 20 MHz.
4. Area in which AM transmission can be received is much larger than that in case of FM transmission.

Disadvantages of amplitude modulation :

1. Amplitude modulation suffers from noise.
2. Quality of audio signal is poor.
3. Efficiency of AM transmission is low.

25. What is phase modulation ? Draw the modulated wave for phase modulation of sine wave carrier by a sine wave signal.

Phase modulation. It is the process by which the phase angle of the high frequency carrier wave changes in accordance with the instantaneous value of the modulating signal.

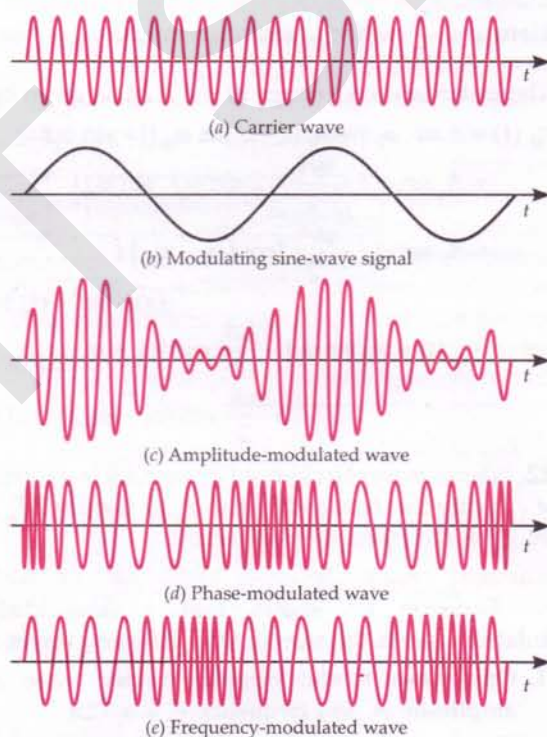


Fig. 15.11 Different types of modulation.

Fig. 15.11(a) shows a high frequency carries waves, Fig. 15.11(b) shows a modulating sine wave signal and Fig. 15.11(d) shows the phase modulated wave. Comparing Fig. 15.11(d) with Fig. 15.11(b), we find the frequency of the phase modulated (PM) wave changes in accordance with the phase of the modulating signal. Frequency is maximum when the phase angle is zero, it decreases when the phase angle increases and becomes minimum when the phase angle becomes π . Frequency again increases and becomes maximum when the phase angle is 2π .

26. What is frequency modulation ? Draw voltage waveform for a frequency modulated wave. Give its important features.

Frequency modulation. It is the process of changing the frequency of the carrier voltage in accordance with the instantaneous value of the modulating voltage. The original frequency of the carrier signal is called *resting frequency* or *centre frequency* (f_c) and is the allotted frequency of the transmitter. When the audio signal is applied, the carrier frequency changes. The amount by which the frequency of the carrier wave changes or shifts above or below the resting frequency is called *frequency deviation* (δ).

Frequency modulated wave. Fig. 15.11(a) shows a carrier wave of high frequency, Fig. 15.11(b) shows modulating (audio) signal of low frequency and Fig. 15.11(c) shows the resultant frequency modulated wave. In frequency modulation, the frequency of the carrier wave is varied in accordance with the amplitude of the modulating signal whereas the amplitude of the carrier wave remains unchanged. So in an F.M. wave, the information is carried as the variations in its frequency.

The frequency of the modulated carrier increases as the modulating amplitude increases and decreases as the modulating amplitude decreases. When the modulating amplitude has maximum positive value, the modulated carrier has a maximum frequency and when the modulating amplitude has a maximum negative value, the modulated carrier has a minimum frequency. When the modulating amplitude is zero, the modulated carrier has the original carrier frequency f_c .

Important features of frequency modulated wave :

1. The amplitude of the modulated signal remains same as that of the carrier wave but its frequency changes in accordance with the modulating signal.
2. The amount of frequency deviation is proportional to the instantaneous value of the modulating signal i.e., $\delta \propto m(t)$. Thus louder the audio signal, greater is the frequency change in the modulated carrier.
3. The rate of frequency deviation is equal to the modulating frequency.

27. Define the terms frequency deviation and modulation index for F.M. wave. Give some important features of the frequency spectrum of an F.M.

Frequency deviation : The maximum amount by which the frequency of the modulated wave changes or shifts above or below the carrier frequency is called *frequency deviation* (δ). Thus

$$\delta = f_{\max} - f_c = f_c - f_{\min}$$

Modulation index : The modulation index (μ_f) for an F.M. wave is defined as the ratio of the maximum frequency deviation to the modulating frequency.

The frequency spectrum of the F.M. wave is much more complex than the A.M. wave and has following important features :

1. The F.M. wave consists of the carrier and an infinite number of sidebands of frequencies $(f_c \pm f_m)$, $(f_c \pm 2f_m)$, $(f_c \pm 3f_m)$,
2. The number of sidebands with significant amplitudes depends on a modulation index μ_f . When the frequency deviation is increased by keeping modulating frequency fixed, the number of sidebands increases. When the modulating frequency is increased by keeping frequency deviation fixed, the number of sidebands decreases.
3. The sidebands are separated from the carrier by f_m , $2f_m$, $3f_m$, ..., and thus have a recurrence frequency of f_m .
4. The sidebands at equal distances from the carrier have equal amplitudes i.e., the sidebands are distributed symmetrically about the carrier frequency f_c , as shown in Fig. 15.12.

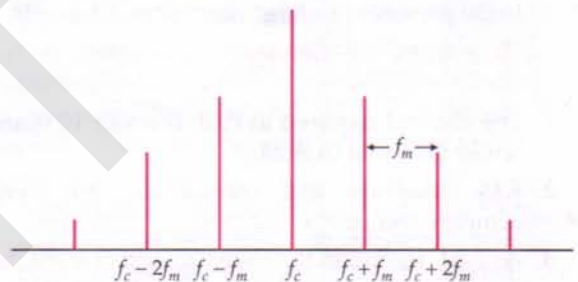


Fig. 15.12 Frequency spectrum of sinusoidally modulated F.M. signal for $\mu_f = 1$.

28. Write an expression for the bandwidth in frequency modulation.

Bandwidth in frequency modulation. In a frequency modulated wave, the information (audio signal) is contained in the sidebands. But the various sidebands are separated from each other by the modulating frequency f_m .

$\therefore$ Bandwidth = $2n \times$ modulating frequency = $2n \times f_m$
where n is the number of significant sidebands in the F.M. wave.

29. Give the advantages and disadvantages of frequency modulation over amplitude modulation.

Advantages of frequency modulation over amplitude modulation :

1. F.M. transmission is highly efficient because all the transmitted power is useful but in A.M. transmission, most of the power goes waste in transmitting the carrier alone.

- Since amplitude remains constant in F.M., so amplitude limiters can be used to reduce noise in F.M.
- Depth of amplitude modulation cannot be increased beyond 100% without causing distortion. There is no such limit in F.M. Hence noise in F.M. can be reduced further by increasing deviation δ .
- In F.M., standard frequency allocations provide a guard band between commercial F.M. stations. This reduces adjacent channel interference.
- In F.M., space wave propagation is used. So its radius of operation is slightly more than the line of sight. It makes possible to operate several transmitters in the same frequency range with negligible interference.
- In the VHF or UHF bands where F.M. operates, there is less noise than in HF or MF bands where A.M. operates.
- F.M. transmission gives high fidelity reception due to the presence of a large number of sidebands.

Disadvantages of frequency modulation over amplitude modulation :

- The channel required in F.M. is about 10 times wider than that in A.M.
- F.M. receivers and transmitters are very complex and costly.
- As F.M. reception is limited to line of sight, so its area of reception is much smaller than that for A.M.

Table 15.2 Frequency ranges for commercial radio and TV broadcast

Nature of broadcast	Frequency band
AM radio	0.5 – 20 MHz
FM radio	88 – 108 MHz
VHF TV	47 – 230 MHz
UHF TV	470 – 960 MHz

15.11 PRODUCTION OF AMPLITUDE MODULATED WAVE

30. With the help of block diagram, explain the principle of generating an amplitude modulated wave from a baseband signal.

Principle of a production of amplitude modulated wave. Fig. 15.13(a) shows a block diagram of simple square-law modulator for obtaining an AM wave. This modulator requires a means for adding modulating and carrier signals, a non-linear element and a bandpass

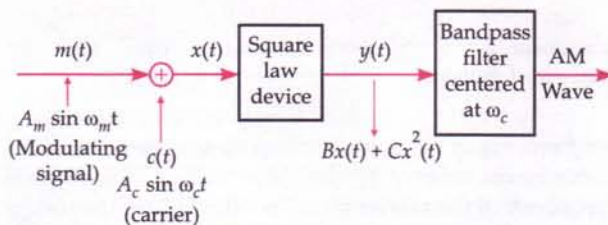


Fig. 15.13 (a) Block diagram of a square-law modulator for obtaining AM signal.

filter for extracting the desired modulated wave. The non-linear element used in a square-law modulator may be a suitably biased diode or a transistor.

Here the modulating signal $A_m \sin \omega_m t$ is added to the carrier signal $A_c \sin \omega_c t$ to produce the input signal,

$$x(t) = A_m \sin \omega_m t + A_c \sin \omega_c t$$

This signal is passed through the square-law device which gives the output,

$$y(t) = Bx(t) + Cx^2(t)$$

where B and C are constants. Therefore,

$$\begin{aligned} y(t) &= B[A_m \sin \omega_m t + A_c \sin \omega_c t] \\ &\quad + C[A_m \sin \omega_m t + A_c \sin \omega_c t]^2 \\ &= BA_m \sin \omega_m t + BA_c \sin \omega_c t \\ &\quad + C[A_m^2 \sin^2 \omega_m t + A_c^2 \sin^2 \omega_c t \\ &\quad + 2A_m A_c \sin \omega_m t \sin \omega_c t] \end{aligned}$$

Using the relations

$$\sin^2 A = (1 - \cos 2A)/2$$

and $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$, we get

$$\begin{aligned} y(t) &= BA_m \sin \omega_m t + BA_c \sin \omega_c t \\ &\quad + \frac{C}{2}(A_m^2 + A_c^2) - \frac{CA_m^2}{2} \cos 2\omega_m t - \frac{CA_c^2}{2} \cos 2\omega_c t \\ &\quad + CA_m A_c \cos(\omega_c - \omega_m)t - CA_m A_c \cos(\omega_c + \omega_m)t \end{aligned}$$

This equation has a d.c. term $\frac{C}{2}(A_m^2 + A_c^2)$ and sinusoidal terms of frequencies ω_m , $2\omega_m$, ω_c , $2\omega_c$, $\omega_c - \omega_m$ and $\omega_c + \omega_m$. When this signal is passed through a bandpass filter (A bandpass filter rejects low and high frequencies and allows a band of frequencies to pass) centred at ω_c , rejects d.c. signal and the sinusoidal signals of frequencies ω_m , $2\omega_m$ and $2\omega_c$, and allows the frequencies ω_c , $\omega_c - \omega_m$ and $\omega_c + \omega_m$. Consequently the output of bandpass filter is

$$\begin{aligned} y(t) &= BA_c \sin \omega_c t + C A_m A_c \cos(\omega_c - \omega_m)t \\ &\quad - CA_m A_c \cos(\omega_c + \omega_m)t \end{aligned}$$

Like any AM wave, clearly the modulated wave consists of a carrier wave of frequency ω_c and two sidebands of frequencies $\omega_c - \omega_m$ and $\omega_c + \omega_m$.

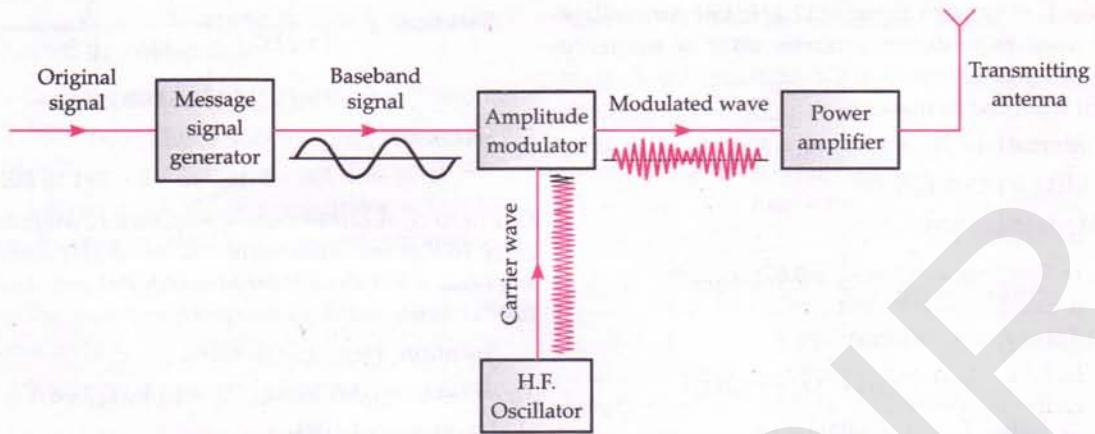


Fig. 15.13 (b) Block diagram of a transmitter of AM wave.

Fig. 15.13(b) shows a block diagram for the transmitter of an AM wave. The information signal is fed to the message signal generator. The message signal generator is just a suitable transducer which converts original signal into an electric signal, called baseband signal. A high frequency oscillator generates the carrier wave. The modulator superimposes the baseband signal over the carrier wave and gives out the modulated wave. The signal is passed through a power amplifier to increase its power and then fed to a transmitting antenna of appropriate size for radiation into space.

31. With the help of a simple circuit diagram, briefly explain the production of amplitude modulated carrier wave.

Production of amplitude modulated carrier wave.

The amplitude of an amplitude modulated carrier waves increases or decreases in accordance with the instantaneous value of the modulating or message signal. A simple circuit for producing amplitude modulated wave is shown in Fig. 15.14.

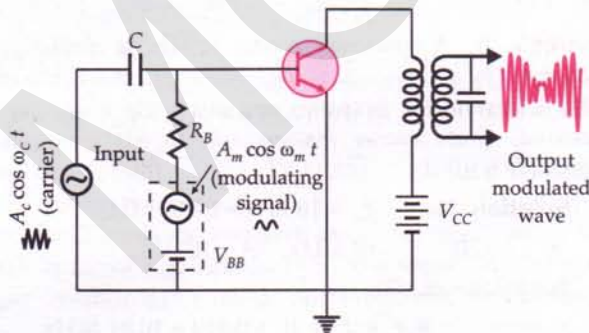


Fig. 15.14 Circuit for an amplitude modulator (Base modulator).

The circuit shown is simply a CE amplifier for the carrier wave signal. The modulating signal $m(t) = A_m \sin \omega_m t$ is applied on the base. Thus the base

biasing voltage is not constant d.c. but is the sum of a constant d.c. voltage V_{BB} and the modulating voltage $A_m \sin \omega_m t$. As the base biasing changes, the amplification produced also changes. The output voltage is a carrier signal varying in amplitude in accordance with the biasing modulating voltage. This gives an amplitude modulated wave.

Examples based on Amplitude Modulation

Formulae Used

1. Modulation factor, $\mu = \frac{A_m}{A_c}$ or $\mu = \frac{A_m}{A_c} \times 100\%$
2. If $A_{\max}$ and $A_{\min}$ are the maximum and minimum amplitudes of the carrier wave, then

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \times 100\%$$
3. Modulating voltage, $m(t) = A_m \sin \omega_m t$
4. Carrier voltage, $c(t) = A_c \sin \omega_c t$
5. Instantaneous voltage of A.M. wave is

$$c_m(t) = A_c (1 + \mu \sin \omega_m t) \sin \omega_c t$$

$$= A_c \sin \omega_c t - \frac{\mu A_c}{2} \cos (\omega_c + \omega_m) t$$

$$+ \frac{\mu A_c}{2} \cos (\omega_c - \omega_m) t$$
6. Component frequencies of A.M. wave are :
 (a) Carrier frequency $= f_c$
 (b) USB $= f_c + f_m$ (c) LSB $= f_c - f_m$
7. Bandwidth $= (f_c + f_m) - (f_c - f_m) = 2f_m$

Units Used

Voltages A_m , A_c , $A_{\max}$ and $A_{\min}$ are in volts. Modulation factor μ has no unit. All frequencies are in Hz.

Example 1. A message signal of 12 kHz and peak voltage 20 V is used to modulate a carrier wave of frequency 12 MHz and peak voltage 30 V. Calculate the (i) modulation index (ii) side-band frequencies. [CBSE OD 08]

Solution. Here $A_m = 20$ V, $A_c = 30$ V, $f_m = 12$ kHz, $f_c = 12$ MHz = 12000 kHz

(i) Modulation index,

$$\mu = \frac{A_m}{A_c} = \frac{20}{30} = 0.67.$$

(ii) USB = $f_c + f_m = 12000 + 12$

$$= 12012 \text{ kHz} = \mathbf{12.012 \text{ MHz.}}$$

$$\text{LSB} = f_c - f_m = 12000 - 12$$

$$= 11988 \text{ kHz} = \mathbf{11.988 \text{ MHz.}}$$

Example 2. A sinusoidal carrier voltage of 100 V is amplitude modulated by a sinusoidal voltage of frequency 10 kHz resulting in maximum modulated carrier amplitude of 120 V. Calculate the modulation factor.

Solution. Here $A_c = 100$ V, $A_{\max} = 120$ V

$$\begin{aligned} \therefore \mu &= \frac{A_m}{A_c} \\ &= \frac{A_{\max} - A_c}{A_c} = \frac{120 - 100}{100} = \mathbf{0.2.} \end{aligned}$$

Example 3. The maximum peak-to-peak voltage of an A.M. wave is 16 mV and the minimum peak-to-peak voltage is 8 mV. What is the modulation factor?

Solution. The situation is shown in Fig. 15.15.

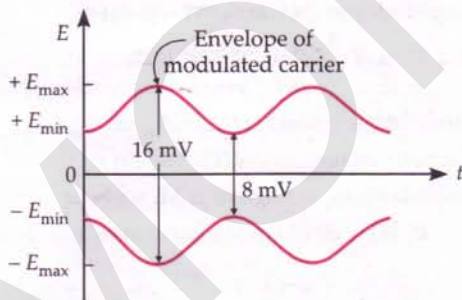


Fig. 15.15

Clearly, $E_{\max} = 16/2 = 8$ mV,

$$E_{\min} = 8/2 = 4 \text{ mV.}$$

$$\therefore m = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}} = \frac{8 - 4}{8 + 4} = \frac{1}{3} = \mathbf{0.33.}$$

Example 4. The tuned circuit of the oscillator in a simple A.M. transmitter uses a coil of 40μ H and shunt capacitor of value 1 nF. If the oscillator output is modulated by audio frequencies upto 5 kHz, what is the frequency range occupied by sidebands?

$$\begin{aligned} \text{Solution. } f_c &= \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{40 \times 10^{-6} \times 1 \times 10^{-9}}} \\ &= 7.96 \times 10^5 \text{ Hz} = \mathbf{796 \text{ kHz}} \end{aligned}$$

Frequency range of side bands

$$= 796 - 5 \text{ to } 796 + 5 = \mathbf{791 \text{ to } 801 \text{ kHz.}}$$

Example 5. A carrier wave of frequency 10 MHz and peak value 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Calculate the frequency and amplitude of the two sidebands and draw the spectrum.

Solution. Here $f_c = 10$ MHz,

$$f_m = 5 \text{ kHz} = 0.005 \text{ MHz, } E_c = 10 \text{ V, } E_m = 6 \text{ V}$$

Frequency of USB

$$= f_c + f_m = 10 + 0.005 = \mathbf{10.005 \text{ MHz}}$$

Frequency of LSB

$$= f_c - f_m = 10 - 0.005 = \mathbf{9.995 \text{ MHz}}$$

$$m = \frac{E_m}{E_c} = \frac{6}{10} = 0.6$$

$\therefore$ Amplitude of USB or LSB

$$= \frac{mE_c}{2} = \frac{0.6 \times 10}{2} = \mathbf{3 \text{ V}}$$

The frequency spectrum for the A.M. wave is shown in Fig. 15.16.

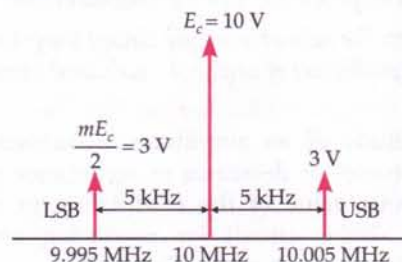


Fig. 15.16

Example 6. A sinusoidal carrier voltage is amplitude modulated by a sinusoidal voltage of 10 kHz to a depth of 30%. Calculate the frequency and amplitude of the two sidebands if the carrier frequency is 10 MHz and its amplitude is 40 V.

Solution. Here $f_m = 10$ kHz = 0.010 MHz,

$$\mu = 30\% = 0.30, f_c = 10 \text{ MHz, } A_c = 40 \text{ V}$$

Frequency of USB

$$= f_c + f_m = 10 + 0.010 = \mathbf{10.01 \text{ MHz}}$$

Frequency of LSB

$$= f_c - f_m = 10 - 0.010 = \mathbf{9.99 \text{ MHz}}$$

Amplitude of each sideband

$$= \frac{\mu A_c}{2} = \frac{0.30 \times 40}{2} = \mathbf{6.0 \text{ V.}}$$

Example 7. A sinusoidal carrier voltage of frequency 1200 kHz is amplitude modulated by a sinusoidal voltage of frequency 20 kHz resulting in maximum and minimum modulated carrier amplitudes of 110 V and 90 V respectively. Calculate (i) the frequency of lower and upper sidebands (ii) the unmodulated carrier amplitude (iii) the modulation index and (iv) the amplitude of each sideband.

Solution. (i) Lower sideband frequency

$$= f_c - f_m = 1200 - 20 = 1180 \text{ kHz}$$

Upper sideband frequency

$$= f_c + f_m = 1200 + 20 = 1220 \text{ kHz}$$

(ii) Unmodulated carrier amplitude

$$A_c = \frac{A_{\max} + A_{\min}}{2} = \frac{110 + 90}{2} = 100 \text{ V}$$

(iii) Modulation index,

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{110 - 90}{110 + 90} = 0.1$$

Amplitude of each sideband

$$= \frac{\mu A_c}{2} = \frac{0.1 \times 100}{2} = 5 \text{ V.}$$

Example 8. A sinusoidal voltage amplitude modulates another sinusoidal voltage of amplitude 2 kV to result in two sidebands, each of amplitude 200 V. Find the modulation index.

Solution. Amplitude of each sideband $= \frac{\mu A_c}{2}$.

$$\therefore 200 \text{ V} = \frac{\mu \times 2 \text{ kV}}{2} = \frac{\mu \times 2000 \text{ V}}{2}$$

$$\text{or } \mu = \frac{2 \times 200}{2000} = 0.2.$$

Example 9. An audio signal given by, $m(t) = 30 \sin(2\pi \times 2500 t)$ volts is used to amplitude modulate a carrier wave given by $c(t) = 60 \sin(2\pi \times 200,000 t)$ volts. Find (i) percent modulation, (ii) components of modulated wave and (iii) the amplitude of each sideband.

Solution. Here

$$m(t) = A_m \sin 2\pi f_m t = 30 \sin(2\pi \times 2500 t) \text{ volts}$$

$$\text{and } c(t) = A_c \sin 2\pi f_c t = 60 \sin(2\pi \times 200,000 t) \text{ volts.}$$

$$\text{Clearly, } A_m = 30 \text{ V, } f_m = 2500 \text{ Hz,}$$

$$A_c = 60 \text{ V, } f_c = 200,000 \text{ Hz}$$

$$(i) m = \frac{A_m}{A_c} \times 100 = \frac{30}{60} \times 100 = 50\%.$$

(ii) The three components of the modulated carrier wave are :

$$(a) f_c = 200,000 \text{ Hz} = 200 \text{ kHz.}$$

$$(b) \text{ USB} = 200,000 + 2500 = 202,500 \text{ Hz} = 202.5 \text{ kHz.}$$

$$(c) \text{ LSB} = 200,000 - 2500$$

$$= 197,500 \text{ Hz} = 197.5 \text{ kHz.}$$

Example 10. A bandwidth of 5 MHz is available for A.M. transmission. If the maximum audio signal frequency used for modulating the carrier is not to exceed 5 kHz, how many stations can be broadcast within this band simultaneously without interfering with each other?

Solution. Bandwidth required for each station

$$= 2 \times \text{Maximum frequency of audio signal}$$

$$= 2 \times 5 = 10 \text{ kHz}$$

Number of stations which can be broadcast within bandwidth of 5 MHz

$$= \frac{5 \text{ MHz}}{10 \text{ kHz}} = \frac{5000 \text{ kHz}}{10 \text{ kHz}} = 500.$$

Example 11. A 1000 kHz carrier is amplitude modulated by an audio signal with frequency span of 100 – 5000 Hz. Determine (i) width or frequency range of each sideband (ii) maximum and minimum frequencies of USB (iii) maximum and minimum frequencies of LSB and (iv) bandwidth of the channel.

Solution. (i) Width of each side band

$$= (f_c - f_m^{\min}) - (f_c - f_m^{\max}).$$

$$= f_m^{\max} - f_m^{\min} = 5000 - 100 = 4900 \text{ Hz.}$$

(ii) Maximum frequency of USB

$$= f_c + f_m^{\max} = 1000 + 5 = 1005 \text{ kHz}$$

Minimum frequency of USB

$$= f_c + f_m^{\min} = 1000 + 0.1 = 1000.1 \text{ kHz.}$$

(iii) Maximum frequency of LSB

$$= f_c - f_m^{\min} = 1000 - 0.1 = 999.9 \text{ kHz}$$

Minimum frequency of LSB

$$= f_c - f_m^{\max} = 1000 - 5 = 995 \text{ kHz.}$$

(iv) Bandwidth of the channel

$$= \text{USB (max)} - \text{LSB (min)} = 1005 - 995 = 10 \text{ kHz.}$$

Clearly, bandwidth of the channel is twice the highest modulating frequency.

Example 12. In a broadcasting station, a 2500 kHz carrier is amplitude modulated by an audio signal of frequency range, 50 – 15000 Hz. What are the frequency ranges of USB and LSB? What bandwidth of RF amplifier is required to handle the output?

Solution. Frequency range of modulating signal is 50 – 15000 Hz

$$\text{i.e., } 0.05 - 15 \text{ kHz and } f_c = 2500 \text{ kHz}$$

Frequency range of USB is from $f_c + 0.05$ to $f_c + 15$ i.e., from 2500.05 kHz to 2515 kHz.

Frequency range of LSB is from $f_c - 15$ to $f_c - 0.05$ i.e., from 2499.95 to 2485 kHz.

Required bandwidth = USB (max) – LSB (min)

$$= 2515 - 2485 = 30 \text{ kHz.}$$

Example 13 An amplitude modulated wave is represented as

$$c_m(t) = 5(1 + 0.6 \cos 6280 t) \sin 211 \times 10^4 t, \text{ volts.}$$

- What are the minimum and maximum amplitudes of the A.M. wave?
- What frequency components are contained in the modulated wave?
- What are the amplitudes of the components?

Solution. Given the A.M. wave,

$$c_m(t) = 5(1 + 0.6 \cos 6280 t) \sin 211 \times 10^4 t, \text{ volts.}$$

Comparing with the standard A.M. wave,

$$c_m(t) = A_c(1 + m \cos \omega_m t) \sin \omega_c t,$$

we get $A_c = 5 \text{ V}, \mu = 0.6$

Modulating frequency,

$$f_m = \frac{\omega_m}{2\pi} = \frac{6280}{2\pi} = 1 \text{ kHz}$$

Carrier frequency,

$$f_c = \frac{\omega_c}{2\pi} = \frac{211 \times 10^4}{2\pi} = 336 \text{ kHz}$$

(i) Minimum amplitude of A.M. wave

$$= A_c - \mu A_c = 5 - 0.6 \times 5 = 2 \text{ V}$$

Maximum amplitude of A.M. wave

$$= A_c + \mu A_c = 5 + 0.6 \times 5 = 8 \text{ V.}$$

(ii) Frequency components of the A.M. wave are

$$f_c - f_m, f_c, f_c + f_m \text{ i.e., } 336 - 1, 336, 336 + 1$$

or **335 kHz, 336 kHz, 337 kHz.**

(iii) The amplitudes of the three components are

$$\frac{\mu A_c}{2}, A_c, \frac{\mu A_c}{2}$$

$$\text{i.e., } \frac{0.6 \times 5}{2}, 5, \frac{0.6 \times 5}{2} \text{ or } 1.5 \text{ V, } 5 \text{ V, } 1.5 \text{ V.}$$

Problems For Practice

- A modulated carrier wave has maximum and minimum amplitudes of 800 mV and 200 mV. What is the percentage modulation? **[Ans. 60%]**
- A message signal of frequency 10 kHz and peak voltage 10 V is used to modulate a carrier of frequency 1 MHz and peak voltage 20 V. Determine: (i) the modulation index, and (ii) the sidebands produced. **[CBSE D 13C]**

[Ans. (i) 0.50 (ii) 1010 kHz, 990 kHz]

- A sinusoidal carrier voltage of frequency 1 MHz and amplitude 100 V is modulated by a sinusoidal voltage of frequency 5 kHz producing 50% modulation. Calculate the frequency and amplitude of upper and lower sidebands.

[Ans. 1005 kHz, 995 kHz, 25 V]

- A sinusoidal carrier voltage of amplitude 80 V and frequency 800 kHz is amplitude modulated by a sinusoidal voltage of frequency 15 kHz resulting in minimum carrier amplitude of 72 V. Find (i) the sideband frequencies and (ii) modulation index.

[Ans. (i) 785 kHz, 815 kHz (ii) 0.1]

- An audio signal given by $15 \sin 2\pi(2000t)$ amplitude modulates a sinusoidal carrier wave $60 \sin 2\pi(100,000t)$. Determine (i) percent modulation (ii) frequencies of modulating signal and carrier and (iii) frequency spectrum of the modulated wave.

[Ans. (i) 25% (ii) 2000 Hz, 100,000 Hz

(iii) 100 kHz, 102 kHz, 98 kHz]

- A 100 kHz bandwidth is to accommodate 10 A.M. broadcasts simultaneously. What is the maximum modulating frequency permissible for each station?

(Ans. 5 kHz)

HINTS

- Here $A_{\max} = 800 \text{ mV}$, $A_{\min} = 200 \text{ mV}$

$$\therefore \mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \times 100 = \frac{800 - 200}{800 + 200} \times 100 = 60\%.$$

- USB = 1 MHz + 5 kHz = **1005 kHz**

$$\text{LSB} = 1 \text{ MHz} - 5 \text{ kHz} = \textbf{995 kHz}$$

Amplitude of each sideband

$$= \frac{\mu A_c}{2} = \frac{0.5 \times 100}{2} = 25 \text{ V.}$$

- (i) LSB = 800 - 15 = **785 kHz**

$$\text{USB} = 800 + 15 = \textbf{815 kHz.}$$

$$(ii) \mu = \frac{A_m}{A_c} = \frac{A_c - A_{\min}}{A_c} = \frac{80 - 72}{80} = 0.1.$$

- Here $A_m = 15 \text{ V}$, $A_c = 60 \text{ V}$

$$(i) \mu = \frac{A_m}{A_c} = \frac{15}{60} = 0.25$$

$$\text{Percent modulation} = 0.25 \times 100 = 25\%.$$

$$(ii) f_m = 2000 \text{ Hz, } f_c = 100,000 \text{ Hz}$$

(By comparing with standard equation).

(iii) Component frequencies are

$$(a) f_c = 100,000 = \textbf{100 kHz.}$$

$$(b) f_c + f_m = 100,000 + 2000 \\ = 102,000 \text{ Hz} = \textbf{102 kHz.}$$

$$(c) f_c - f_m = 100,000 - 2000 \\ = 98,000 \text{ Hz} = \textbf{98 kHz.}$$

- If f is the maximum permissible modulation frequency, then

$$\frac{100 \text{ kHz}}{2f} = 10 \text{ or } f = \textbf{5 kHz.}$$

15.12 ANTENNA

32. What is an antenna ? What role does it play in a communication system ?

Antenna. An antenna is basically a small length of a conductor that is used to radiate or receive electromagnetic waves. It acts as a conversion device. At the transmitting end, it converts high frequency current into electromagnetic waves. At the receiving end, it transforms electromagnetic waves into electrical signal that is fed to the input of the receiver. Several types of antenna are used in communication.

33. What is a dipole antenna ? What should be the length of a dipole antenna ?

Dipole antenna. A simple dipole antenna is just a piece of a conductor of length much smaller than the wavelength of the carrier signal. Sometimes rounded structures called lobes, are attached to it to increase its length. The length of the dipole antenna should be such that it is in resonance with the carrier frequency. Usually, the length of a dipole antenna is taken equal to $\lambda/2$, where λ is the wavelength of the carrier wave. A dipole antenna is omni-directional and is used to transmit radiowaves.



Fig. 15.17 A dipole antenna.

34. Briefly describe a dish antenna. Where is it used ? Define gain of a dish antenna.

Dish antenna. A dish antenna is a highly directional antenna used for the transmission and reception of UHF and microwaves. As shown in Fig. 15.18, its active component is a small dipole placed at the focus of a parabolic reflector or spherical dish. In the receiver mode, the dish collects the electromagnetic radiations and focuses them on the dipole which converts them into electrical signals. In the transmitter mode, the dipole converts the electrical signal into electromagnetic waves and directs them on the reflector which are then transmitted as a parallel beam.



Fig. 15.18 A dish antenna.

The gain of a dish antenna is defined by

$$P = 6 \left(\frac{D}{\lambda} \right)^2$$

where D = diameter of the boundary of the reflector

λ = wavelength of electromagnetic radiations

Dish antennas are used in radars and satellite communication. The microwave signal from a satellite is received by a cable operator through a dish antenna.

15.13 RECEIVER

35. With the help of a schematic diagram, briefly explain the role played by a receiver in a communication system.

Receiver. The function of a receiver is to recover the original message or data from the modulated signal after its propagation through the communication channel. The basic process involved is the separation of the modulating signal from the carrier wave. This process is called demodulation. Fig. 15.19 shows a schematic representation of a receiver.

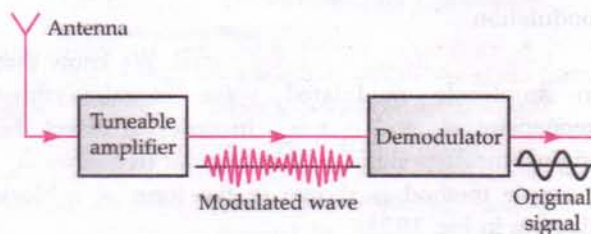


Fig. 15.19 Schematic representation of a receiver.

At the receiving station, the antenna picks up the modulated wave radiated from the transmitter. By a tuneable amplifier, the desired signal is selected (and unwanted signals are rejected) and then amplified. Finally, it is fed to a demodulator or detector which recovers the original signal from the modulated wave.

36. With the help of a block diagram, briefly explain the function of a receiver for detection of an AM wave.

Receiver for detection of an amplitude modulated wave. Fig. 15.20 shows the block diagram for a typical receiver used for detecting an amplitude modulated wave. At the receiving station, the antenna picks up the modulated wave radiated from the transmitter. By a tuneable amplifier, the signal is selected (and unwanted signals rejected) and then amplified. To facilitate further processing, the carrier frequency is changed to a lower frequency by a device called intermediate frequency (IF) stage. Then the signal is fed to a demodulator/detector which recovers the message signal from the modulated wave. To further strengthen the detected signal, it is fed to an amplifier.

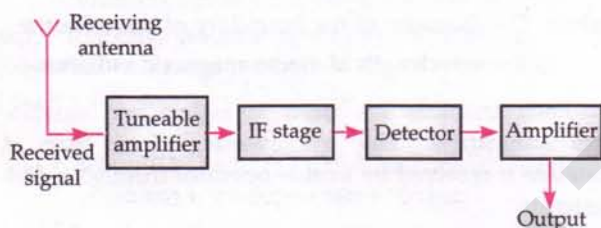


Fig. 15.20 Block diagram of a receiver for an AM wave.

15.14 DEMODULATION

37. What is demodulation? Draw a block diagram of a detector for an amplitude modulated signal explaining briefly the function of each of its components.

Demodulation or detection. The process of recovering the audio signal from the modulated wave is known as demodulation or detection. It is the reverse process of modulation.

Elementary detector for AM signal. We know that an amplitude modulated wave contains three frequencies ω_c , and $\omega_c \pm \omega_m$. In order to detect the original message signal $m(t)$ of angular frequency ω_m , a simple method is shown in the form of a block diagram in Fig. 15.21.

The modulated signal, as shown in Fig. 15.21(a) is passed through a rectifier which gives a series of positive half cycles of radio frequency pulses, as shown in Fig. 15.21(b). The envelope of these pulses is the message signal $m(t)$. In order to detect $m(t)$, the signal is passed through an envelope detector which may consist of a parallel combination of a low valued capacitor C and a resistor R .

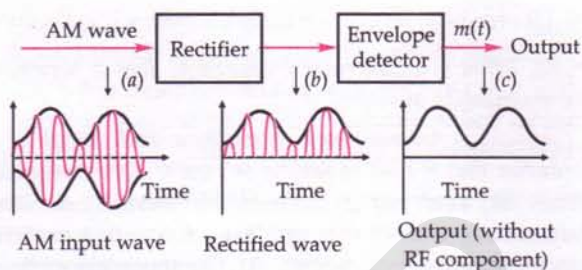


Fig. 15.21 Block diagram of a detector for AM signal. A voltage or current is plotted along vertical axis.

38. With the help of a suitable circuit diagram, briefly explain the working of a junction diode as an amplitude demodulator.

Junction diode as a detector. Fig. 15.22 shows a circuit diagram for a junction diode as detector for an amplitude modulated wave.

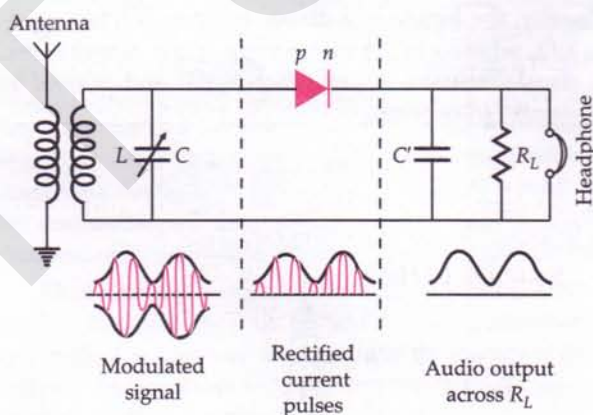


Fig. 15.22 Junction diode as a detector.

The input circuit is a parallel combination of inductance L and variable capacitor C . It is called *tuned circuit*. By adjusting the frequency of this circuit, the desired modulated radio signal is resonantly selected from the different signals picked up by the antenna. Diode rectifies this signal. So the output of the diode is a series of positive half cycles of radio frequency current pulses. The peaks of these pulses vary in accordance with the audio signal. To recover the audio signal, the rectified output is fed to a parallel combination of a low valued capacitor C' and a resistor R_L . The capacitor C' offers a low reactance $\left(X_C = \frac{1}{2\pi f C'}\right)$ to the high frequency carrier wave and a high reactance to low frequency audio wave. So the capacitor C' acts a

by-pass for high frequency carrier waves while the low frequency audio wave appears across the resistor R_L . This sends current through a headphone to reproduce original audio signal.

For exact recovery of the original message, the following condition must be satisfied :

$$\frac{1}{f_c} < R_L C < \frac{1}{f_m}$$

Examples based on Length of Antenna and Demodulation Process

Formulae Used

1. Length of a dipole antenna, $l = \frac{\lambda}{2} = \frac{c}{2f}$
2. Condition for satisfactory detection by a diode,

$$\frac{1}{f_c} < RC < \frac{1}{f_m}$$

Units Used

Wavelength λ is in metre, frequencies f_c and f_m in hertz, resistance R in Ω and capacitance C in farad.

Example 14. Calculate the length of a half wave dipole at (i) 30 MHz (ii) 300 MHz and (iii) 3000 MHz. What interference do you draw from the results ? [CBSE Text]

Solution. (i) Here $f = 30 \times 10^6$ Hz,
 $c = 3 \times 10^8$ ms⁻¹.

$$\therefore \lambda = \frac{c}{f} = \frac{3 \times 10^8}{30 \times 10^6} = 10 \text{ m}$$

Length of dipole = $\lambda/2 = 5$ m.

(ii) Here $f = 300 \times 10^6$ Hz

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ m}$$

Length of dipole = $\lambda/2 = 0.5$ m.

(iii) Here $f = 3000 \times 10^6$ Hz

$$\therefore \lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ ms}^{-1}}{3000 \times 10^6 \text{ s}^{-1}} = 0.1 \text{ m}$$

Length of dipole = $\lambda/2 = 0.05 \text{ m} = 5 \text{ cm}$

Clearly, the length of dipole decreases with the increase in the frequency of carrier waves.

Example 15. In a diode AM-detector, the output circuit consists of $R = 1 \text{ k}\Omega$ and $C = 10 \text{ pF}$. A carrier signal of 100 kHz is to be detected. Is it good ? If yes, then explain why ? If not, what value would you suggest ? [CBSE OD 07C]

Solution. For demodulation, $\frac{1}{f_c} \ll RC$

$$\frac{1}{f_c} = \frac{1}{100 \times 10^3} = 10^{-5} \text{ s}$$

$$RC = 10^3 \times 10 \times 10^{-12} \text{ s} = 10^{-8} \text{ s}$$

We see that $1/f_c$ here is not less than RC . Therefore it is not good.

Let us take $C = 1 \mu\text{F}$. Then $RC = 10^3 \times 10^{-6} \text{ s} = 10^{-3} \text{ s}$

This value is much larger than $1/f_c$. Thus a capacitor of $1 \mu\text{F}$ provides a satisfactory circuit for a detector.

Problems For Practice

1. Calculate the length of a half wave dipole antenna at : (i) 1 MHz (ii) 100 MHz and (iii) 1000 MHz. What conclusion you draw from the results ?

[Ans. (i) 150 m (ii) 1.5 m (iii) 15 cm ; the length of dipole antenna decreases with the increase in frequency of carrier wave]

2. In an amplitude modulation diode detector, the output circuit consists of $R = 1 \text{ M}\Omega$ and $C = 1 \text{ pF}$. Calculate the carrier frequency it can detect.

(Ans. $f_c \gg 1 \text{ MHz}$)

HINTS

1. Proceed as in Example 14.
2. Here $RC = 10^6 \times 10^{-12} \text{ s} = 10^{-6} \text{ s}$

For demodulation, $\frac{1}{f_c} \gg RC$

$$\text{or } f_c \gg \frac{1}{10^{-6}} = 10^6 \text{ Hz or } f_c \gg 1 \text{ MHz.}$$

15.15 COMMUNICATION CHANNELS AND THEIR BANDWIDTHS

39. What is a communication channel ? Define its bandwidth. Briefly describe the various communication channels employed in communication. On what factors does the quality of transmission depend in each case ? How do we make the choice of communication channel ?

Communication channel. The physical path between the transmitter and receiver is called the communication channel or the transmission medium of communication system.

Bandwidth of a communication channel. The bandwidth of a communication channel is defined as the difference between the highest and lowest frequencies that the channel allows to pass through it (i.e., its passband)

The bandwidth of a communication channel must be sufficiently large (wide) to pass all significant

information frequencies. In other words, *the bandwidth of a communication channel must be equal to or greater than the bandwidth of the information signal*. For example, if a cable television transmission has a passband from 500 kHz to 5000 kHz, it has a bandwidth of 4500 kHz or 4.5 MHz.

The transmission media can be divided broadly into *two* categories :

1. Guided media. Such media are used for point-to-point communication. Important examples of guided media are :

- (i) Twisted pair
- (ii) Coaxial cable
- (iii) Optical fibre

The quality of transmission in guided media is determined mainly by the nature of the media.

2. Unguided media. Free space is an example of unguided medium. In unguided media, the spectrum or frequency band of the signal produced by the transmitting antenna is more important than the medium in determining the quality of transmission.

Depending on the nature of communication channel, the communication is classified broadly into *three* categories :

1. Line communication. It invokes point-to-point contact between transmitter and receiver. It occurs through guided media such as twisted pair and coaxial cable.

Coaxial cable is a widely used wire medium, which offers a bandwidth of 750 MHz. These cables are normally operated below 18 GHz ($1 \text{ GHz} = 10^{12} \text{ Hz}$).

2. Optical communication. It makes use of a light beam in carrying an information from one point to another through a guided medium like optical fibre.

Optical communication using fibres is carried in the frequency range (of 1 THz to 1000 THz, Microwaves to Ultraviolet waves, $1 \text{ THz} = 10^{12} \text{ Hz}$). An optical fibre can have a transmission bandwidth above 100 GHz.

3. Space communication. Here electromagnetic waves of different frequencies are used to carry the information through the physical space acting as the transmission medium. Radio, television and satellite communications are all space communications.

Communication through free space using radio waves takes place over a wide range of frequencies : from a few hundreds of kHz to a few GHz. This range of frequencies is further subdivided into bands and allocated for various services as shown in Table 15.3.

Table 15.3 Some important wireless communication frequency bands

Service	Frequency bands	Comments
Standard AM broadcast	540-1600 kHz	
FM broadcast	88-108 MHz	
Television	54-72 MHz	VHF (very high frequencies)
	76-88 MHz	TV
	174-216 MHz	
	420-890 MHz	UHF (ultra high frequencies) TV
Cellular Mobile Radio	896-901 MHz	Mobile to base station
	840-935 MHz	Base station to mobile
Satellite Communication	5.925-6.425 GHz	Uplink
	3.7-4.2 GHz	Downlink

Choice of Communication Channel. The choice of the communication channel depends on the nature of the signal or the frequency band to be transmitted.

The *International Telecommunication Union (ITU)* presently controls the system of allocating frequencies and services within the overall frequency spectrum.

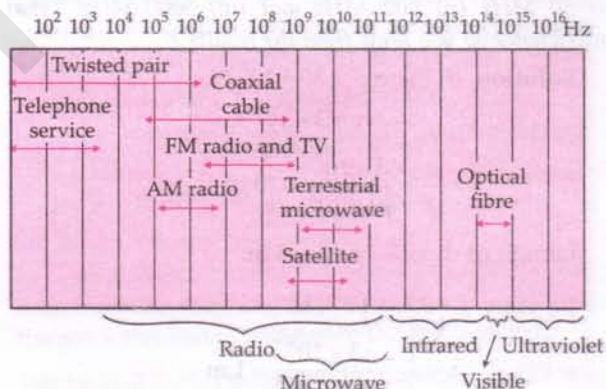


Fig. 15.23 Spectrum of the electromagnetic waves showing the frequency ranges used in different communication systems.

Examples based on

Bandwidths of Communication Channels

Formulae Used

$$\text{Number of channels} = \frac{\text{Total bandwidth of channel}}{\text{Bandwidth needed per channel}}$$

Units Used

Bandwidth is in hertz (Hz).

Example 16. A microwave telephone link operating at the central frequency of 10 GHz (or $\lambda = 3$ cm) has been established. If 2% of this is available for microwave communication channel, then how many telephone channels can be simultaneously granted if each telephone is allotted a bandwidth of 8 kHz. [CBSE OD 07C]

Solution. Microwave frequency used

$$= 10 \text{ GHz} = 10^{10} \text{ Hz}$$

Total bandwidth of the channel = 2% of 10^{10} Hz

$$= 2 \times 10^8 \text{ Hz}$$

Bandwidth per channel = 8 kHz = 8×10^3 Hz

Number of telephone channels which can be accommodated

$$= \frac{\text{Total bandwidth of the channel}}{\text{Bandwidth per channel}}$$

$$= \frac{2 \times 10^8}{8 \times 10^3} = 2.5 \times 10^4.$$

Example 17. Assume that light of frequency 4.5×10^{14} Hz is used in an optical communication system. If 2% of the frequency bandwidth is used, how many T.V. channels can be accommodated in this bandwidth? The bandwidth needed for T.V. transmission is 4.5×10^6 Hz / channel. [CBSE D 03C]

Solution. Optical frequency used

$$= 4.5 \times 10^{14} \text{ Hz}$$

Total bandwidth of the channel

$$= 2\% \text{ of } 4.5 \times 10^{14} \text{ Hz} = 9 \times 10^{12} \text{ Hz}$$

Bandwidth per TV channel = 4.5×10^6 Hz

Number of TV channels which can be accommodated

$$= \frac{\text{Total bandwidth of the channel}}{\text{Bandwidth needed per TV channel}}$$

$$= \frac{9 \times 10^{12}}{4.5 \times 10^6} = 2 \times 10^6 \text{ channels.}$$

Example 18. Consider an optical communication system operating at $\lambda \approx 800$ nm. Suppose, only 1% of the optical source frequency is the available channel bandwidth for optical communication. How many channels can be accommodated for transmitting (a) audio-signals requiring a bandwidth of 8 kHz, and (b) Video TV signals requiring an approximate bandwidth of 4.5 MHz. [CBSE OD 06]

Solution. Optical wavelength used,

$$\lambda = 800 \text{ nm} = 8 \times 10^{-7} \text{ m}$$

Optical source frequency,

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{8 \times 10^{-7}} = 3.75 \times 10^{14} \text{ Hz}$$

Total bandwidth of channel

$$= 1\% \text{ of } 3.75 \times 10^{14} \text{ Hz} = 3.75 \times 10^{12} \text{ Hz}$$

Number of channels

$$= \frac{\text{Total bandwidth of channel}}{\text{Bandwidth needed per channel}}$$

(a) Number of channels for audio signal

$$= \frac{3.75 \times 10^{12}}{8 \times 10^3} = 4.7 \times 10^8.$$

(b) Number of channels for video TV signal

$$= \frac{3.75 \times 10^{12}}{4.5 \times 10^6} = 8.3 \times 10^5.$$

It may be noted that in case of optical communication, the number of channels which can be simultaneously transmitted is very large.

Problems For Practice

- The TV signals have a bandwidth of 4.7 MHz. What is the number of channels that can be accommodated in a bandwidth of 4700 GHz?

(Ans. 10^6 channels)

HINTS

- Number of channels

$$= \frac{\text{Total bandwidth of the channel}}{\text{Bandwidth needed per channel}}$$

$$= \frac{4700 \text{ GHz}}{4.7 \text{ MHz}} = \frac{4700 \times 10^9}{4.7 \times 10^6} = 10^6.$$

15.16 SPACE COMMUNICATION : PROPAGATION OF EM WAVES

40. What is space communication? Mention the different types in which electromagnetic waves can propagate from one point to another.

Propagation of electromagnetic waves : Space communication. The term space communication refers to the sending, receiving and processing of information through physical space as the communication medium. In this type of communication, information is carried from one place to another by electromagnetic waves of high frequency called carrier waves. The frequencies used in space communication lie in the range 10^4 Hz to 10^{11} Hz and the corresponding wavelengths lie in the range 1 mm to 30 km. These waves travel through the open space with the speed of light. The earth's atmosphere helps in the propagation of these waves from one place to another on the surface of the earth. Radio, television and satellite communications are all space communications.

Different types of space communication can be understood through a simple illustration. Consider two persons playing with a ball in a closed room. One

person (transmitter) throws the ball to the other person (receiver). Here the ball acts as the message or information to be transmitted. The ball can be sent to the receiver in three ways :

- (i) by rolling along the ground,
- (ii) throwing directly to the receiving person, and
- (iii) by throwing towards the roof from where it is then reflected towards the receiving person.

Similarly, an electromagnetic wave radiated from an antenna can be transmitted through space in three ways :

- (i) along the ground (*ground wave*),
- (ii) directly in a straight line through intervening tropospheric space (*space wave* or *tropospheric wave* or *surface wave*), and
- (iii) Upwards in the sky followed by reflection from the ionosphere (*sky wave* or *ionospheric wave*)

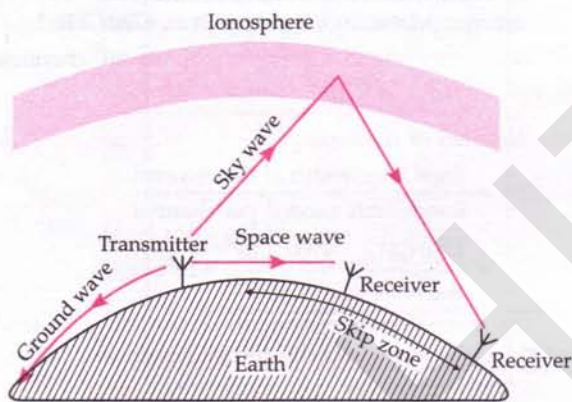


Fig. 15.24 Three modes of space communication.

The three modes of propagation of electromagnetic waves through space are shown in Fig. 15.24. We shall discuss these modes of propagation in the next few sections.

15.17 GROUND WAVE PROPAGATION

41. What is ground wave propagation ? Why does the energy of a ground wave decrease during its propagation along the earth's surface ? Why is ground wave transmission of signals restricted to a frequency of 1500 kHz ?

Ground wave propagation. A radiowave that can travel directly from one point to another following the surface of the earth is called a ground wave. Ground wave propagation is possible only when the transmitting and receiving antenna are close to the surface of the earth.

A ground wave is guided by the earth and bends round its curvature because of the electrical discontinuity existing between the earth and the atmosphere.

As the *em* wave glides over the earth surface, its electric field component induces charges in the earth's surface. As the ground wave travels, the induced charges also travel in the earth setting up a surface current. This induced current experiences some resistance at the earth's surface. So as the ground wave passes over the earth's surface, it is weakened due to the partial energy absorption by the earth.

Moreover, a ground wave also gets attenuated due to *diffraction effect*. When the wave propagates over the earth's surface, it gradually tilts over. At some distance from the transmitter, the wave 'lies down and dies'.

Due to the above energy losses, the *ground waves* are not suited for long range transmission. Further, these losses are frequency dependent and are higher for higher frequencies. Hence ground wave propagation is possible at low frequencies (500 kHz to 1500 kHz) or for radio broadcast at long wavelengths (> 200 m). This region of the AM band is called *medium wave band*.

The maximum range of ground wave propagation depends on two factors :

- (i) The frequency of the transmitted wave, and
- (ii) the power of the transmitter.

To radiate signals with high efficiency, the antennas should have a size comparable to the wavelength λ (at least $\approx \lambda/4$) of the transmitted signal.

42. Give some advantages and disadvantages of ground wave propagation.

Advantages of ground wave propagation :

1. Given enough transmitting power, ground waves can be used to communicate between any two locations on the earth.
2. Ground waves are relatively unaffected by changing atmospheric conditions.

Disadvantages of ground wave propagation :

1. Ground waves require a relatively high transmission power.
2. As ground waves are limited to VLF, LF and MF signals, they require large antennas.
3. Ground losses vary with surface material and composition.

15.18 SPACE OR TROPOSPHERIC WAVE PROPAGATION

43. What is Space wave propagation ?

Space wave propagation. The television frequencies lie in the range 100–220 MHz. In this transmission, the reception is possible only if the receiving antenna directly intercepts the signal. If a radiowave

transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna, it is called a space wave. The space wave propagation is also called **tropospheric** or **line of sight propagation**.

The **range of space wave propagation** is limited by

- (i) the **line of sight distance** i.e., the distance at which the transmitting and receiving antenna can see each other, and
- (ii) the **curvature of the earth**.

44. Deduce an expression for the distance upto which the T.V. signals can directly be received from a T.V. tower of height h .

Range of a transmitting antenna. Let T.V. signals be transmitted from an antenna of height $PQ = h_T$. Due to the curvature of the earth, no direct signals are possible beyond the points S or T , as shown in Fig. 15.25. Let $PS = PT = d$.

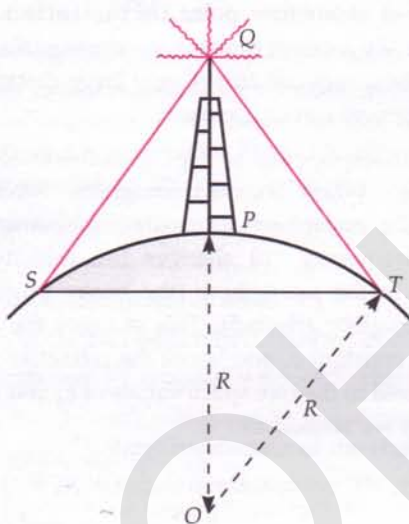


Fig. 15.25 Range of transmitting antenna.

From right-angled $\triangle OTQ$, we get

$$OQ^2 = OT^2 + QT^2$$

Here

$$OQ = R + h_T$$

$$QT = PT = d$$

$$OT = R = \text{Radius of the earth}$$

$$\therefore (R + h_T)^2 = R^2 + d^2$$

$$\text{or } R^2 + h_T^2 + 2h_TR = R^2 + d^2$$

$$\text{or } d^2 = h_T^2 + 2h_TR = 2h_TR \left[1 + \frac{h_T}{2R} \right]$$

But $h_T \ll R$, therefore, $d^2 = 2h_TR$ or $d = \sqrt{2h_TR}$.

45. Write an expression for the maximum line of sight (LOS) distance between two antennas for space wave propagation.

Range of space wave propagation between two antennas on earth's surface. A space wave travels in a straight line from transmitting antenna to the receiving antenna i.e., it is a line of sight (LOS) communication. Due to the curvature of the earth, direct space waves get blocked at some points. Thus the earth presents a horizon to space wave propagation commonly called **radio horizon**.

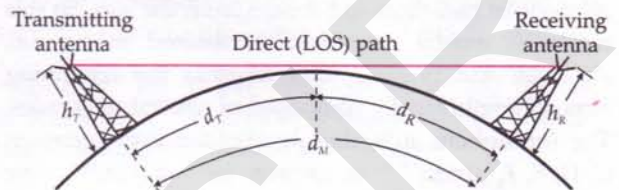


Fig. 15.26 Line of sight communication by space waves.

Figure 15.26 shows the effect of antenna height on the radio horizon. The line-of-sight radio horizon for a single antenna at sea level is given by

$$d = \sqrt{2Rh}$$

where d = distance to radio horizon

h = antenna height above sea level

R = Radius of the earth

The maximum line-of-sight distance d_M , as shown in Fig. 15.26 between the transmitting and receiving antennas is given by

$$d_M = d_T + d_R = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where d_T = radio horizon for transmitting antenna

d_R = radio horizon for receiving antenna

h_T = height of transmitting antenna

h_R = height of receiving antenna

Clearly, if a signal is to be received beyond the horizon, then the height of receiving antenna must be large enough so as to intercept line-of-sight waves directly.

Moreover, the power transmitted from a transmitter decreases nearly as the inverse square of the distance. So the signal becomes weaker with the increasing distance.

Hence the range of T.V. transmission can be increased

- (i) by increasing the height of the transmitting and receiving antennas.
- (ii) by using **repeater** transmitting stations.

15.19 SKY WAVE OR IONOSPHERIC PROPAGATION

46. What is ionosphere ? Explain its importance in communication.

Ionosphere. It is the outermost region of the earth's atmosphere which extends from about 60–350 km from the earth's surface. Due to low pressure in this part of atmosphere, ionisation is produced by ultra-violet radiation and X-rays from the sun. So this region is mostly composed of ionised matter i.e., electrons and positive ions whereas the remaining atmosphere is mostly composed of neutral molecules. The ionosphere is further subdivided into layers as C, D, E, F_1 and F_2 , as shown in Fig. 15.27. The corresponding electron densities in different layers are also indicated.

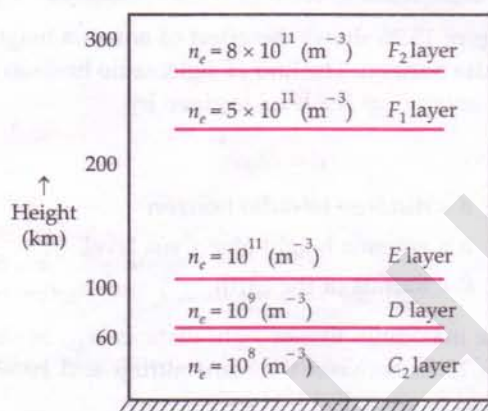


Fig. 15.27 Different layers of ionosphere.

The ionosphere plays an important role in space communication. The radiowaves of the HF frequency band having frequency range 3–30 MHz cannot

penetrate through the ionosphere. They are reflected back towards the earth. This region of the AM band is called shortwave band. Above frequency of 40 MHz, the ionosphere bends the electromagnetic waves and does not reflect them back towards the earth. Fig. 15.28 shows the effect of ionosphere on electromagnetic waves with different frequencies.

47. What is sky propagation ? Describe the mechanism of reflection of a radiowave from the ionosphere. What is critical frequency ?

Sky wave propagation. A radiowave directed towards the sky and reflected by the ionosphere towards the desired location of the earth is called a sky wave. Radio waves of frequencies between 2 MHz to 20 MHz can be reflected by the ionosphere. This region of the AM band is called **short wave band**. These waves radiated from a certain point and after being reflected by the ionosphere can be received at another point on the surface. This is known as **sky wave** or **ionospheric propagation**. In this way, radio waves can travel very large distances and can even travel round the earth.

Mechanism of reflection of a radiowave from the ionosphere. When an electromagnetic wave passes through the ionosphere, its oscillating electric field (or angular frequency ω) changes the velocity of the electrons in the ionosphere (the heavy positive ions being negligibly affected). This changes the effective dielectric constant ϵ' , and hence the refractive index n' , as compared to the free space values of ϵ_0 and n_0 . These quantities are related as

$$n' = \sqrt{\epsilon' \mu_0} = \sqrt{\epsilon_0 \mu_0 \left(1 - \frac{Ne^2}{\epsilon_0 m \omega^2} \right)}$$

or

$$n' = n_0 \left(1 - \frac{Ne^2}{\epsilon_0 m \omega^2} \right)^{1/2}$$

where e is the electronic charge, m is the mass of the electron and N is the electron density in the ionosphere.

Clearly, $n' < n_0$ i.e., the refractive index of the ionosphere is less than its free space value of n_0 . Thus the ionosphere behaves as a rarer medium. Consequently, the em wave gradually turns away from the normal as it penetrates the ionosphere. As we go high in the ionosphere, the value of N increases and hence n' decreases. The refraction or bending of the beam continues till it reaches **critical angle** and after which it gets reflected back, as shown in Fig. 15.29.

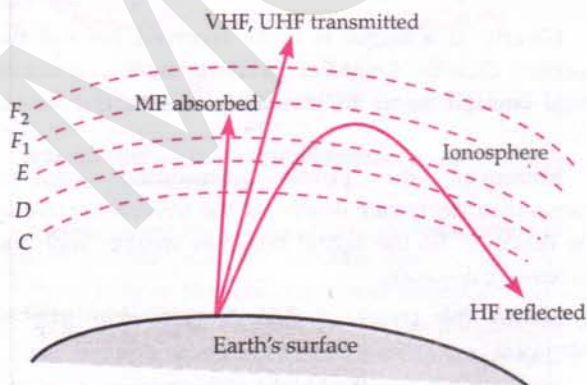


Fig. 15.28 Effect of ionosphere on different e.m.waves.

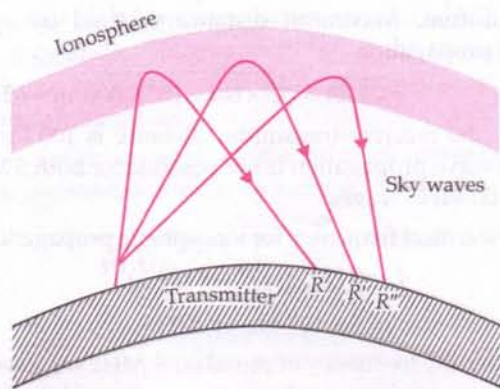


Fig. 15.29 Sky wave or ionospheric propagation.

Obviously, the different frequencies ω are reflected from different regions of the ionosphere which have different values of N . That is why the different points on earth receive signals reflected from different depths of the ionosphere. If the frequency ω is too high, then after a certain value, the density N may never be so high as to produce enough bending for attaining critical angle or condition of reflection. This value is called **critical frequency**. If the maximum electron density of the ionosphere is $N_{\max}$ per m^{-3} , then the critical frequency f_c is given by

$$f_c \approx 9 (N_{\max})^{1/2}$$

The value of f_c ranges from about 5 to 10 MHz. Frequencies higher than f_c cross the ionosphere and do not return back to the earth.

Some important facts of the different layers of the atmosphere and their interaction with the electromagnetic waves are summarised in Table 15.4.

Table 15.4 Different layers of atmosphere and their interaction with the propagating EM waves

Name of the stratum (Layer)	Approximate height over earth's surface	Exists during	Frequencies most affected
Troposphere	10 km	Day and night	VHF (up to several GHz)
D (part of Stratosphere)	65-75 km	Day only	Reflects LF, absorbs MF and HF to some degree
E (part of Stratosphere)	100 km	Day only	Helps surface waves, reflects HF
E_1 (part of Mesosphere)	170-190 km	Day time, merges with F_2 at night	Partially absorbs HF waves yet allowing them to reach F_2
F_2 (Thermosphere)	300 km at night, 250-400 km during day time	Day and night	Efficiently reflects HF waves, particularly at night

48. Define the following terms in connection with sky wave propagation : (i) Critical frequency (ii) Maximum usable frequency (iii) Skip distance and (iv) Fading.

Some definitions in connection with sky wave propagation.

(i) **Critical frequency**. It is that highest frequency of radio wave which when sent straight (normally) towards the given layer of ionosphere gets reflected from ionosphere and returns to the earth. Above this frequency, the radiowave will penetrate the ionosphere and is not reflected by it. It is given by

$$f_c = 9 (N_{\max})^{1/2}$$

where $N_{\max}$ = the maximum electron density of the ionosphere.

(ii) **Maximum usable frequency (MUF)**. It is defined as the highest frequency of the radiowave which when sent at certain angle towards the given layer of ionosphere, gets reflected and returns to the earth.

$$\text{Thus } \text{MUF} = \frac{f_c}{\cos \theta} = f_c \times \sec \theta$$

where θ is the angle between the normal and the incident ray.

(iii) **Skip distance**. It is defined as the smallest distance from a transmitter, measured along the earth's surface, to the point where the sky wave of a fixed frequency but less than critical frequency returns to the earth after reflection from the ionosphere.

The reception within the skip distance is possible only by ground wave propagation.

(iv) **Fading**. It is defined as the variation in the strength of a signal at a receiver due to interference of waves. Fading is more at higher frequencies. It results into errors in data transmission and retrieval.

Examples based on

Propagation of Radiowaves and Range of TV Transmission

Formulae Used

1. Critical frequency for sky wave propagation,

$$f_c = 9(N_{\max})^{1/2}$$

where $N_{\max}$ is the maximum electron density in the ionosphere. Frequencies below f_c are reflected back to earth.

2. Ground wave propagation is limited to 1500 kHz.
3. Frequencies above 20 MHz are carried through satellite communication.
4. Waves of UHF/VHF regions can propagate through space wave or tropospheric wave.
5. The range of TV transmission,

$$d = \sqrt{2hR},$$

where h = height of antenna and

R = radius of earth.

6. Population covered = Population density $\times$ area
7. Maximum distance covered in LOS communication,

$$d_M = d_T + d_R = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Units Used

Frequency f_c is in hertz, electron density $N_{\max}$ in m^{-3} , and distance d , height h and radius R are all in metre.

Example 19. A radio can tune to any station in 7.5 MHz to 12 MHz band. What is the corresponding wavelength band?

Solution. Frequency of 7.5 MHz belongs to SW band. The corresponding wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.5 \times 10^6} = 40 \text{ m}$$

Frequency of 12 MHz belongs to HF band, the corresponding wavelength is

$$\lambda = \frac{3 \times 10^8}{12 \times 10^6} = 25 \text{ m}$$

Corresponding to the given frequency band, the wavelength band is 25 m – 40 m band.

Example 20. A ground receiver station is receiving a signal at (a) 5 MHz and (b) 100 MHz, transmitted from a ground transmitter at a height of 300 m located at a distance of 100 km. Identify whether it is coming via space wave or sky wave propagation or satellite transponder. (Given the value of radius of the earth is 6400 km and maximum electron density, $N_{\max} = 10^{12} \text{ m}^{-3}$).

[NCERT ; CBSE OD 05]

Solution. Maximum distance covered by space wave propagation

$$= \sqrt{2Rh} = \sqrt{2 \times 6.4 \times 10^6 \times 300} \text{ m} = 62 \text{ km}$$

As the receiver-transmitter distance is 100 km, so space wave propagation is not possible for both 5 MHz and 100 MHz waves.

The critical frequency for ionospheric propagation is

$$f_c = 9(N_{\max})^{1/2} = 9(10^{12})^{1/2} \\ = 9 \times 10^6 \text{ Hz} = 9 \text{ MHz}$$

Since the frequency of signal (a) 5 MHz $< f_c$, so this signal comes via ionospheric propagation while signal (b) of 100 MHz comes via satellite transmission.

Example 21. Frequencies higher than 10 MHz are found not to be reflected by the ionosphere on a particular day at a place. Calculate the maximum electron density of the ionosphere. [NCERT]

Solution. Here $f_c = 10 \text{ MHz} = 10^7 \text{ Hz}$

$$\text{Now } f_c = 9(N_{\max})^{1/2} \text{ or } f_c^2 = 81 N_{\max}$$

$$\therefore N_{\max} = \frac{f_c^2}{81} = \frac{(10^7)^2}{81} = 1.23 \times 10^{12} \text{ m}^{-3}.$$

Example 22. On a particular day, the maximum frequency reflected from the ionosphere is 10 MHz. On another day, it was found to increase to 11 MHz. Calculate the ratio of the maximum electron densities of the ionosphere on the two days. Point out a plausible explanation for this. [NCERT]

Solution. Here $f_c = 10 \text{ MHz}$, $f'_c = 11 \text{ MHz}$,

$$\text{But } f_c = 9(N_{\max})^{1/2} \text{ and } f'_c = 9(N'_{\max})^{1/2}$$

$$\therefore \frac{N'_{\max}}{N_{\max}} = \left(\frac{f'_c}{f_c}\right)^2 = \left(\frac{11}{10}\right)^2 = 1.21$$

The degree of ionisation of the ionosphere due to solar radiation may vary from day to day. This changes the refractive index and hence the frequency f_c .

Example 23. A TV transmitting antenna is 125 m tall. How much service area can this transmitting antenna cover, if the receiving antenna is at the ground level? Radius of earth = 6400 km. [CBSE F 08]

Solution. Here $h_T = 125 \text{ m}$, $R = 6400 \times 10^3 \text{ m}$

$$d = \sqrt{2h_T R} = \sqrt{2 \times 64 \times 10^5 \times 125} \\ = 40 \times 10^3 \text{ m} = 40 \text{ km}$$

$$\text{Area covered} = \pi d^2 = 3.14 \times (40)^2 = 5024 \text{ km}^2.$$

Example 24. What should be the height of a transmitting antenna if the TV telecast is to cover a radius of 128 km? [CBSE F 94C]

Solution. Here $d = 128 \text{ km} = 128 \times 10^3 \text{ m}$

$$\text{Diameter of the earth} = 1.28 \times 10^7 \text{ m}$$

$$\begin{aligned} \therefore R &= 0.64 \times 10^7 \text{ m} \\ \text{As } d &= \sqrt{2hR} \\ \therefore h &= \frac{d^2}{2R} = \frac{(128 \times 10^3)^2}{2 \times 0.64 \times 10^7} \\ &= \frac{128 \times 128}{128} \times 10^1 = 1280 \text{ m.} \end{aligned}$$

Example 25. By what percentage will the transmission range of a TV tower be affected when the height of the tower is increased by 21% ? [CBSE D 09]

Solution. Original transmission range,

$$d = \sqrt{2hR}$$

New height,

$$h' = h + 21\% \text{ of } h = 1 + \frac{21}{100}h = 1.21h$$

New transmission range,

$$d' = \sqrt{2h'R}$$

$$\therefore \frac{d'}{d} = \sqrt{\frac{h'}{h}} = \sqrt{1.21} = 1.1$$

Increase in range,

$$\begin{aligned} \frac{\Delta d}{d} \times 100 &= \frac{d' - d}{d} \times 100 \\ &= \left(\frac{d'}{d} - 1 \right) \times 100 = (1.1 - 1) \times 100 = 10\%. \end{aligned}$$

Example 26. What will be the required height of a T.V. tower which can cover the population of 60.3 lakhs if average population density around the tower is 1000 km^{-2} ? [Radius of earth = $6.4 \times 10^6 \text{ m}$] [CBSE Sample Paper 03]

Solution. Area covered $\times$ population density
= Population covered

$$\therefore \pi d^2 \times (1000 \text{ km}^{-2}) = 60.3 \text{ lakhs}$$

$$\pi d^2 \times 1000 (1000 \text{ m})^{-2} = 60.3 \times 10^5$$

$$d^2 = \frac{60.3 \times 10^5 \times 1000}{\pi} = 1.92 \times 10^9$$

Height of T.V. tower,

$$h = \frac{d^2}{2R} = \frac{1.92 \times 10^9}{2 \times 6.4 \times 10^6} = 150 \text{ m.}$$

Example 27. The TV transmission tower at a particular station has a height of 160 m. (a) What is its coverage range ? (b) How much population is covered by the transmission, if the average density around the tower is 1200 km^{-2} ? (c) By how much should the height be increased to double its coverage range ? Given radius of the earth = 6400 km. [CBSE F 03]

Solution. Here, height of tower, $h = 160 \text{ m}$

Population density, $p = 1200 \text{ km}^{-2}$

Radius of the earth, $R = 6400 \text{ km} = 64 \times 10^5 \text{ m}$

(a) Coverage range,

$$\begin{aligned} d &= \sqrt{2hR} = \sqrt{2 \times 160 \times 64 \times 10^5} \\ &= 32000 \sqrt{2} = 45248 \text{ m} \approx 45 \text{ km.} \end{aligned}$$

(b) Population covered,

$P = \text{Population density} \times \text{Area covered}$

$$= p \times \pi d^2 = 1200 \times \pi \times (45)^2 = 76.37 \text{ lakhs.}$$

(c) To double the coverage range, the height h' must be such that

$$2d = \sqrt{2h'R} \quad \text{or} \quad 2\sqrt{2hR} = \sqrt{2h'R}$$

or

$$h' = 4h = 4 \times 160 = 640 \text{ m}$$

Height of the tower to be increased

$$= h' - h = 640 - 160 = 480 \text{ m.}$$

Example 28. A transmitting antenna at the top of a tower has a height 32 m and that of the receiving antenna is 100 m. What is the maximum distance between them for satisfactory communication in LOS mode ? Given radius of earth $6.4 \times 10^6 \text{ m}$ [NCERT ; CBSE D 10]

Solution. Here $h_T = 32 \text{ m}$, $h_R = 100 \text{ m}$,

$$R = 6.4 \times 10^6 \text{ m}$$

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

$$d_M = \sqrt{2 \times 64 \times 10^5 \times 32} + \sqrt{2 \times 64 \times 10^5 \times 100} \text{ m}$$

$$= 64 \times 10^2 \times \sqrt{10} + 8 \times 10^3 \times \sqrt{10} \text{ m}$$

$$= 144 \times 10^2 \sqrt{10} \text{ m} = 45.5 \text{ km}$$

Problems For Practice

- Frequencies higher than 9 MHz are found not to be reflected by the ionosphere on a particular day at a place. Calculate the maximum electron density of the ionosphere. (Ans. 10^{12} m^{-3})
- The electron density of a layer of ionosphere is 10^{12} m^{-3} . Calculate the critical frequency of reflection of radiowaves from this layer. (Ans. 9 MHz)
- A TV tower has a height of 500 m at a given place. Calculate its coverage range, if the radius of the earth is 6400 km. [CBSE OD 04] (Ans. 80 km)
- A TV tower has height of 100 m. How much population is covered by TV broadcast ? Given radius of the earth = $6.4 \times 10^6 \text{ m}$ and average density of population = 10^3 km^{-2} . [CBSE F 03] (Ans. 40 lakhs)
- On a particular day, the maximum frequency reflected from the ionosphere is 10 MHz. On another day, it was found to decrease to 8 MHz. Calculate the ratio of the maximum electron densities of the ionosphere on the two days. [CBSE D 06] (Ans. 25 : 16)

6. A transmitting antenna at the top of a tower has a height of 45 m and the height of the receiving antenna is 80 m. What is the maximum distance between them, for satisfactory communication in LOS mode? (Radius of earth = 6400 km)

[CBSE OD 13]

HINTS

1. Use $N_{\max} = \frac{f_c^2}{81}$.
2. Use $f_c = 9(N_{\max})^{1/2}$.
3. $d = \sqrt{2hR} = \sqrt{2 \times 500 \times 6400 \times 1000}$
 $= 80,000 \text{ m} = 80 \text{ km}$.
4. Height of tower, $h = 100 \text{ m}$
 Radius of earth, $R = 6.4 \times 10^6 \text{ m}$
 Average population density, $p = 1000 \text{ km}^{-2}$
 Radius of the area covered by TV broadcast,
 $d = \sqrt{2Rh}$
 $\therefore$ Area of the range is
 $A = \pi d^2 = \pi \times 2Rh = \pi \times 2 \times 6.4 \times 10^6 \times 100 \text{ m}^2$
 $= 40.02 \times 10^8 \text{ m}^2 = 40.02 \times 10^2 \text{ km}^2$
 Population covered
 $= pA = 1000 \times 40.02 \times 10^2$
 $= 40.02 \times 10^5 = 40 \text{ lakh}$.
5. $f_c = 10 \text{ MHz}$, $f'_c = 8 \text{ MHz}$
 $\frac{N_{\max}}{N'_{\max}} = \left(\frac{f_c}{f'_c}\right)^2 = \left(\frac{10}{8}\right)^2 = 25 : 16$
6. Here, $h_T = 36 \text{ m}$, $h_R = 49 \text{ m}$, $R = 6400 \text{ km} = 64 \times 10^5 \text{ m}$
 Maximum distance for satisfactory communication in LOS mode,
 $d_R = \sqrt{2Rh_T} + \sqrt{2Rh_R}$
 $= \sqrt{2 \times 64 \times 10^5 \times 45} + \sqrt{2 \times 64 \times 10^5 \times 80} \text{ m}$
 $= 8 \times 3 \times 10^3 + 8 \times 4 \times 10^3$
 $= 56 \times 10^3 \text{ m} = 56 \text{ km}$.

15.20 INTERNET

Introduction. It is a global network of computers linked by high-speed datalines and wireless systems. It allows communication and sharing of information among all computers connected through the vast network. Billions of people use it worldwide. It was started in 1960's and opened for public use in 1990's. It has witnessed tremendous growth in the last few years and is still expanding fast. Its importance is described beautifully in the following words :

The Internet will continue to change everything by offering a level of connectivity that was unimaginable only a few years ago and every house, business and school will want to be hooked to that incredible global database.

– Bill Gates

In fact, the word Internet is a short form of INTER-NETwork which is the interconnected network of all worldwide servers.

Networking of computers : The way Internet works.

A network is a collection of interlinked computers by means of a communication system. Thus two or more computers are said to be networked if they are capable of exchanging information between them. This information can be shared either through connecting wires or by some wireless mean of communication such as Wi-Fi.

The hierarchy of networks leading to an Internet is as follows :

(i) **Localised Area Network (LAN).** Small computer networks that are confined to a localised area (generally upto 1 km) or an institution (e.g., an office, a building, a factory or a school) are called local area networks or LANs. The key purpose of a LAN is to serve its users in resource sharing. LAN users can share data, information, programs, hard-disks, modems, etc. A LAN is built up by connecting all or some computers (called workstations) of an institution to main computer called server.

Thus a computer connected to a network and is seeking to share the resources of the network is called a workstation or a node. The computer that facilitates the sharing of data, software and hardware (e.g., printers, modems, etc.) on the network is called a server.

The block diagram of a Local Area Network is shown in Fig. 15.30.

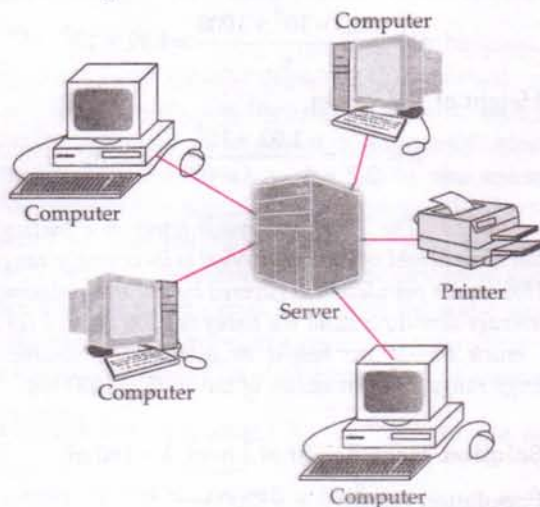


Fig. 15.30 A Local Area Network (LAN).

(ii) **Wide Area Network (WAN).** Various LANs of a country are connected together through their servers by means of telephone lines or satellites as shown in Fig. 15.31. Such a network of computers which spreads across a country/countries/continents is called a Wide Area Network (WAN). The WANs link computers to facilitate fast and efficient exchange of information at lesser costs and higher speeds.

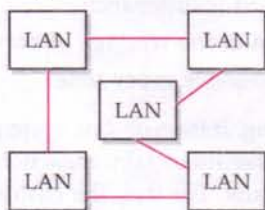


Fig. 15.31 A Wide Area Network (WAN).

Differences between a LAN and A WAN

	LAN	WAN
1.	Interconnects an institution just spreading over few kilometres.	Interconnects networks of countries/continents.
2.	Operates at a high total data rate of atleast several mbps.	Operates at a rate less than 1 mbps (Mega bits per second).
3.	Completely owned by a single organisation.	Owned by several organisations under the control of information ministry of any country.
4.	Very low error rates because of short distances involved.	Comparatively higher error rates because of large distances involved.

(iii) **Internet.** The Internet is worldwide network of computer networks. It interlinks various WANs together through gateways to the **Internet backbone**. A gateway is a device that connects dissimilar networks. A backbone is central interconnecting structure that connects different WANs together through the gateways. Fig. 15.32 show a block diagram of an Internet.

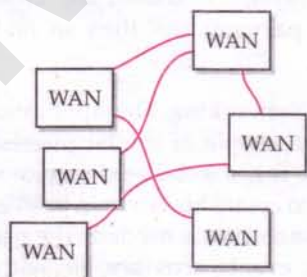


Fig. 15.32 Formation of Internet

All information related to a local network is stored in server computers of a LAN. The servers of different LANs act as a channel for information exchange between computers connected to a LAN and also servers of other networks. Every computer that extracts information from a server is called a **client computer**.

How does an Internet function ?

- At the source computer, the message or file/document to be sent to another computer is first divided into very small parts called packets. A **packet** generally contains an information.
- Each packet is given a number serialwise e.g., 1, 2, 3, etc.
- All these packets are then sent to the address of destination computer.
- The destination computer receives the packets in random manner. If a packet is lost, it is demanded again.
- The packets are reassembled in the order of their numbers and the original message / file / document is obtained.

The first network of computers called *Advanced Research Projects Agency NETwork (ARPANET)* was developed by US Department of Defence in 1969. In mid 80s, another federal agency, the *National Science Foundation*, created a new high-capacity network called NSFnet, which was more capable than ARPANET. In 1990s, the internetworking of ARPANET, NSFnet and other private networks resulted into the present day Internet. The entire world uses a *common set of rules for Internet Communication among computers called protocols*. Nowadays, standard sets of protocols called *Transmission Control Protocol/Internet Protocol* are used for exchange of information through internet.

(i) **Transmission Control Protocol (TCP).** This part is responsible for dividing the file/message into packets on the *source computer*. It also reassembles the received packets at the *destination* or recipient computer.

(ii) **Internet Protocol (IP).** This part is responsible for handling the address of the destination computer so that each packet is routed (sent) to its proper destination.

The exchange of information on Internet is very fast because electronic signals (messages) of computers are communicated through electromagnetic waves which travel with the speed of light. We can connect our computers through various Internet Service Providers (ISP) by paying a prescribed fee. Many mobile network companies also act as ISP.

WWW. World Wide Web. On Internet, information is provided/available through webpages that may contain text, images, videos, etc. Any one can provide specific information on Internet by making a couple of webpages containing that information. Such a set of webpages together is called a **website**. Each website has a unique address in a special format called *Uniform Resource Locator*. One can move from one webpage to another through a system called *Interlinked hypertext documents*. In this system one webpage is linked to another webpage by providing hyperlinking (a way of highlighting) to any text, image or video. This set of protocols that allows us to access any document on the Internet through interlinking of webpages is called **WWW** or **World Wide Web**. WWW is based upon clients and servers. A **web browser** is a WWW client that navigates through the worldwide web and displays web pages. **Web server** is a WWW server that responds to requests made by web browsers.

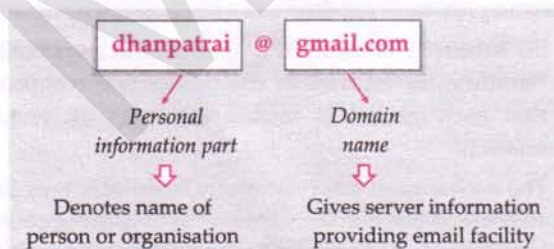
Applications of Internet. The Internet finds use in every field of modern life. Some of the uses of Internet are as follows :

(i) **Internet surfing.** Moving from one webpage / website to another on the Internet is called Internet surfing. Thus one can easily search and view information on any topic of interest on the net.

(ii) **e-mail : Electronic mail.** It is the cheapest and fastest way of sending messages through the Internet. E-mailing allows exchange of texts written on computers, images and videos. For availing this facility, one needs to create personal email account with an **email-Id** (identity) or **email address**. These email-Ids are password protected so that no one else can misuse it. Internet Service Providers (ISP) provide these email-Ids through which people can identify and communicate with us through the Internet.

Every email-Id has two parts separated by a sign @ called at the rate of.

For example,



E-mail messages are almost instantaneously delivered to the addressee because of communication *via* high speed electromagnetic waves. Moreover, these

messages get stored in an email account and can be viewed at any time we desire. Emailing has become the most popular tool of communication. One can send animations, audio, video, etc. *via* email.

Advantages of e-mailing :

1. Fast delivery
2. Low cost
3. Easy record maintenance
4. Reduction of the wastage of paper stationary
5. Easily accessible every time.

(iii) **e-Banking.** It is an on-line system which allows customers to make financial transactions on a website operated by a bank. For this, the customer must have an account with that bank, have internet access and must register with the bank for this facility. Then the bank provides the login number and password for unique identification of the customer. The account holder can avail on-line facilities such as checking the status of the balance, check book requisition, loan, recurring, credit card, debit card, etc.

(iv) **e-Shopping : e-Commerce.** E-commerce is the collection of tools and practices involving internet technologies that allow a company to create, maintain and optimise business relations with customers and other businesses. The customers can view images and receive all the information about various products or services of companies through their websites. Buying products through the product selling websites is called e-shopping. The buyers have options of paying cash on delivery or making on-line payments.

(v) **e-Booking : e-Ticketing.** This application of Internet is useful for reservation in travel and tourism industry. One can make on-line booking of flight tickets, railway tickets, hotels, holiday packages, insurance and other services. To book on e-ticket, the customer has to open the webpage of the Airline Company or Indian Railways. On entering the travel preference, he gets a view of the available flights / trains. After fixing the choice, the customer has to make on-line payment and then an on-line ticket is issued to him.

(vi) **Social Networking.** This application provides a platform to the people of similar interests to build a social network. It is a web-based service which allows an individual to create his/her own profile, list of users with whom connection is needed. The users can share ideas, pictures, events, activities, etc. with their group. Facebook, Twitter, Whatsapp, Google+, etc. are some common social networking sites.

15.21 MOBILE TELEPHONY

Introduction. Mobile telephone is one of the most popular means of communication being used these days. Today's mobile phone is like a handy computer equipped with the Internet. Communicating through mobile phones has greatly changed our lifestyle.

A mobile phone is a low power transceiver (a combination of transmitter/receiver in a single package) which can wirelessly send and receive radio frequency signals. It operates in the UHF range of frequencies around 800-900 MHz. In a mobile phone, the outgoing and incoming signals have different frequencies, this makes possible for two persons to talk and listen simultaneously.

Applications of mobile phones :

1. Make and receive phone calls
2. Make task or to-do lists
3. Keep track of appointments and set reminders
4. Use the built-in calculator for simple calculations
5. Send or receive e-mail
6. Get information (news, entertainment, stock quotes) from the internet.
7. Play games
8. Listen radio/music and watch TV
9. Send text messages
10. Take photos and videos, etc.

Working principle of a mobile phone. Mobile telephony is based on cellular radio network technology. Here the entire service area is divided into numerous smaller parts, called **cells**. To completely cover the given physical area, hexagonal cells are best suited. At the centre of each cell is a **base station** which consists of a computer and a radio antenna which can receive and transmit signals to and from all mobile phones present within the cell. Each base station is connected to a switching office called *mobile telephone switching office* (MTSO), which coordinates communication between all the base stations and *telephone central office* (TCO).

Every base station (BS) has a working range of minimum 1.5 to 2 km and maximum 48 to 56 km or more area around it. When a mobile phone is switched on, MTSO records its location by identifying the cell in which it is present. When a mobile telephone physically leaves a cell, its base station notices the telephone's signal fading away and asks all the surrounding base stations how much power they are getting from it. The base station (BS) then transfers ownership to the new cell getting the strongest signal *i.e.*, the cell where the phone is now located. MTSO then changes the channel carrying the call. This process is called **handoff/handover**. MTSO is the nerve centre of the system while the base stations are just radio relays. As mobile phones work on cell division of physical areas, so they are also known as cell phones.

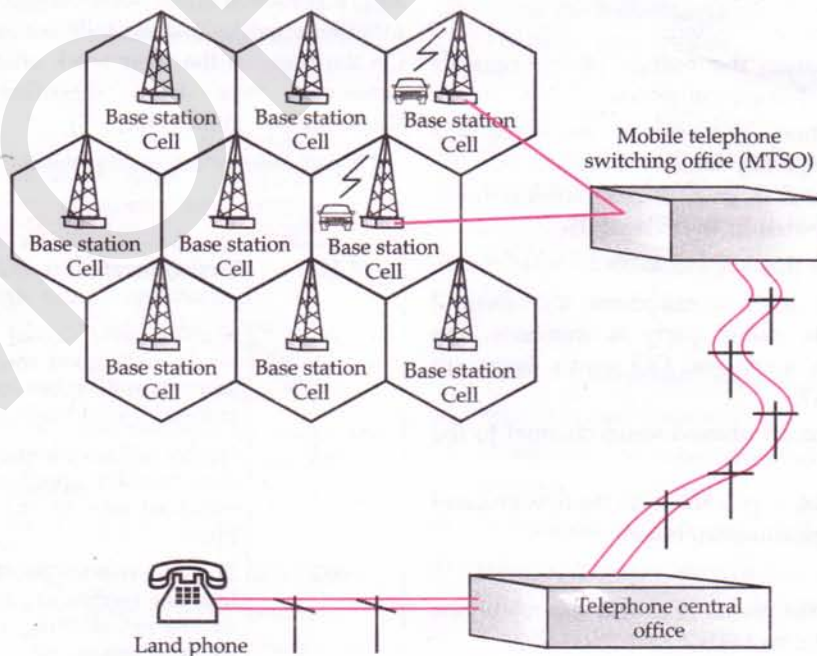


Fig. 15.33 A typical cellular system.

Mobile phone numbering system : SIM CARD.

To give a unique identification to every mobile phone, a **SIM** (Subscriber Identity Module) is inserted in every mobile phone. A SIM card is a small IC (integrated circuit) chip with a unique SIM number and a mobile phone number. Each SIM has a unique 32-bit electronic serial number (ESN) and a unique 10-digit (represented by 34 bits) mobile identification number (MIN). A SIM card has memory (for data and applications), a processor and the ability to interact with the user. All SIM cards are issued by mobile operator companies and their information is provided to MTSO. After SIM verification, MTSO activates the mobile number of the user. In India, all mobile numbers have the prefix 9, 8 or 7. As per National numbering plan 2003, the 10 digit mobile number is split as follows :

XXXX - NNNNNN
 Network operator digits Subscriber number digits



Fig. 15.34 A SIM card.

In India, the Telecom Regulatory Authority of India (TRAI) was established in 1997 by an act of Parliament to regulate the use of mobile phones system.

The Process of Call Management in Mobile Phones

Transmitting a call :

1. The user enters the mobile phone number (MIN) of the destination phone.
2. The user's phone generates an em wave of a particular frequency which carries the number to be called and its own identification number and is transmitted to the closest BS.
3. The BS relays this MIN to MTSO.
4. MTSO sends MIN to telephone CO (central office). If the called party is available, the connection is made and CO relays the result back to MTSO.
5. MTSO assigns an unused voice channel to the call.
6. The user's mobile gets tuned to the new channel and communication can begin.

Receiving a Call :

1. When a mobile phone is called, the telephone CO sends MIN to MTSO.
2. MTSO searches for location of that phone by sending query signal to each cell (paging).

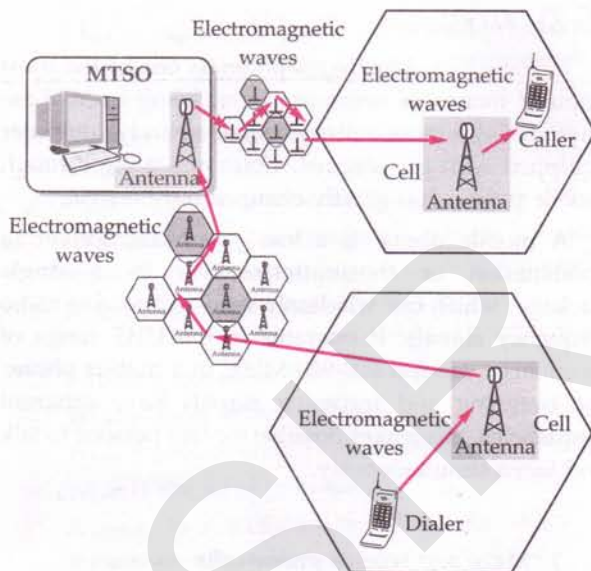


Fig. 15.35 Process of mobile phone call from dialer to caller through em waves.

3. When the called phone is found, the MTSO transmits the ringing signal to that phone.
4. When the caller answers, MTSO assigns a voice channel to the caller, voice communication can begin.

Mobile Network Generations

Due to the increasing popularity of mobile phones and advancement of technology, the efficiency of mobile networks is drastically increasing day by day. On the basis of their work efficiency, the mobile networks have been classified into different 'Generations' (abbreviation G).

A summary of different generations is given below :

Generations	Brief Description
1G	First generation mobile networks based on analogue radio signals.
2G	Narrow band digital signal based networks with good sound quality, better security and higher total capacity. They provided worldwide connectivity.
3G	Newer cellular networks with increased data transfer speed of 384 kbits/s for efficient use of Internet on mobile phone.
4G	These networks provide a high speed Internet facility on mobile phones for surfing net, chatting, viewing television, listening music, etc.

In India, 2G, 3G and 4G are operational.

15.22 GLOBAL POSITIONING SYSTEM (GPS)

Introduction. The Global Positioning System (GPS) is a space-based satellite navigation system that provides location and time information in all weather conditions, anywhere on or near the earth where there is unobstructed line of sight to four or more GPS satellites. The system provides critical capabilities to military, civil and commercial users around the world. It is maintained by United States Government and is freely accessible to any one with a GPS receiver.

Components. The GPS consists of three segments : space, control and user segments as shown in Fig. 15.36.

(i) *Space segment (GPS satellites).* This segment consists of 24 satellites deployed on six orbits around the earth at the altitude of approximately 20,000 km (four GPS satellites per orbit). These satellites revolve around the earth at 12-hour-intervals.

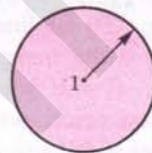
(ii) *Control segment (Ground control stations).* Ground control stations (five unmanned monitor stations and one Master Control Station) constantly monitor, control and maintain satellite orbits to make sure that the deviations of the satellites from the orbit as well as GPS timing are within the tolerance level.

(iii) *User segment (GPS receivers).* This segment consists of civilian or military users and their GPS receivers. The number of simultaneous users is limitless.

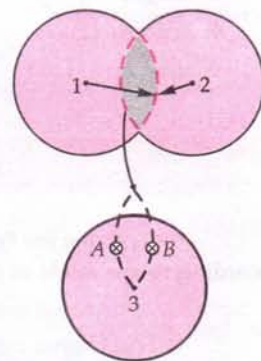
Working principle of a GPS device. The orbits of the satellites are so arranged that a GPS receiver on the earth can receive signals from atleast four satellites at any given time. A position on earth can be definitely determined by measuring the distances to three

satellites simultaneously. For this GPS receivers decode the information sent by the satellite. The satellites used to define the current position are equipped with atomic clocks, allowing an extremely precise time measurement. The signals sent by the satellite contain encoded information on the exact moment of transmission. A receiver on the earth contains a memory with the coordinates of each satellite. From the time difference between the sending and receiving of the electromagnetic signal, the distance of the satellite from the receiver can be determined ($s = c \times t$).

In space, if your distance from a satellite is known, you are somewhere on a sphere of that radius.



If 2 satellite distances are known, the only locations that are at right distance from both satellites form a circle (dotted line) where the two surfaces intersect.



If a third satellite's distance is known, you must be on its sphere as well as intersection circle of the first two. Now we have two positions A and B that can lie on the three spheres at the same time.

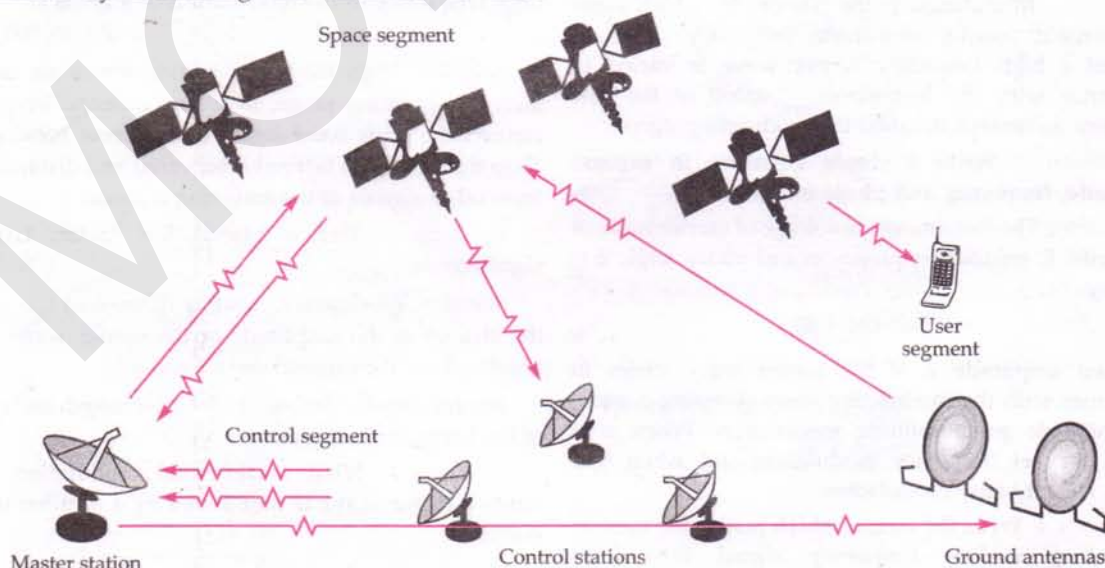
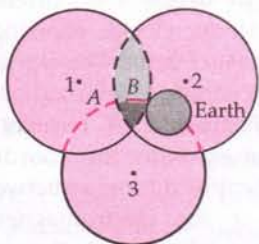


Fig. 15.36 Three segments of GPS.

The earth is a 4th sphere. One position A is in space. So only one position B is on the earth.



The time measured by clocks used on GPS receivers is not as precise as the time measured by atomic clocks used on GPS satellites. Hence a fourth satellite is used to reduce the margin of error in position accuracy. Clearly, if the measurement involves a constant unknown error, we need to know four unknown variables in 3D space :

- ❖ longitude (X)
- ❖ height (Z)
- ❖ latitude (Y)
- ❖ time error (Δt)

It therefore follows that in three dimensional space, four satellites are needed to determine a position accurately.

Applications of GPS :

Global positioning system has many day-to-day applications :

1. It helps in navigation on water, air and land.
2. It assists in map designing of a location.
3. It helps automatic vehicle movements (without man).
4. One can measure speed of moving objects using this technology.
5. One can locate change in position of glaciers, mountains heights.
6. It assists in keeping standard time world over.
7. It assists in tracking animals and birds and studying their movements by attaching GPS devices to their bodies.
8. It assists in airplane traffic movement.
9. It assists visually impaired in location identification.

VERY SHORT ANSWER CONCEPTUAL PROBLEMS

Problem 1. Name the type of communication systems according to the mode of the transmission. [CBSE D 05]

Solution. (i) Analog communication system.
(ii) Digital communication system.

Problem 2. What is meant by the term modulation ? [CBSE D 03, 09]

Solution. Modulation is the process by which some characteristic, usually amplitude, frequency or phase angle of a high frequency carrier wave is varied in accordance with the instantaneous value of the low frequency audio signal, called the modulating signal.

Problem 3. Write a single equation to express amplitude, frequency and phase modulation.

Solution. The instantaneous voltage of carrier wave of amplitude E , angular frequency ω and phase angle ϕ is given by

$$e = E \sin (\omega t + \phi)$$

When amplitude E of the carrier wave varies in accordance with the modulating wave (keeping ω and ϕ constant), we get amplitude modulation. When ω is varied, we get frequency modulation and when ϕ is varied, we get phase modulation.

Problem 4. Write the factors which justify the need of modulating a low frequency signal into high frequencies before transmission. [CBSE OD 15]

Solution. Modulation is needed

- (i) to transmit a low frequency signal to a distant place so that it may not die out in the way itself
- (ii) for protecting the waveform of the signal and
- (iii) to keep the height of antenna small.

Problem 5. Why do we need carrier waves of very high frequency in the modulation of signals ?

[CBSE D 09, OD 09, 10]

Solution. High frequency carrier waves are used to increase operating range, to reduce antenna length and convert the wide band signal into narrow band signal. Then the signal can be easily recovered and distinguished from other signals at the receiving station.

Problem 6. Define modulation factor. Give its significance. [CBSE F 12]

Solution. Modulation factor is defined as the ratio of the change in the amplitude of the carrier wave to the amplitude of the original carrier wave.

Modulation factor determines the strength and quality of the transmitted signal.

Problem 7. What is the total modulation index when a carrier wave is modulated by a number of sine waves ?

Solution. $m_{\text{total}} = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$

Problem 8. How does the amplitude of the carrier wave vary in AM ?

Solution. In AM, the amplitude of the high frequency carrier wave varies proportionally to the instantaneous value of the amplitude of the audio (modulating) signal.

Problem 9. At what rate does the amplitude of the carrier wave change in amplitude modulation ?

Solution. Amplitude of the carrier wave in AM changes at the frequency f_m of the modulating signal.

Problem 10. What happens when modulation factor exceeds unity in AM ?

Solution. When m exceeds unity, the carrier wave is *overmodulated*. The negative half cycle of the modulating signal is clipped and distortion will occur during reception.

Problem 11. What are the three frequencies contained in an AM wave ? What are LSB and USB ?

Solution. The three frequencies contained in an AM wave are : $f_c - f_m$, f_c and $f_c + f_m$.

LSB (lower sideband) = $f_c - f_m$

USB (upper sideband) = $f_c + f_m$.

Problem 12. Write an expression for the bandwidth of an AM transmission ?

Solution. Bandwidth of AM transmission
= $2 \times$ Maximum frequency of the modulating signal

Problem 13. Which is better for high fidelity reception : FM or AM ?

Solution. FM transmission gives high fidelity due to the presence of a large number of sidebands.

Problem 14. Which is more efficient mode of transmission : FM or AM ?

Solution. FM transmission is more efficient because all the transmitted power is useful but in AM transmission most of the power goes waste in transmitting the carrier alone.

Problem 15. What is function of amplitude limiters used in FM receivers ?

Solution. They remove the variations in amplitude of the modulated wave caused by noise.

Problem 16. What is meant by sensitivity of a radio receiver ?

Solution. The ability of a radio receiver to amplify weak signals is called its sensitivity.

Problem 17. What is meant by selectivity of a radio receiver ?

Solution. The ability of a radio receiver to reject unwanted signals is called its selectivity.

Problem 18. In a carrier frequency of 100 kHz and a modulating frequency of 5 kHz, what is the bandwidth of AM transmission ?

Solution. 10 kHz.

Problem 19. Why do we need a higher bandwidth for transmission of music compared to that for commercial telephonic communication ?

[CBSE D 09C]

Solution. Speech signals contain frequencies between 300 Hz to 3100 Hz. Such signals require a small bandwidth of 2800 Hz for telephonic communication. Audio signals have frequencies between 20 Hz to 20 kHz. So, the transmission of a good music requires a higher bandwidth of about 20 kHz.

Problem 20. What should be the length of the dipole antenna for a carrier wave of frequency 3×10^8 Hz ?

[CBSE OD 03 ; D 05]

Solution. Length of a dipole antenna

$$= \frac{\lambda}{2} = \frac{c}{2\nu} = \frac{3 \times 10^8}{2 \times 3 \times 10^8} = 0.5 \text{ m}$$

Problem 21. An audio signal of frequency ν_a is to be transmitted as an electromagnetic wave

(i) directly as such.

(ii) through its use as the modulating signal on a carrier wave of frequency ν_c .

State the ratio of the size of the transmitting antenna, that can properly sense the time variation of the signal, in terms of ν_a and ν_c .

[CBSE Sample Paper 11]

Solution. Size of antenna $\propto \frac{1}{\text{Frequency}}$

$\therefore$ Required ratio = $\nu_c : \nu_a$

Problem 22. Why is the transmission of signals through a coaxial cable not possible for frequencies greater than 20 MHz ?

[CBSE F 04]

Solution. For frequencies greater than 20 MHz, dielectric loss becomes quite high.

Problem 23. Give one example each of a 'system' that uses the (i) sky wave (ii) space wave mode of propagation.

[CBSE Sample Paper 13]

Solution. (i) Short broadcast services use sky wave propagation.

(ii) TV broadcast/microwaves links/satellite communication use space wave propagation.

Problem 24. Why is ground wave transmission of signals restricted to a frequency of 1500 kHz ?

[CBSE D 2000 ; OD 01, 04 ; 13C]

Solution. As a ground wave passes over the surface of the earth, it is weakened due to the absorption of energy by the earth. These energy losses increase with the increase in frequency. Hence ground wave propagation can be sustained only upto a frequency of 1500 kHz. Higher frequencies are highly damped.

Problem 25. Why are short wave bands used for long distance transmission of signals ?

[CBSE OD 02, 04]

Solution. Radio waves of short wave bands can be easily reflected by the ionosphere. So they are used in long distance transmission.

Problem 26. Why is the transmission of signals using sky waves, restricted to frequencies upto 30 megahertz ?

[CBSE D 99]

Solution. This is because ionosphere cannot reflect e.m. radiations having frequency greater than 30 MHz.

Problem 27. Give the reason why transmission of TV signals via sky waves is not possible.

[CBSE D 02 ; F 08, 09]

Solution. Television frequencies lie in the range 100-220 MHz which cannot be reflected by the ionosphere. So sky wave propagation method is not used in TV transmission.

Problem 28. It is necessary to use satellites for long distance T.V. transmission. Why ?

[CBSE D 96C ; Himachal 04]

Solution. TV signals being of high frequency are not reflected by the ionosphere. Also, ground wave transmission is possible only upto a limited range. That is why satellites are used for long distance TV transmission.

Problem 29. What is the essential requirement for transmitting a microwave from one point to another on the earth ?

Solution. For microwave transmission, the transmitting and receiving antenna must be in the line of sight.

Problem 30. What is switching circuit in a telephone system ?

Solution. In a locality, all the telephones are connected to the telephone exchange. When a calling subscriber dials in the called subscriber, the switching circuit interconnects their telephones.

Problem 31. What is the origin of the word modem ?

Solution. The word modem is the contraction of the terms modulator and demodulator.

Problem 32. What is the function of a FAX ?

[CBSE Sample Paper 03]

Solution. It is used to produce an exact copy of a document or picture at a distant place.

Problem 33. Name the factor which decides the quality of reproduced document sent by a FAX. [CBSE D 03]

Solution. Resolution capacity of the scanner.

Problem 34. What is the difference between telegraphy and teleprinting ?

Solution. In telegraphy, the information is transmitted in the form of codes, say dots and dashes. Teleprinting is the advanced form of telegraphy in which text is transmitted to distant places.

SHORT ANSWER CONCEPTUAL PROBLEMS

Problem 1. Modulation is used to :

- reduce the bandwidth used.
- separate the transmissions of different users.
- ensure that intelligence may be transmitted to long distances.
- allow the use of practical antennas.

Solution. (a), the use of high frequency carrier wave reduces the bandwidth.

Problem 2. AM is used for broadcasting because

- it is more noise immune than other modulation systems.
- it requires less transmitting power compared with other systems.
- its use avoids receiver complexity.
- no other modulation system can provide the necessary bandwidth faithful transmission.

Solution. (c), AM is used for broadcasting because its use avoids receiver complexity. Only a diode and a capacitor are sufficient to separate the audio signal from the AM wave.

Problem 3. We do not choose to transmit an audio signal by just directly converting it to an e.m. wave of the same frequency. Give reasons for the same.

[CBSE Sample Paper 08]

Solution. (i) Due to their low frequency ranges, audio signals radiated directly from an antenna die out after travelling some distance.

(ii) Height of antenna required is impractically large.

(iii) Audio signals transmitted from different stations get inseparably mixed up.

Problem 4. Complete the following block diagram depicting the essential elements of a basic communication system :

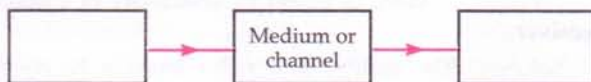


Fig. 15.37

Name the two basic modes of communication. Which of these modes is used for telephonic communication ?

[CBSE Sample Paper 11]

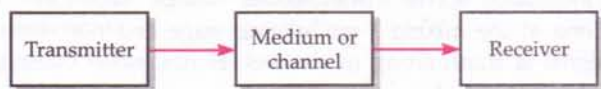
Solution.

Fig. 15.38

Two basic modes of transmission are

- (i) point-to-point and
- (ii) broadcast mode.

Point-to-point mode is used for telephonic communication.

Problem 5. Explain briefly the following terms used in communication system :

- (i) Transducer (ii) Repeater (iii) Transmitter
- (iv) Bandpass filter (v) Amplification

[CBSE OD 12 ; F 11 ; D 12, 13C]

Solution. (i) *Transducer.* It is a device which converts energy from one form to another form.

(ii) *Repeater.* It is a combination of receiver, amplifier and transmitter. It picks up a signal from the transmitter, amplifies and transmits it to the receiver sometimes with a change of carrier frequency.

(iii) *Transmitter.* It is a device which processes the incoming message signal into a form suitable for transmission through a channel and for its subsequent transmission.

(iv) *Bandpass filter.* A bandpass filter blocks lower and higher frequencies and allows only band of frequencies (centred around carrier frequency) to pass through it.

(v) *Amplification.* It is the process of increasing the amplitude and hence the strength of an electrical signal by using an electric circuit (consisting of at least one transistor) called the amplifier.

Problem 6. Distinguish between analog and digital communications. [CBSE D 05 ; OD 15]

Solution. Analog communication makes use of analog electronic circuit and analog signal in such a way that the output voltage varies continuously in accordance with the input signal. Digital communication makes use of an electronic circuit that can handle only digital signals.

Problem 7. We do not choose to transmit an audio signal by just directly converting it to an e.m. wave of the same frequency. Give reasons for the same.

[CBSE Sample Paper 08]

Solution.

- (i) Due to their low frequency ranges, audio signals radiated directly from an antenna die out after travelling some distance.

- (ii) Height of antenna required is impractically large.

- (iii) Audio signals transmitted from different stations get inseparably mixed up.

Problem 8. Explain, why high frequency carrier waves are needed for effective transmission of signals.

[CBSE D 09, 12 ; OD 10]

Solution. Need for modulating a signal :

- (i) To reduce length of transmitting antenna to practical heights ($l \approx \lambda / 4$).
- (ii) To increase the power radiated by an antenna ($P \propto 1/\lambda^4$) so as to increase the range of transmission.
- (iii) To convert a wide band signal into a narrow band signal.
- (iv) To transmit simultaneously a large number of audio signals from the same antenna without getting them mixed up.

Problem 9. Why are broadcast frequencies (of carrier waves) sufficiently spaced in amplitude modulated waves ? [CBSE F 13]

Solution. Different audio signals fall in the same spectral range. To transmit them simultaneously, different broadcasting stations are allotted different slots of carrier frequencies so that a single receiver can tune into these frequencies without any interference, confusion or overlap.

Problem 10. Define the term modulation. Name three different types of modulation used for a message signal using a sinusoidal continuous carrier wave. Explain the meaning of any one of them. [CBSE D 06]

Solution. **Modulation.** It is the process of varying some characteristic (amplitude, frequency or phase angle) of the high frequency carrier wave in accordance with the instantaneous value of the low frequency message signal.

Types of modulation :

- (i) Amplitude modulation (ii) Frequency modulation
- (iii) Phase modulation.

Amplitude modulation. In this process the modulating wave (message signal) is superimposed on a high frequency carrier wave such that the amplitude of the modulated wave varies as the amplitude of the modulating wave. Frequency of the modulated wave is equal to the frequency of the carrier wave.

Problem 11. Draw a block diagram of a simple modulator for obtaining an AM signal. Briefly explain how amplitude modulation is achieved. [CBSE D 08, 09]

Or

Draw a block diagram of a simple modulator for obtaining an amplitude modulated (AM) signal explaining the function of each of its components. [CBSE D 08C]

Solution. The block diagram of a simple modulator for obtaining AM signal is shown below :

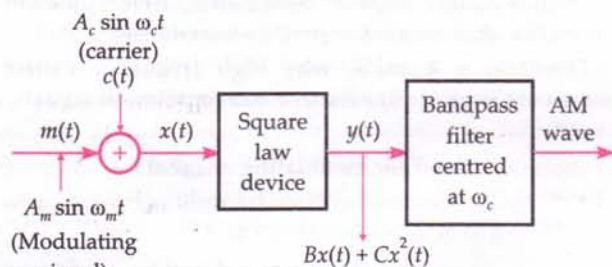


Fig. 15.39

Here the modulating signal is added to the carrier signal to produce the input signal,

$$x(t) = A_m \sin \omega_m t + A_c \sin \omega_c t$$

This signal is passed through a square law device which gives the output,

$$y(t) = Bx(t) + Cx^2(t)$$

Here B and C are constants. When this signal is passed through a bandpass filter centred at ω_c , the filter rejects the sinusoidal signals of frequencies ω_m , $2\omega_m$ and $2\omega_c$ and retains the frequencies ω_c , $\omega_c - \omega_m$ and $\omega_c + \omega_m$. The output of the bandpass filter is thus an AM wave.

Role of bandpass filter in amplitude modulation. The bandpass filter rejects d.c. signal, low frequencies (ω_m , $2\omega_m$), high frequency ($2\omega_c$) and retains the band of frequencies centred around ω_c (ω_c , $\omega_c \pm \omega_m$).

Problem 12. A radio broadcast is transmitted using amplitude modulation at a carrier frequency of 680 kHz. Explain the meaning of each of the italicised words.

Solution. When the message signal is superimposed on a carrier wave of frequency 680 kHz, the amplitude of the resultant modulated wave varies in proportion with the instantaneous value of the message signal. The amplitude of the modulated wave varies at the frequency (f_m) of the message signal while its frequency remains equal to that of the carrier wave (680 Hz).

Problem 13. Why is an FM signal less susceptible to noise than an AM signal ?

[CBSE OD 06]

Or

Why is an AM signal likely to be more noisy than an FM signal upon transmission through channel ?

[Exemplar Problem]

Solution. In AM, the carrier wave's instantaneous voltage is varied by modulating waves voltage. On transmission, noise signals can also be added and receiver assumes noise a part of the modulating signal. However

in FM, the carriers' frequency is changed as per modulating waves instantaneous voltage. This can be done at the mixing / modulating stage and not while signal is transmitting in channel. Hence, noise doesn't affect FM signal.

Problem 14. Define the term 'modulation index' for an AM wave. What would be the modulation index for an AM wave for which the maximum amplitude is 'a' while the minimum amplitude is 'b' ?

[CBSE SP 08]

Solution. Modulating index for an AM wave is defined as the ratio of change in the amplitude of the carrier wave to the amplitude of the original carrier wave.

$$\begin{aligned} \mu &= \frac{\text{Change in the amplitude of carrier wave}}{\text{Amplitude of the original carrier wave}} \\ &= \frac{A_m}{A_c} \end{aligned}$$

When a and b are the maximum and minimum amplitudes of the AM wave, then

$$A_m = \frac{a-b}{2}$$

and

$$A_c = a - A_m = a - \frac{a-b}{2} = \frac{a+b}{2}$$

$$\therefore \mu = \frac{A_m}{A_c} = \frac{(a-b)/2}{(a+b)/2} = \frac{a-b}{a+b}$$

Problem 15. (i) Define modulation index. (ii) Why is the amplitude of modulating signal kept less than the amplitude of carrier wave ?

[CBSE D 11]

Solution. (i) Refer answer to the above question.

(ii) When A_m exceeds A_c , the carrier is over-modulated ($\mu > 1$), which causes distortion during reception. That is why A_m is kept less than A_c .

Problem 16. Define modulation index. Give its physical significance.

For an amplitude modulated wave, the maximum amplitude is found to be 10 V while the minimum amplitude is 3 V. Determine the modulation index μ .

[CBSE F 12]

Solution. For modulation index, refer answer to Problem 14. Modulation index determines the strength and quality of the transmitted signal.

Maximum amplitude, $A_{\max} = 10\text{ V}$

Minimum amplitude, $A_{\min} = 3\text{ V}$

Modulation index,

$$\begin{aligned} \mu &= \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \\ &= \frac{10 - 3}{10 + 3} = \frac{7}{13} = 0.54 \end{aligned}$$

Problem 17. In the given block diagram of a receiver, identify the boxes labelled as X and Y and write their functions. [CBSE OD 12 ; D 13]

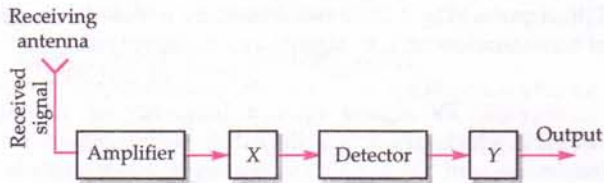


Fig. 15.40

Solution. X = Intermediate frequency (IF) stage.

Y = Amplifier

IF stage changes the carrier frequency to a (standard) lower frequency.

Amplifier strengthens the detected signal which is not strong enough to be made use of directly.

Problem 18. Draw the waveforms for

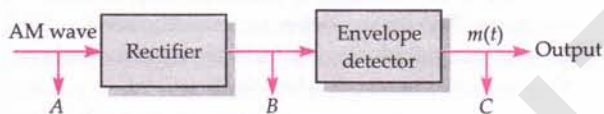


Fig. 15.41

the (i) input AM wave at A, (ii) output, B, of the rectifier and (iii) output signal, C, of the envelope detector. [CBSE OD 13C]

Solution.

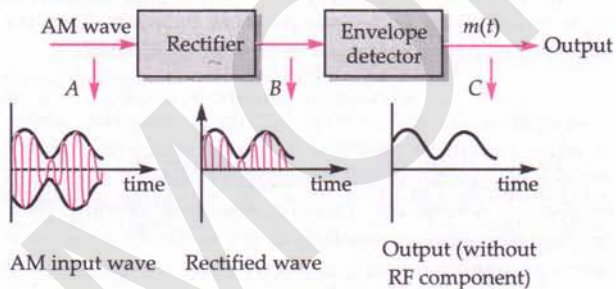


Fig. 15.42

Problem 19. How do we make the choice of a communication channel? A message signal has a bandwidth of 5 MHz. Suggest a possible communication channel for its transmission. [CBSE 03]

Solution. The choice of communication channel depends both on the nature of the signal and the nature of the communication of medium.

(i) For guided media, the medium itself is important.

(ii) For unguided media, the frequency band of signal is important.

For the transmission of a message signal of bandwidth 5 MHz, the suitable communication channels are (i) Co-axial cable and (ii) AM radio.

Problem 20. Name the three different modes of propagation of electromagnetic waves. Explain, using a proper diagram the mode of propagation used in the frequency range from a few MHz to 40 MHz. [CBSE D 12]

Or

Mention three different modes of propagation used in communication system. Explain with the help of a diagram how long distance communication can be achieved by ionospheric reflection of radio waves. [CBSE OD 12]

Solution. The different modes of propagation of e.m. waves are : (i) Ground wave propagation (ii) Space wave propagation (iii) Sky wave propagation.

Sky wave propagation is used for the transmission of e.m. waves having frequency range from a few MHz to 40 MHz. As shown in the figure, here a radiowave is directed towards the sky and is reflected by the ionosphere towards the desired location of the earth.

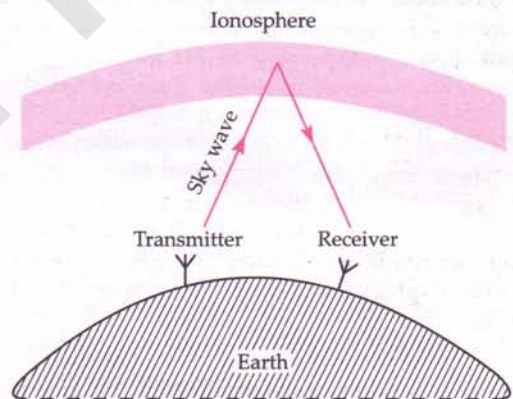


Fig. 15.43 Sky wave propagation.

Problem 21. Which mode of propagation is used by short wave broadcast services having frequency range from a few MHz upto 30 MHz? Explain diagrammatically how long distance communication can be achieved by this mode. Why is there an upper limit to frequency of waves used in this mode? [CBSE OD 10]

Or

What is sky wave communication? Why is this mode of propagation restricted to the frequencies only upto few MHz? [CBSE OD 11]

Solution. Sky wave propagation can be used by short wave broadcast services having frequency range from a few MHz to 30 MHz.

In sky wave propagation, a radiowave is directed towards the sky and is reflected by the ionosphere towards the desired location of the earth, as shown in Fig. 15.43.

In this way, radiowaves can travel very large distances and can even bend round the earth.

Sky wave propagation is restricted to frequencies upto 30 MHz because ionosphere cannot reflect e.m. radiations having frequencies greater than 30 MHz.

Problem 22. What is a ground wave? Why short wave communication over long distance is not possible via ground waves? [CBSE OD 90]

Solution. A wave that travels directly from one point to another following the surface of the earth is called ground or surface wave.

In ground wave propagation, loss of energy due to interaction with matter increases with the increase in frequency (or decrease in wavelength). Wavelengths less than 200 m (or frequencies > 1500 kHz) get heavily damped. So short wave communication via ground waves is not possible.

Problem 23. Explain, using a proper diagram the mode of propagation used in the frequency range above 40 MHz. [CBSE D 12]

Solution. Above 40 MHz, the mode of propagation used is via space waves. A space wave travels in a straight line from the transmitting antenna to the receiving antenna. Space waves are used for line of sight (LOS) communication as well as satellite communication.

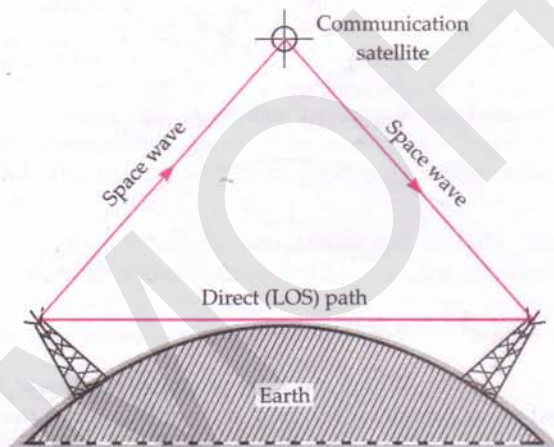


Fig. 15.44 Space wave propagation.

Problem 24. What is the range of frequencies used in satellite communication? What is common between these waves and light waves? [CBSE D 10]

Solution. The following two frequency bands are used for satellite communication:

5.9 – 6.4 Hz for uplink

3.7 – 4.2 GHz for downlink.

The waves used for satellite communication and light waves are both electromagnetic waves which travel through free space with a speed of $3 \times 10^8 \text{ ms}^{-1}$.

Problem 25. Sky waves are not used in transmitting T.V. signals. Why? State two factors by which the range of transmission of T.V. signals can be increased.

[CBSE D 01, 01C; OD 13C]

Solution. TV signals have a frequency of 100 to 220 MHz which cannot be reflected by the ionosphere. So transmission of TV signals via sky wave is not possible. Range of TV transmission can be increased by using (i) tall antenna and (ii) geostationary satellites.

Problem 26. Why do we require a satellite for long distance TV transmission? A fax message is to be sent from Delhi to Washington via a geostationary satellite. Calculate the minimum time delay between the dispatch and its getting received. (Take height of the geostationary satellite = 36,000 km) [CBSE Sample Paper 03; F 05]

Solution. TV signals being of high frequency range, 100–220 MHz, cannot be reflected by ionosphere. Also ground wave transmission is possible only upto a limited range. That is why satellites are used for long distance TV transmission. The transponders in a satellite receive these signals and transmit them towards different ground stations.

Numerical: Minimum distance,

$$s = 2h = 2 \times 36000 \text{ km} = 72000 \text{ km}$$

$$\text{Time delay} = \frac{s}{c} = \frac{2 \times 36000 \times 1000 \text{ m}}{3 \times 10^8 \text{ ms}^{-1}} = 0.245 \text{ s}$$

Problem 27. (i) Why is communication using line of sight mode limited to frequencies above 40 MHz?

(ii) A transmitting antenna at the top of a tower has a height 36 m and the height of the receiving antenna is 49 m. What is the maximum distance between them for satisfactory communication in line of sight mode?

[CBSE D 08, 10]

Solution. (i) EM waves having frequencies above 40 MHz cannot be reflected by the ionosphere. The sky wave propagation is not possible. Moreover, such high frequency waves are heavily absorbed during their propagation over the surface of the earth. Their ground wave propagation is not possible. Such waves can only propagate by line of sight mode.

(ii) Here, $h_T = 36 \text{ m}$, $h_R = 49 \text{ m}$,

$$R = 6400 \text{ km} = 64 \times 10^5 \text{ m}$$

Maximum distance for satisfactory communication in LOS mode,

$$\begin{aligned} d_R &= \sqrt{2Rh_T} + \sqrt{2Rh_R} \\ &= \sqrt{2 \times 64 \times 10^5 \times 36} + \sqrt{2 \times 64 \times 10^5 \times 49} \text{ m} \\ &= 8 \times 6 \times 10^2 \sqrt{20} + 8 \times 7 \times 10^2 \sqrt{20} \text{ m} \\ &= 10400 \sqrt{20} \text{ m} = 10400 \times 4.472 \text{ m} \\ &= 46510.2 \text{ m} \approx 46.5 \text{ km} \end{aligned}$$

Problem 28. Give reasons for the following :

- (i) For ground wave transmission, size of antenna (l) should be comparable to wavelength of signal i.e., $l = \lambda / 4$.
- (ii) Audio Signals, converted into an e.m. wave, are not directly transmitted as such.
- (iii) The amplitude of modulating signal is kept less than the amplitude of carrier wave.

[CBSE F 13]

Solution. (i) When the size of the antenna is comparable to wavelength of the signal, the time variations of the signal are properly sensed by the antenna.

(ii) An e.m. wave, of audio signal frequency, would have a very high wavelength. It would, therefore, need an antenna, whose size would be practically unattainable.

(iii) The amplitude of modulating signal is kept less than the amplitude of carrier wave to avoid distortion.

Problem 29. Give some limitations of amplitude modulation.

Solution. (i) Amplitude modulation suffers from noise.

(ii) Quality of audio signal is poor.

(iii) Efficiency of AM transmission is low.

Problem 30. Give some advantages of AM over FM transmission.

Solution. (i) FM transmission needs a much wider channel, typically 200 kHz.

(ii) Reception of FM signal is limited to the line of sight.

(iii) Transmitting and receiving components used in FM transmission are costly and complex.

Problem 31. Give some advantages of FM over AM transmission.

Solution. (i) Efficiency of FM transmission is quite high.

(ii) FM transmission gives high fidelity reception due to the presence of a large number of sidebands.

(iii) Noise level is very low in FM transmission.

(iv) Frequency allocation in FM allows for a guard band which reduces adjacent channel interference.

Problem 32. Give some advantages of digital communication.

Solution.

(i) Digital signals can be easily received.

(ii) Digital signals do not get distorted by noise.

(iii) Digital signals can be coded.

(iv) Digital signals can be reproduced more accurately.

Problem 33. Distinguish between a FAX and E-mail.

Solution. **FAX.** It is used for facsimile/exact reproduction of document (picture, letter, map, etc) at the receiving end. The document is first converted into digital data form. This process, called scanning, is accomplished by using an optical scanner. The digital data representing the document is then transmitted to the destination by using a suitable medium. At the receiving end, the digital data is used to reconstruct the original document.

E-mail. The messages sent through a computer network are called electronic mails (E-mails). The messages produced using word processing are transmitted over the worldwide network, called internet. The messages get stored in a computer, called a mail server. Anyone connected to the internet can contact the mail server to check, whether it is holding any mail for him/her.

HOTS

Problems on Higher Order Thinking Skills

Problem 1. A schematic arrangement for transmitting a message signal (20 Hz to 20 kHz) is given below :

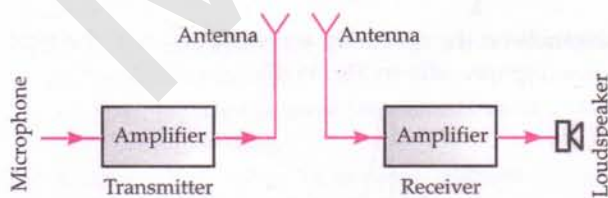


Fig. 15.45

Give two drawbacks from which this arrangement suffers. Describe briefly with the help of a block diagram the

alternative arrangement for the transmission and reception of the message signal.

[CBSE OD 03]

Solution. The drawbacks of the given arrangement are :

(i) Audio signals cannot be efficiently radiated and do not propagate well in space.

(ii) Simultaneous transmission of signals by different transmitters can cause confusion at the receiver end due to overlapping of frequencies.

The above difficulties can be overcome by translating the message signal into radio frequency range before transmission and recover it back at the

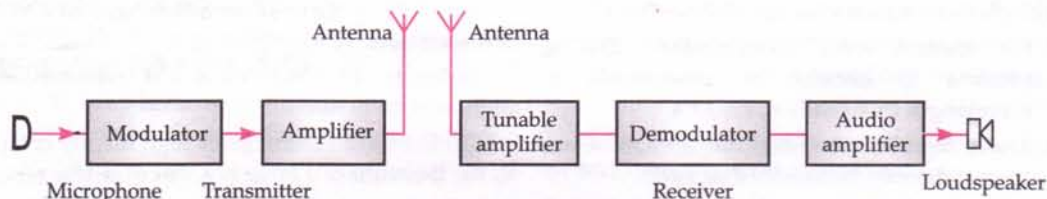


Fig. 15.46

receiver. Different transmitting stations can be allotted slots in radio frequency range and a single receiver can then tune into these transmitters without any confusion or overlap as shown in Fig. 15.46.

Problem 2. A (sinusoidal) carrier wave $c(t) = A_c \sin \omega_c t$ is amplitude modulated by a (sinusoidal) message signal, $m(t) = A_m \sin \omega_m t$. Write the equation of the (amplitude) modulated signal. Use this equation to obtain the values of the frequencies of all the sinusoidal waves present in the modulated signal. [CBSE SP 15]

Solution. The equation of the (amplitude) modulated signal is

$$c_m(t) = [(A_c + A_m \sin \omega_m t)] \sin \omega_c t$$

This can be rewritten as

$$c_m(t) = [A_c(1 + \mu \sin \omega_m t)] \sin \omega_c t$$

where $\mu = A_m / A_c =$ modulation index.

$$\begin{aligned} c_m(t) &= A_c \sin \omega_c t + \frac{\mu A_c}{2} \cdot 2 \sin \omega_m t \cdot \sin \omega_c t \\ &= A_c \sin \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t] \end{aligned}$$

Clearly, there are three sinusoidal waves present in the amplitude modulated signal.

Their frequencies are :

$$f_1 = \frac{\omega_c}{2\pi}, \quad f_2 = \frac{\omega_c - \omega_m}{2\pi} \quad \text{and} \quad f_3 = \frac{\omega_c + \omega_m}{2\pi}$$

Problem 3. Why cannot the (i) ground waves, (ii) space waves and (iii) sky waves be sustained for long distance communications beyond 10 to 20 MHz ?

What is meant by critical frequency for sky wave propagation ? Write an expression for the critical frequency in terms of the maximum electron density of the ionosphere. What is the range of variation of frequency for this critical frequency ? [CBSE F 04]

Solution. (i) In ground waves, conduction losses due to ground become quite high at high frequencies.

(ii) Space waves have limited line of sight distance.

(iii) At frequencies beyond 10 to 20 MHz, sky waves penetrate ionosphere and are not reflected back.

In sky wave propagation, critical frequency is the maximum frequency beyond which total internal reflection of a radiowave does not take place from a given depth of ionosphere. It is given by $f_c = 9 (N_{\max})^{1/2}$, where $N_{\max}$ is maximum electron density of the ionosphere in per m^3 . The value of f_c ranges from 5 to 10 MHz.

Problem 4. An optical communication system, having an operating wavelength, λ (in metres), can use only $x\%$ of its source frequency as its channel bandwidth. The system is to be used for transmitting TV signals requiring a bandwidth of F hertz. How many channels can this system transmit simultaneously ?

Assuming all other factors to remain constant, show graphically, the dependence of the number of channels that can be transmitted simultaneously, on the operating wavelength of the system. [CBSE D 04C]

Solution. Optical source frequency $= \frac{c}{\lambda}$ Hz

$$\text{Bandwidth of channel} = x\% \text{ of } \frac{c}{\lambda} = \frac{cx}{100 \lambda}$$

Number of channels,

$$\begin{aligned} n &= \frac{\text{Total bandwidth of channel}}{\text{Bandwidth needed per channel}} \\ &= \frac{cx / 100 \lambda}{F} = \frac{cx}{100 \lambda F} \end{aligned}$$

As $n \propto \frac{1}{\lambda}$, the dependence of the number of channels on the operating wavelength λ is of the type shown graphically in Fig. 15.47.

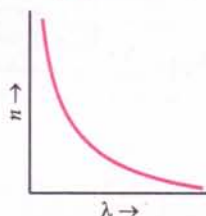


Fig. 15.47

Problem 5. What does the term LOS communication mean? Name the types of waves that are used for this communication. Which of the two-height of transmitting antenna and height of receiving antenna can affect the range over which this mode of communication remains effective?

[CBSE Sample Paper 08]

Solution. A space wave travels in a straight line from transmitting antenna to the receiving antenna. It is called line of sight (LOS) communication. TV frequencies of the range 100 – 220 MHz and microwave links use this mode of communication.

The maximum line-of-sight distance between the transmitting and receiving antennas is given by

$$d_m = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Clearly, the range of LOS communication depends both on the height of transmitting antenna (h_T) and the height of receiving antenna (h_R).

Problem 6. Is it necessary for the transmitting antenna and the receiving antenna to be of the same height for line of sight communication? Find an expression for maximum line of sight distance d_m between these two antennas of heights h_T and h_R .

[CBSE Sample Paper 11]

Solution. No, it is not necessary to have the same height for both transmitting and receiving antenna.

For transmitting antenna of height h_T ,

$$(h_T + R_e)^2 = x^2 + R_e^2$$

$$\therefore h_T \ll R_e$$

$\therefore$

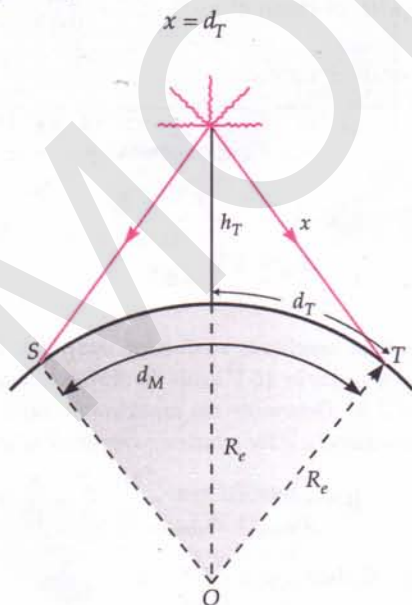


Fig. 15.48

$$\therefore (h_T + R_e)^2 = d_T^2 + R_e^2$$

$$\text{or } h_T^2 + R_e^2 + 2h_TR_e = d_T^2 + R_e^2 \quad (h_T^2 \text{ is negligible})$$

$$\Rightarrow d_T = \sqrt{2R_e h_T}$$

d_T is also called the radio horizon of the transmitting antenna.

The maximum line-of-sight distance d_M between the two antennas is

$$d_M = \sqrt{2R_e h_T} + \sqrt{2R_e h_R}$$

where h_R is the height of the receiving antenna.

Problem 7. What is space wave propagation? Which two communication methods make use of this mode of propagation? If the sum of the heights of transmitting and receiving antennas in line of sight of communication is fixed at h , show that the range is maximum when the two antennas have a height $h/2$ each.

[CBSE Sample Paper 08]

Solution. If a radiowave transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna, the propagation is called space wave propagation. Television broadcast and satellite communication make use of this mode of propagation.

The range of line-of-sight communication between two antennas is given by

$$r = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Given $h_T + h_R = h$ and let $h_T = H$, then $h_R = h - H$

$$\therefore r = \sqrt{2R} [\sqrt{H} + \sqrt{h - H}]$$

For r to be maximum,

$$\frac{dr}{dH} = \sqrt{2R} \left[\frac{1}{2\sqrt{H}} + \frac{1}{2\sqrt{h-H}} (-1) \right] = 0$$

$$\text{or } \frac{1}{2\sqrt{H}} - \frac{1}{2\sqrt{h-H}} = 0$$

$$\text{or } H = h - H$$

$$\text{or } H = \frac{h}{2}$$

Hence for maximum range,

$$h_T = h_R = h/2.$$

GUIDELINES TO NCERT EXERCISES

15.1. Which of the following frequencies will be suitable for beyond the horizon communication using sky waves ?

- (a) 10 kHz (b) 10 MHz
(c) 1 GHz (d) 1000 GHz

Ans. (b) 10 MHz. Frequency of 10 kHz will require very large radiating antenna while frequencies 1 GHz and 1000 GHz will penetrate the ionosphere and cannot be reflected by it.

15.2. Frequencies in the UHF range normally propagate by means of :

- (a) Ground waves (b) Sky waves
(c) Surface waves (d) Space waves

Ans. (d) Space waves. Frequencies in the UHF (0.3 to 3 GHz) range normally propagate by means of space waves. Their sky wave reflection from ionosphere is not possible.

15.3. Digital signals (i) do not provide a continuous set of values (ii) represent values as discrete steps (iii) can utilize binary system and (iv) can utilize decimal as well as binary systems. Which of the following is true ?

- (a) (i) and (ii) only
(b) (ii) and (iii) only
(c) (i), (ii) and (iii) but not (iv)
(d) All of (i), (ii), (iii) and (iv)

Ans. (c). Digital signals cannot utilize decimal system which represents a continuous set of values.

15.4. Is it necessary for a transmitting antenna to be at the same height as that of the receiving antenna for line of sight communication ? A TV transmitting antenna is 81 m tall. How much service area it can cover if the receiving antenna is at the ground level ? [CBSE F 15]

Ans. No, it is not necessary for a transmitting antenna to be at the same height as that of the receiving antenna for LOS communication. The heights should be such that the receiving antenna is capable of directly intercepting the signals radiated by the transmitting antenna.

Service area covered by the transmitting antenna,

$$\begin{aligned} A &= \pi d_T^2 = \pi (2Rh_T) \\ &= \frac{22}{7} \times 2 \times 6.4 \times 10^6 \times 81 \text{ m}^2 \\ &= 3258 \text{ km}^2. \end{aligned}$$

15.5. A carrier wave of peak voltage 12 V is used to transmit a message signal. What should be the peak voltage of the modulating signal in order to have a modulation index of 75% ? [CBSE OD 10]

Ans. As $\mu = \frac{A_m}{A_c}$

$$\therefore A_m = \mu A_c = \frac{75}{100} \times 12 \text{ V} = 9 \text{ V}.$$

15.6. A modulating signal is a square wave as shown below :



Fig. 15.49

The carrier wave is given by $c(t) = 2 \sin(8\pi t)$ volts.

- (i) Sketch the amplitude modulated waveform
(ii) What is the modulation index ?

Ans. (i) The amplitude modulated wave is shown below :

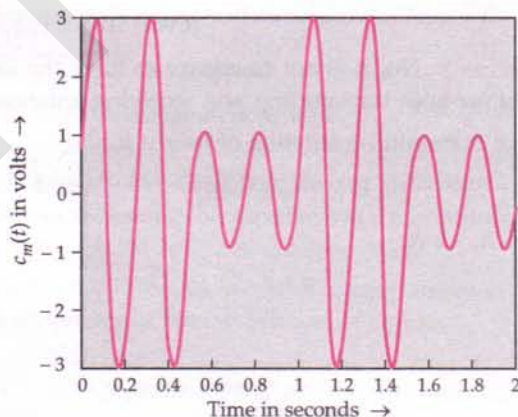


Fig. 15.50

(ii) $\mu = \frac{A_m}{A_c} = \frac{1 \text{ V}}{2 \text{ V}} = 0.5.$

15.7. For an amplitude modulated wave, the maximum amplitude is found to be 10 V while the minimum amplitude is found to be 2 V. Determine the modulation index, μ . What would be the value of μ if the minimum amplitude is zero volt ?

Ans. $\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{10 - 2}{10 + 2} = \frac{8}{12} = \frac{2}{3}.$

If $A_{\min} = 0$, then $\mu = \frac{A_{\max}}{A_{\max}} = 1.$

This is true for all values of $A_{\max}$.

15.8. Due to economic reasons, only the upper sideband of an AM wave is transmitted, but at the receiving station, there is a facility for generating the carrier. Show that if a device is available which can multiply two signals, then it is possible to recover the modulating signal at the receiver station.

Ans. Let, for simplicity, the received signal be

$$A_1 \cos (\omega_c + \omega_m) t$$

The carrier signal available at the receiving stations is

$$A_c \cos \omega_c t$$

Multiplying the two signals, we get

$$\begin{aligned} A_1 A_c \cos (\omega_c + \omega_m) t \cos \omega_c t \\ = \frac{A_1 A_c}{2} [\cos (2\omega_c + \omega_m) t + \cos \omega_m t] \end{aligned}$$

If this signal is passed through a low pass filter, we can record the modulating signal $\frac{A_1 A_c}{2} \cos \omega_m t$.

Text Based Exercises

TYPE A : VERY SHORT ANSWER QUESTIONS (1 mark each)

1. What is meant by communication ? [Himachal 06]
2. Name the three basic units of any communication system. [Haryana 05, 06]
3. What is a communication channel or transmission medium ?
4. Define the term "transducer" for a communication system. [CBSE Sample Paper 13]

Or

What is the function of a transducer ?

[Haryana 05 ; CBSE F 11]

5. Identify the parts X and Y in the following block diagram of a generalised communication system :

[CBSE D 08C]

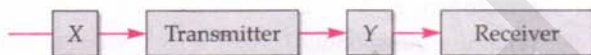


Fig. 15.51

6. What is an analog communication system ? [Himachal 06]
7. What is a digital communication system ?
8. Name the type of communication in which the signal is a discrete and binary coded version of the message or information. [CBSE D 06]
9. Define bandwidth of a signal.
10. What is channel bandwidth ? [Punjab 05C]
11. What do you mean by noise in communication ? [Punjab 06C]
12. What is attenuation ? [CBSE OD 10C, 14C]
13. What is meant by the term baseband ?

14. A device X can convert one form of energy into another. Another device Y can be regarded as a combination of a transmitter and a receiver. Name the devices X and Y. [CBSE Sample Paper 11]
15. Define the term amplification.
16. Define range of a communication system.
17. What is a repeater ? What is its function ? [CBSE D 10 ; F 11]
18. What does the term channel refer to ?
19. Give the approximate bandwidth of speech signals.
20. Write the bandwidths generally used for AM and FM radio transmission.
21. What bandwidth is required by video signals for transmission of pictures ?
22. What bandwidth is required for TV transmission ?
23. What is meant by the term modulation ? [Punjab 05C ; Haryana 06 ; CBSE D 03]
24. What is a modulating signal ?
25. What do you mean by base band signal ? [Punjab 05C]
26. What is a modulated wave ?
27. What are carrier waves ? [Punjab 06]
28. Distinguish between amplitude modulation and frequency modulation. [CBSE OD 15C]
29. Define modulation index. [CBSE F 12, OD 14C]
30. What is the importance of modulation index ? [Punjab 06 ; CBSE F 12]
31. A carrier wave of peak voltage 20 V is used to transmit a message signal. What should be the peak voltage of the modulating signal, in order to have modulation index 80% ? [CBSE OD 08]

32. What percentage of AM wave power is carried by sidebands for $m = 1$?
33. In an AM wave with 100% modulation, what fraction of the total transmitted power is carried by each sideband?
34. Which basic mode of communication is used for telephonic communication? [CBSE OD 15]
35. Write an expression for the bandwidth in frequency modulation.
36. What should be the length of the dipole antenna for a carrier wave of frequency 3×10^8 Hz? [CBSE OD 03 ; D 05]
37. What does the term 'demodulation' in a communication system mean? [CBSE OD 14C]
38. What is a modem? [Haryana 06 ; Punjab 06C]
39. What is the origin of the word modem?
40. Name the device which can represent digital data by analog signals and vice-versa. [CBSE D 05C]
41. Name the process by which exact reproduction of a document at a distant place can be received. [CBSE OD 05C]
42. Give any one difference between FAX and Email systems of communication. [CBSE OD 06 ; Haryana 06]
43. Name an appropriate communication channel needed to send a signal of band-width 100 kHz over a distance of 8 km. [CBSE D 05]
44. How does the effective power radiated by an antenna vary with wavelength? [CBSE OD 05 ; D 14C]
45. What is the main function of a modem? [CBSE F 05]
46. What type of modulation is required for television broadcast? [CBSE F 03]
47. What type of modulation is required for commercial broadcast of voice signals? [CBSE F 03]
48. What should be the frequency of carrier wave with reference to message signal for the process of modulation? [CBSE F 03]
49. Name the type of radio wave propagation involved when TV signals, broadcast by a tall antenna, are intercepted directly by the receiver antenna. [CBSE D 04C]
50. Name the mode of propagation of radiowaves which travel in straight line from the transmitting antenna to the receiving antenna. [CBSE OD 08C ; F 10C]
51. What is the cut-off frequency beyond which the ionosphere does not reflect electromagnetic radiations? [CBSE OD 94C]
52. What is the range of wavelength of television broadcast? [CBSE F 95]
53. What is a geostationary satellite? [CBSE D 94C]
54. How is an artificial satellite helpful in telecommunication? [CBSE D 94C]
55. What is a ground wave? [Punjab 04, 06C ; CBSE F 05]
56. What is a sky wave? [Punjab 04, 05C ; CBSE F 05]
57. What is space wave propagation? [CBSE D 09]
58. What is ground wave propagation? [CBSE D 09]
59. What is sky wave propagation? [CBSE D 09]
60. Suggest two methods by which the range of transmission by a TV tower can be increased. [Punjab 04 ; CBSE D 99C]
61. Show, by drawing a diagram, how sky wave propagation of radio waves takes place. [CBSE F 94]
62. Define the term 'critical frequency' in relation to sky wave propagation of electromagnetic waves. [CBSE OD 06]
63. Name the waves relevant in telecommunication.
64. Name the type of communication that uses carrier signals having frequencies in the range 10^{12} Hz to 10^{16} Hz.
65. Name the two types of orbits (other than the geostationary orbit) used in satellite communication. [CBSE OD 04C]
66. What type of satellite generally is required for remote sensing?
67. What is a transponder? [CBSE D 05C]
68. What is remote sensing?
69. Give two applications of remote sensing.
70. Name the type of waves which are used for line of sight (LOS) communication. What is the range of their frequencies? [CBSE OD 13]
71. The carrier wave of a signal is given by $c(t) = 3\sin(8\pi t)$ volt. The modulating signal is a square wave as shown. Find its modulation index. [CBSE D 14]

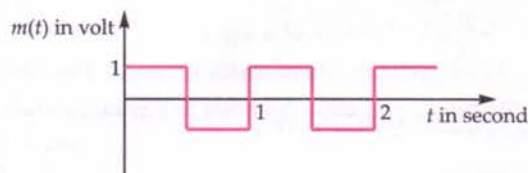


Fig. 15.52

72. The figure 15.53 shows the block diagram of a generalized communication system. Identify the element labelled 'X' and write its function. [CBSE D 14]

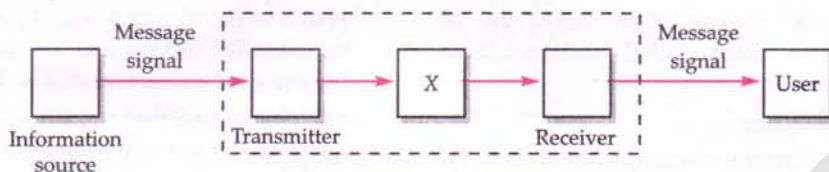


Fig. 15.53

73. Given a block diagram of a generalized communication system.

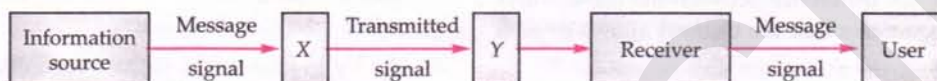


Fig. 15.54

Identify the boxes 'X' and 'Y' and write their functions.

[CBSE OD 15C]

74. How are side bands produced ? [CBSE D 15]
75. Why is the frequency of outgoing and incoming signals different in a mobile phone ? [CBSE D 15C]
76. Explain briefly two commonly used applications of the 'Internet'. [CBSE OD 15]
77. Name the network within an institution, made by connecting all or some of their computers. [CBSE Sample Paper 16]
78. What is Internet ?
79. What is a LAN ?
80. What is a WAN ?
81. What is a server computer ?
82. Name the two early computer networks.
83. Give full forms of :
(i) ARPANET (ii) NSFnet (iii) TCP (iv) IP
84. Give two uses of email ?
85. Define : (i) e-Banking (ii) e-shopping (iii) e-Booking (iv) Social Networking.
86. What is WWW ?
87. What is the range of operations for mobile phone ?
88. What is the role of MTSO in a mobile telephone network ?
89. What is handoff/handover ?
90. What are ESN and MIN in a SIM CARD ?
91. Which mobile network generations are in operations today in India ?
92. Define GPS.
93. What makes the space segment of GPS ?
94. What makes the control segment of GPS ?
95. What is the limit of number of simultaneous user segment of GPS ?
96. In a GPS system, what is the minimum number of satellites required so that exact location can be determined in 3D-space ?
97. Name four variables used to determine accurate position in a GPS system.

Answers

- Communication is the process by which information is transferred faithfully from one point (source) to another (destination) in an intelligible form.
- A communication system consists of
 - a transmitter,
 - a transmission medium and
 - a receiver.
- The physical path between the transmitter and receiver is called the communication channel or the transmission medium.
- A transducer is a device which converts energy from one form to another. For example, a microphone converts a sound signal into an electrical signal.
- X = Information source and
Y = Communication channel.

6. A communication system which makes use of analog signals is called analog communication system *e.g.*, telex.
7. A communication system which makes use of digital signals is called digital communication system *e.g.*, fax.
8. Digital communication.
9. Bandwidth of a signal is the range over which the frequencies in a signal vary. It is equal to the difference between the highest and lowest frequencies present in a signal.
10. The bandwidth of a communication channel is defined as the difference between the highest and lowest frequencies that the channel allows to pass through it.
11. The unwanted signals which get mixed up with the information signal during its propagation through the communication channel constitute noise.
12. Attenuation is the loss of strength of a signal during its propagation through the communication channel.
13. Baseband refers to the band of frequencies of the original signal as is produced by the source of information.
14. (a) Transducer
(b) Repeater.
15. Amplification is the process of increasing the amplitude and hence the strength of an electrical signal by using an electronic circuit (consisting of atleast one transistor).
16. Range is defined as the largest distance between the source and the destination upto which a signal can be received with sufficient strength.
17. A repeater is a combination of a transmitter, an amplifier and a receiver which picks up a signal from the transmitter, amplifies and retransmits it to the receiver sometimes with a change of carrier frequency.
Multiple repeaters help in extending the range of the communication system.
18. The term channel refers to the frequency range allocated to a particular transmission station.
19. The frequency range of speech signals is 300 Hz to 3100 Hz. So they require a bandwidth of 2800 Hz (3100 Hz – 300 Hz) for telephonic conversation.
20. (i) Bandwidth used for AM radio transmission is 10 kHz.
(ii) Bandwidth used for FM radio transmission is 150 kHz.
21. 4.2 MHz
22. 6 MHz
23. Modulation is the process by which some characteristic, usually amplitude, frequency or phase angle of a high frequency carrier wave is varied in accordance with the instantaneous value of the low frequency audio signal, called the modulating signal.
24. The low frequency information signal is called a modulating signal.
25. The modulating signal *i.e.*, the signal to be transmitted is called a base band signal.
26. The resultant wave obtained by the superposition of the modulating wave and the carrier wave is called modulated wave.
27. A high frequency signal used to carry low frequency information signal to larger distances is called a carrier wave.
28. In amplitude modulation, the amplitude of the carrier wave changes in accordance with the modulating signal while in frequency modulation, the frequency of the carrier wave changes in accordance with the modulating signal.
29. Modulation index is defined as the ratio of the change in the amplitude of the carrier wave to the amplitude of the original carrier wave. It is also known as modulation factor.
30. Modulation factor determines the strength and quality of the transmitted signal. The greater the value of modulation index, the stronger and clearer will be the audio signal.
31. As $\mu = \frac{A_m}{A_c}$
 $\therefore A_m = \mu A_c = \frac{80}{100} \times 20 = 16 \text{ V}$
32. 33.3%
33. One-sixth of the total transmitted power.
34. Point to point communication mode is used for telephonic communication.
35. In FM transmission,
Bandwidth = $2n \times$ modulating frequency
 $= 2n \times f_m$
where n is the number of significant bands in the FM wave.
36. Length of a dipole antenna
 $= \frac{\lambda}{2} = \frac{c}{2\nu} = \frac{3 \times 10^8}{2 \times 3 \times 10^8} = 0.5 \text{ m}$
37. The process of recovering the information signal from the modulated wave at the receiver end is known as demodulation or detection.

38. A modem is a device that connects a transmitting digital device to a receiving digital device through a telephone line.
39. The word modem is the contraction of the terms modulator and demodulator.
40. Modem.
41. Facsimile telegraphy (FAX).
42. Fax is the electronic reproduction of a document or picture at a distant place via telephone line. Email is the message sent and received through a computer network.
43. Twisted pair wire.
44. Power radiated by antenna $\propto \left(\frac{1}{\lambda}\right)^2$
45. A modem modulates the signal at the transmitting end and demodulates it at the receiving end.
46. AM for video signal and FM for voice signal.
47. Frequency modulation.
48. Radio frequency or greater. The carrier frequency must be greater than twice the maximum frequency present in the message signal.
49. Space wave propagation or line of sight communication.
50. This mode of propagation is called space wave propagation or line of sight (LOS) propagation.
51. 40 MHz
52. The e.m. waves used in T.V. broadcast have wavelength range from 1.36 m to 3 m.
53. A satellite which revolves around the earth in its equatorial plane with a period of revolution of 24 hours is called a geostationary satellite. It appears stationary to an observer on the earth.
54. Due to its great height above the earth, an artificial satellite can send the signal to a wide area on the earth.
55. If a radiowave from the transmitting antenna reaches the receiving antenna either directly or after reflection from the ground, it is called a ground wave.
56. If a radiowave from the transmitting antenna reaches the receiving antenna after reflection from the ionosphere, it is called a sky wave.
57. In space wave propagation, a radiowave transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna.
58. In ground wave propagation, a radiowave travels directly from one point to another following the surface of the earth.
59. A radiowave of frequency range 2 MHz to 20 MHz can be directed towards the sky and reflected by the ionosphere towards a desired location of the earth. This mode of propagation is called sky wave propagation.
60. (i) By increasing the height of T.V. tower.
(ii) By increasing the height of receiving antenna so that it can directly intercept the signal from the transmitting antenna.
61. In sky wave propagation, a radio wave is directed towards the sky and is reflected by the ionosphere towards the desired location on the earth. See Fig. 15.43.
62. Critical frequency is the highest frequency of the radiowave, which when sent normally towards a given layer of the ionosphere, gets reflected from the ionosphere and returns to the earth. It is given by
$$f_c = 9 (N_{\max})^{1/2},$$
 where $N_{\max}$ is the maximum electron density of the ionosphere layer.
63. Microwaves.
64. Optical communication.
65. (i) Polar orbit ; (ii) Highly elliptical orbit.
66. Sun-synchronous satellite.
67. A transponder is a device fitted on the satellite which receives the signal and retransmits it after amplification.
68. Remote sensing is the technique of obtaining information about an object, area, or phenomenon, acquired by a sensor that is not in direct contact with the target of investigation.
69. (i) Spying work for military purposes.
(ii) In ground-water surveys.
70. Space waves are used for the line of sight (LOS) communication. This communication is possible with electromagnetic waves of frequencies above 40 MHz.
71. Modulation index $= \frac{A_m}{A_c} = \frac{1.5}{3} = 0.5.$
72. X = communication channel. It connects the transmitter to the receiver.
73. X : Transmitter. It converts the message signal into a suitable form for transmission into channel.
Y : Channel. It sends the signal to the receiver.
74. Sidebands are produced due to the superposition of carrier waves of frequency ω_c over modulating/audio signal of frequency ω_m .
75. To avoid overlapping of the two signals.
76. Applications of internet are : e-mail, e-banking, chatting, file transfer, e-shopping, e-ticketing, surfing, etc.
77. LAN.

TYPE B : SHORT ANSWER QUESTIONS (2 or 3 marks each)

1. Draw the block diagram of a generalised communication system. [CBSE F 11 ; D 14C]
2. Distinguish between 'point to point' and 'broadcast' communication modes. Give one example of each. [CBSE OD 14C]
3. Write three important factors which justify the need of modulating a message signal. Show diagrammatically how an amplitude modulated wave is obtained when a modulating signal is superimposed on a carrier wave.

[CBSE OD 11; D 12, 13]

4. Explain briefly with the help of diagrams the terms :
(i) amplitude modulation,
(ii) frequency modulation.

Which of these (a) gives better quality transmission, (b) has a larger coverage ?

[CBSE D 04 C]

5. Figure 15.55 shows a block diagram of a transmitter. Identify the boxes 'X' and 'Y' and write their functions. [CBSE F 12]

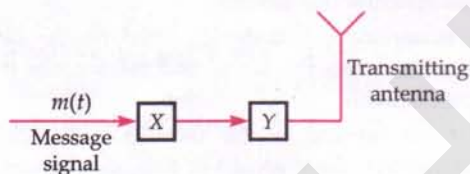


Fig. 15.55

6. In the block diagram of a simple modulator for obtaining an AM signal, shown in Fig. 15.56, identify the boxes A and B. Write their functions. [CBSE OD 13]

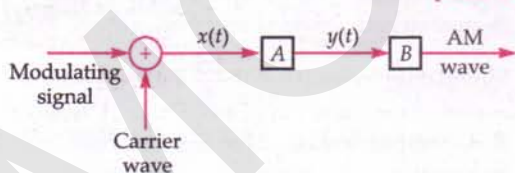


Fig. 15.56

7. Draw a plot of the variation of 'amplitude' versus ' ω ' for an amplitude modulated wave. Hence explain the need for keeping the broadcast frequencies sufficiently spaced out.

[CBSE Sample Paper 11]

8. What is meant by 'detection of a modulated signal' ? Draw block diagram of a detector for AM waves and state briefly, showing the waveforms, how the original message signal is obtained.

[CBSE D 13 C, 14C, 15]

9. What does the process of detection of amplitude modulated wave mean ? Fig. 15.57 shows the block diagram of a detector for AM signals.

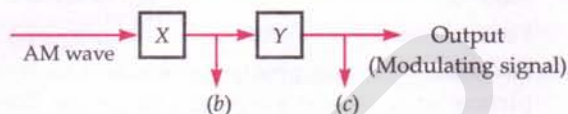


Fig. 15.57

Identify the boxes X and Y in this block diagram and show the waveform corresponding to the positions indicated by arrows (b) and (c).

[CBSE OD 08C]

10. What is a dipole antenna ? What role does it play in the communication system ? How does the length of dipole antenna vary with the frequency of the carrier waves ? [CBSE D 03C]
11. Draw a schematic diagram showing the (i) ground wave (ii) sky wave and (iii) space wave propagation modes for em waves.

Write the frequency range for each of the following :

(i) Standard AM broadcast (ii) Television

(iii) Satellite communication [CBSE D 11]

12. What is ground wave communication ? On what factors does the maximum range of propagation in this mode depend ? [CBSE OD 11]
13. What is space wave communication ? Write the range of frequencies suitable for space wave communication. [CBSE OD 11]
14. Distinguish between 'sky waves' and 'space waves' modes of propagation in communication system.
(a) Why is sky wave mode propagation restricted to frequencies upto 40 MHz ?
(b) Give two examples where space wave mode of propagation is used. [CBSE D 10, 13]
15. Mention the frequency at which T.V. signals are transmitted. Derive an expression for the range upto which signals transmitted by a T.V. tower can be received. [CBSE D 05 C ; OD 2000, 03C]
16. Draw a diagram to show that "taller the antenna, larger the coverage of telecast of a T.V. station". Give its explanation. [CBSE D 2000C, 02C]
17. Name the device used for data transmission from one computer to another. Justify the name. [CBSE D 04]
18. Write the functions of (any two) of the following in communication systems : [CBSE OD 14]
(i) Transmitter (ii) Receiver (iii) Modulator
(iv) Demodulator (v) Transducer (vi) Repeater

19. Write two basic modes of communication. Explain the process of amplitude modulation. Draw a schematic sketch showing how amplitude modulated signal is obtained by superposing a modulating signal over a sinusoidal carrier wave.
[CBSE OD 14]
20. (a) Arrange the following networks in increasing order of the number of computers that may be present in the network :
Internet ; LAN ; WAN
(b) What is the minimum number of satellites that enables a Global Positioning System (GPS) receiver to determine one's longitude/latitude position, i.e., to make a 2D position fix.
[CBSE SP 15]
21. Answer the following questions :
(a) Define 'bandwidth' and describe briefly its importance in communicating signals.
(b) Distinguish between digital and analogue signals.
(c) Write the functions of transducer and repeater.
[CBSE F 15]
22. Distinguish between two types of propagation of electromagnetic waves with respect to (i) frequency range over which they are applicable (ii) communication systems in which they are used.
[CBSE Sample Paper 16]
23. (a) Write the factors which prevent a baseband signal of low frequency to be transmitted over long distances.
(b) What should be done to overcome these factors ? Draw a block diagram to obtain the desired signal.
[CBSE Sample Paper 16]
24. Explain the following terms in relation to the use of internet : (i) Internet surfing (ii) Social networking (iii) E-mail.
[CBSE OD 15C]
25. How does Internet work/function ?
26. How is a WAN different from a LAN ?
27. What is the role of TCP/IP ?
28. Discuss the structure of email Id.
29. How is WWW formed in Internet ?
30. Give four some applications of Internet.
31. State some applications of mobile phones.
32. How do mobile phones work ?
33. Briefly describe the three components of GPS.
34. How does GPS function ?
35. Describe two applications of GPS.

Answers

- See Fig. 15.1 on page 15.2.
- Refer answer to Q. 4 on page 15.3.
- Refer to the solution of Problem 8 on page 15.37 and see Fig. 15.8 on page 15.8.
- Refer to points 20 and 24 of Glimpses.
(a) Frequency modulation gives better quality transmission.
(b) Amplitude modulation has a larger coverage.
- X is an *amplitude modulator*. It superimposes the base band signal over the carrier wave to give the modulated wave.
Y is a *power amplifier*. It increases the power of modulated signal before it is fed to a transmitting antenna for radiation into space.
- A = Square law device ;
B = Band pass filter

Square law device is a non-linear device which receives the input signal,

$$\begin{aligned} x(t) &= \text{message signal} + \text{Carrier signal} \\ &= A_m \sin \omega_m t + A_c \sin \omega_c t \end{aligned}$$

The square law device gives the output

$$y(t) = Bx(t) + Cx^2(t)$$

Here B and C are constants.

Band pass filter rejects dc signal, low frequencies ($\omega_m, 2\omega_m$), high frequency ($2\omega_c$) and retains the band of frequencies centred around ω_c ($\omega_c, \omega_c \pm \omega_m$).

7. A plot of the variation of amplitude A versus ω for AM is shown in Fig. 15.58.

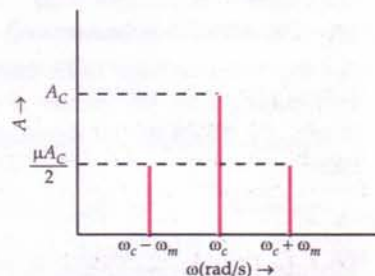


Fig. 15.58

When the broadcast frequencies are sufficiently spaced out, the different stations can operate without interference.

8. Refer answer to Q. 37 and page 15.18.
9. The box X is a *rectifier* and the box Y is an *envelope detector*.

Refer answer to Q. 37 on page 15.18.

10. Refer answer to Q. 33 on page 15.17.

The length of a dipole antenna varies inversely with the frequency of the carrier waves.

11. See Fig. 15.24 on page 15.22.

Service	Frequency band
(i) Standard AM broadcast	500 – 1500 kHz
(ii) Television	100 – 200 MHz
(iii) Satellite communication	5.9 – 6.4 GHz

12. In ground wave propagation, a radiowave travels directly from one point to another following the surface of the earth.

The maximum range of ground wave propagation depends on *two* factors :

- (i) The frequency of transmitted wave, and
- (ii) The power of the transmitter.

13. In space wave propagation, a radiowave transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna. Frequencies in the range 100–220 MHz are suitable for space wave propagation.

14. In sky wave propagation, a radiowave is directed towards the sky and is reflected by the ionosphere towards the desired location. In space wave propagation, a radiowave transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna.

- (a) Radiowaves of frequencies more than 40 MHz penetrate into the atmosphere and cannot be reflected towards desired location.
- (b) Television broadcast, microwave links and satellite communication.

15. Television frequencies lie in the range 100 – 220 MHz. Refer answer to Q. 44 on page 15.23.

16. See Fig. 15.25 on page 15.23. Larger the height of transmitting antenna, greater is the line of sight distance in which the TV signal can be intercepted directly.

17. A modem is used for data transmission from one computer to another. As it functions both as **Modulator** and **Demodulator**, so it is called modem.

18. (i) Transmitter processes the incoming signal into a form suitable for transmission through a channel and for its subsequent transmission.

- (ii) Receiver is a device which recovers the original message signal from the signal received at the output of the communication channel.

- (iii) Modulator is a device which changes some characteristic like amplitude/frequency/phase angle of a high frequency carrier wave in accordance with the instantaneous value of the low frequency message signal through the appropriate superposition.

- (iv) Demodulator is a device which recovers the original information signal from the modulated wave at the receiver end.

- (v) Transducer is a device which converts energy from one form to another.

- (vi) Repeater is a combination of a transmitter, an amplifier and a receiver which picks up a signal from the transmitter, amplifies and retransmits it to the receiver sometimes with a change of carrier frequency.

19. Two basic modes of communication are :

- (i) point-to-point and (ii) broadcast mode.

In amplitude modulation, the amplitude of a high frequency carrier wave changes in accordance with the instantaneous value of the modulating signal. See Fig. 15.8 on page 15.8.

20. (a) LAN, WAN, Internet

- (b) Three

21. (a) Bandwidth is the frequency range over which an equipment operates or the range over which the frequencies in a signal vary.

Importance. The knowledge of bandwidth helps in designing equipments used in communication system for distinguishing different message signals.

- (b) In a digital signal the current or voltage can take only two discrete values. In an analog signal the current or voltage varies continuously with time.

- (c) A transducer converts energy from one form to another. A repeater enhances the range of a communication system.

22.

Method of propagation	Frequency range	Use
Ground wave	500 – 1500 kHz	Standard AM broadcast
Space wave	Above 40 MHz	Television

23. (a) (i) Size of antenna
(ii) Effective power radiated by antenna.
(iii) Mixing up of signals from different transmitter.
(b) Modulation. For block diagram of amplitude modulation, see Fig. 15.13(a) on page 15.12.
24. (i) *Internet surfing*. Visiting, or using, the different websites on the internet.
(ii) *Social networking*. Social networking implies using site like :
- (a) Facebook, Twitter, etc. to share ideas and information with a large number of people.
(b) Using internet for chatting, video sharing etc. among friends and acquaintances.
(iii) *E-mail*. Using internet (rather than the post office) for exchanging (multimedia) communication between different persons and organizations.

■ TYPE C : LONG ANSWER QUESTIONS (5 marks each)

1. Define amplitude modulation. Derive an expression for an amplitude modulated wave. Hence state the sideband frequencies present in the AM wave.
2. What is communication channel ? Briefly describe the various communication channels employed in communication. On what factors, does the quality of transmission depend in each case ? How do we make the choice of communication channel ?
3. With the help of a labelled diagram, describe the three methods by which radiowaves can propagate through the atmosphere.

[CBSE Text]

Answers

1. Refer answer to Q. 21 and Q. 22 on page 15.9.
2. Refer answer to Q. 39 on page 15.20.
3. See Fig. 15.24 on page 15.22 and refer to point 37 of Glimpses.

■ TYPE D : VALUE BASED QUESTIONS (4 marks each)

1. Deepa's uncle wants to talk to his son in USA. He does not have much money to spend on telephone calls. He has a computer at his home. Deepa told her uncle that he could talk to his son with the help of computer and also told him about internet. Her uncle now talks to his son everyday. He thanked Deepa for giving useful advice.
- (a) What according to you are the values displayed by Deepa ?
(b) How does internet work ?
2. Nidhi has to take admission in some professional college. It was last date of admission and Nidhi left her birth certificate at her home. College was very far from the home. She called her brother and he faxed the birth certificate. She got the admission and thanked her brother.
- (a) What value was displayed by Nidhi ?
(b) What value was displayed by her brother ?
(c) What is the function of a Fax ?
3. When Sunita, a class XII student, came to know that her parents are planning to rent out the top floor of their house to a mobile company she protested. She tried hard to convince her parents that this move would be a health hazard. Ultimately her parents agreed.
- (a) In what way can the setting up of transmission tower by a mobile company in a residential colony prove to be injurious to health ?
(b) By objecting to this move of her parents, what value did Sunita display ?
(c) Estimate the range of em waves which can be transmitted by an antenna of height 20 m. (Given radius of the earth = 6400 km).
4. Arnab was talking on his mobile to his friend for a long time. After his conversation was over, his sister Anita advised him that if his conversation was of such a long duration, it would be better to talk through a land line.

[CBSE D 14]

[CBSE OD 14C]

- (a) Why is it considered harmful to use a mobile phone for a long duration ?
- (b) Which values are reflected in the advice of his sister Anita ?
- (c) A message signal of frequency 10 kHz is superposed to modulate a carrier wave of frequency 1 MHz. Determine the sidebands produced.
5. Anuj's mother was having constant headaches. After a medical check-up, she was diagnosed with tumour. Anuj realized there was a telecommunication tower very close to their house. He enquired from the doctor if the radiation from the

tower could have caused the tumour. As the doctor supported his anxiety, he lodged a complaint with the police and ultimately succeeded in getting the tower removed to a distant place away from the residential colony.

- (a) What values were displayed by Anuj ?
- (b) Anuj made a rough estimate about the height of the antenna to be about 20 m from the ground. Calculate the maximum distance upto which radiations from the tower are likely to reach. Use the value of radius of the Earth = 6400×10^3 m. [CBSE D 14C]

Answers

- (a) Caring and creating awareness.

(b) Internet is a global network of computers linked by high-speed data lines and wireless systems. It allows communication and sharing of information between any two or more computers connected through the network.
- (a) Awareness

(b) Understanding

(b) It is used to produce an exact copy of a document or picture at a distant place.
- (a) Electromagnetic radiations emitted by a transmitting antenna can cause

 - Cardiac problem
 - Cancer
 - Giddiness and headache.

(b) Scientific temperament, awareness and social responsibility.

(c) $\text{Range} = \sqrt{2h_T R} = \sqrt{2 \times 20 \times 6400 \times 10^3} \text{ m}$
 $= 16 \times 10^3 \text{ m} = 16 \text{ km}.$

- (a) The ultra high frequency em radiations, continuously emitted by a mobile phone, may harm the system of the human body.

(b) Sister Anita shows :

 - Concern about her brother
 - Awareness about the likely effects of em radiations on human body
 - Sense of responsibility.

(c) The side bands are :

$$(f_c + f_m) \text{ and } (f_c - f_m)$$

or $(1000 + 10) \text{ kHz}$ and $(1000 - 10) \text{ kHz}$
or 1010 kHz and $990 \text{ kHz}.$
- (a) Concern, scientific temperament, keen observer, alertness.

(b) Refer answer to Q. 3(c) above.

COMPETITION SECTION

Communication Systems

GLIMPSES

1. **Communication.** It is the process by which information is transferred faithfully from one point to the other point in an intelligible form.
2. **Message signal.** It is defined as the single-valued function of time (that conveys the information) and which, at every instant of time takes a unique value. Signals can be *analog* or *digital*.
3. **Analog signal.** A signal in which current or voltage varies continuously with time is called analog signal.
4. **Digital signal.** A signal in which current or voltage can take only two discrete values is called a digital signal.
5. **Communication system.** The setup used to transmit information from one point to another is called communication system. A communication system mainly consists of (i) *transmitter*, (ii) *communication channel* and (iii) *receiver*.
6. **Analog communication.** It makes use of analog electronic circuit and analog signal in such a way that the output voltage varies continuously in accordance with the input voltage. For example, telegraphy, telex, TV network, etc.
7. **Digital communication.** It makes use of an electronic circuit that can handle only digital signals. For example, FAX, E-mail, etc.
8. **Transmitter.** It is a setup that transmits the message to the receiving end through a communication channel. It modifies the message signal in a form suitable for transmission and then transmits it. It mainly consists of (i) *message signal generator*, (ii) *modulator* and (iii) *antenna*.
9. **Bandwidth.** It refers to the range over which the frequencies in a signal vary. It is equal to the difference between the highest and lowest frequencies present in a signal.
10. **Baseband.** It refers to the band of frequencies of the original signal as is produced by the source of information.
11. **Transducer.** Any device which converts energy from one form to another is called a transducer. An electrical transducer converts some physical variations in pressure, displacement, temperature, etc. into corresponding variations in an electrical signal.
12. **Noise.** These are unwanted signals which get mixed up with the information signal during its propagation through the communication channel.
13. **Attenuation.** It is the loss of strength of a signal during its propagation through the communication channel.
14. **Amplification.** It is the process of increasing the amplitude and hence the strength of an electrical signal by using an electronic circuit (consisting of atleast one transistor).
15. **Range.** It is defined as the largest distance between the source and the destination upto which a signal can be received with sufficient strength.
16. **Repeater.** A repeater is a combination of a transmitter, an amplifier and a receiver which picks-up a signal from the transmitter, amplifies and retransmits it to the receiver sometimes with a change of carrier frequency.

17. **Nyquist sampling theorem.** An analog signal can be perfectly reconstructed from its samples without any loss of information if it is sampled atleast twice the rate of the maximum frequency component or the bandwidth of the signal. This is called sampling theorem.

$$f_s > 2 f_m$$

where f_s is the sampling frequency and f_m is the maximum frequency of the signal.

18. **Modulation.** A high frequency carrier wave is used to carry the audio signal to large distances. Modulation is the process of changing some characteristic (amplitude, frequency or phase) of high frequency carrier wave in accordance with the instantaneous value of the low frequency audio signal called the modulating signal. Correspondingly, we have three types of modulation : (i) amplitude modulation (ii) frequency modulation and (iii) phase modulation.

19. **Need for modulation.** The low frequency information signals (i) radiated directly from an antenna die out after travelling some distance, (ii) require antenna of impractically large length and (iii) transmitted from different stations get inseparably mixed up. Due to the above reasons, low frequency information signals need to be modulated by a high frequency carrier wave.

20. **Amplitude modulation.** It is the process in which the amplitude of the high frequency carrier wave changes in accordance with the instantaneous value of the modulating signal. Here a voltage proportional to the amplitude of the modulating signal gets added to the carrier amplitude.

21. **Modulation factor.** It is defined as the ratio of the change in the amplitude of carrier wave to the amplitude of original carrier wave. It is also known as degree of modulation or modulation index. If A_m and A_c are the amplitudes of modulating and carrier signals respectively, then the modulation factor is given by

$$\begin{aligned} \mu &= \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} \\ &= \frac{A_m}{A_c} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \end{aligned}$$

where $A_{\max}$ and $A_{\min}$ are the maximum and minimum voltages of AM wave.

22. **Expression for amplitude modulated wave.**

Let modulating signal, $m(t) = A_m \sin \omega_m t$

And carrier signal, $c(t) = A_c \sin \omega_c t$

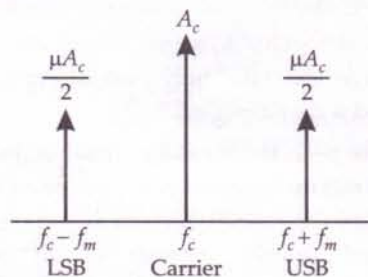
The instantaneous voltage of the A.M. wave is

$$c_m(t) = A_c \sin \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t] \quad \dots(i)$$

23. **Sideband frequencies and bandwidth of A.M. wave.** The equation (i) shows that the A.M. wave is the sum of three sinusoidal waves :

- One the original carrier wave of amplitude A_c and frequency $f_c = \omega_c / 2\pi$.
- The second wave of amplitude $\mu A_c / 2$ and of frequency $(f_c + f_m)$. The sum of carrier frequency and modulating frequency $(f_c + f_m)$ is called upper side band frequency (USB).
- The third wave of amplitude $\mu A_c / 2$ and frequency $(f_c - f_m)$. The difference of carrier frequency and modulating frequency $(f_c - f_m)$ is called lower side band frequency (LSB).

The frequency spectrum of an A.M. is shown below :



The difference between the highest and the lowest frequencies present in the A.M. wave is called its bandwidth.

$$\text{Bandwidth} = (f_c + f_m) - (f_c - f_m) = 2 f_m$$

Hence bandwidth of the A.M. wave is twice the frequency of the modulating signal.

24. **Frequency modulation.** It is the process of changing the frequency of the carrier voltage in accordance with the instantaneous value of the modulating voltage. The original frequency of the carrier signal is called resting frequency or centre frequency (f_c). The amount by which the frequency of the carrier wave changes or shifts above or below the resting frequency is called frequency deviation (Δf).

Clearly, $\Delta f \propto m(t)$

25. **Carrier swing.** The total variation in frequency of F.M. wave from the lowest to the highest is called carrier swing (CS).

Carrier swing

$$= 2 \times \text{Frequency deviation in centre frequency}$$

$$CS = 2 \Delta f$$

26. **Modulation index in F.M.** It is defined as the ratio of frequency deviation to the modulating frequency. So

$$\mu_f = \frac{\text{Frequency deviation}}{\text{Modulating frequency}}$$

$$= \frac{\Delta f}{f_m}$$

27. **Bandwidth in F.M.** If the various sidebands are separated by the modulating frequency f_m , then

$$\text{Bandwidth} = 2n \times \text{modulating frequency}$$

$$= 2n \times f_m$$

where n is the number of significant bands present in the F.M. wave.

28. **Phase modulation.** It is the process of varying the phase angle of the carrier voltage in accordance with the instantaneous value of the modulating voltage.
29. **Pulse modulation.** It is a process in which some parameter of a pulse train is varied in accordance with the message signal or the modulating signal.
30. **Different types of pulse modulation :**

- (i) **Pulse amplitude modulation (PAM).** It is defined as the process in which amplitudes of the regularly spaced rectangular pulses vary in direct proportion to the instantaneous values of continuous message signal.
- (ii) **Pulse duration modulation (PDM).** Here the starting time and amplitude of each pulse are constant but the width or duration of each pulse is made proportional to the instantaneous value of the analog signal.
- (iii) **Pulse position modulation (PPM).** Here the position of a pulse relative to its unmodulated time of occurrence is varied in accordance with the instantaneous value of the message signal.

- (iv) **Pulse code modulation (PCM).** It is a digital process in which the message signal is sampled and the amplitude of each sample is rounded off to the nearest value of a finite set of allowable values and rounded values are coded. The signals are transmitted as a binary code.

31. **Antenna.** An antenna is basically a small length of a conductor that is used to radiate or receive electromagnetic waves. It acts as a conversion device. At the transmitting end, it transforms high frequency current into electromagnetic waves. At the receiving end, it transforms electromagnetic waves into electrical signal that is fed to the input of the receiver. A dipole antenna is omnidirectional, whereas the dish antenna is highly directional. The length of a dipole antenna is usually $\lambda/2$.
32. **Receiver.** Its function is to recover the original message or data from the modulated signal after its propagation through the communication channel. It separates the modulating signal from the carrier wave. The process is called demodulation.
33. **Demodulation.** The process of recovering the message signal from the modulated wave is known as modulation or detection. A junction diode can be used as a detector or modulator.
34. **Modem.** A modem is a device that connects a transmitting digital device to a receiving digital device through telephone line. The name modem is a contraction of the terms modulator and demodulator. In the transmitting mode, a modem acts as a modulator and in the receiving mode, it acts as a demodulator.
35. **Communication channels.** The physical path between the transmitter and receiver is called the communication channel or the transmission medium.

The transmission media can be broadly divided into two categories :

- (i) **Guided media.** Such media are used for point-to-point communication. For example, twisted pair, coaxial cable and optical fibre.

The quality of transmission in guided media is determined mainly by the nature of the media.

(ii) **Unguided media.** Free space is an example of unguided medium. Here the frequency band of the transmitted signal is more important than the medium in determining the quality of transmission.

Depending on the nature of the communication channel, the communication is classified broadly into three categories :

- (a) Line communication
- (b) Optical communication
- (c) Space communication

36. **Bandwidth of a communication channel.** It is defined as the difference between the highest and the lowest frequencies that communication channel allows to pass through it. It must be equal to or greater than the bandwidth of the information signal.

Number of channels

$$= \frac{\text{Total bandwidth of channel}}{\text{Bandwidth needed per channel}}$$

37. **Space communication.** Here electromagnetic waves of different frequencies are used to carry information through the physical space acting as the transmission medium.

It is of three types :

- (i) **Ground wave propagation.** A radiowave that can travel directly from one point to another following the surface of the earth is called a ground wave. As the ground wave passes over this surface, it is attenuated due to the absorption of its energy by the earth. This absorption increases with the increase in frequency. Hence ground wave propagation is possible at low frequencies (500 kHz to 1500 kHz). This region of AM band is called *medium wave band*.
- (ii) **Space wave propagation.** If a radiowave transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna, it is called a space wave. The space wave propagation is also called tropospheric or line of sight propagation. Television frequencies (100 – 220 MHz) are transmitted through this mode.
- (iii) **Sky wave propagation.** A radiowave directed towards the sky and reflected by

the ionosphere towards the desired location of the earth is called a sky wave. Radiowaves of frequencies 2 – 20 MHz can be reflected by the ionosphere. This region of the AM band is called *short wave band*.

38. **Critical frequency.** It is the highest frequency of radiowave which when sent straight (normally) towards the given layer of ionosphere gets reflected from ionosphere and returns to the earth. Above this frequency, the radiowave will penetrate the ionosphere and is not reflected back.

$$f_c = 9 (N_{\max})^{1/2}$$

where $N_{\max}$ = the maximum electron density of the ionosphere.

39. **Skip distance.** It is defined as the smallest distance measured from a transmitter, measured along the earth's surface to a point where the sky wave of fixed frequency but less than critical frequency returns to the earth after reflection from the ionosphere.

40. **Range of TV transmission.** The signals transmitted from antenna of height h can be directly received by a receiver on the surface of the earth within the range given by

$$d = \sqrt{2hR}, \quad R \gg h$$

41. **Range of line-of-sight distance between two antennas.** The maximum line-of-sight distance d_M between transmitting and receiving antennas of heights h_T and h_R is given by

$$d_M = d_T + d_R = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where R is the radius of the earth.

42. **Satellite communication.** It is the mode of communication of a signal between transmitter and receiver through a satellite. A beam of modulated microwave is sent directly towards the satellite from a transmitter on the earth. The signals are received, processed, amplified and retransmitted by a device called *transponder* fitted in the satellite. The signals received on the earth are amplified and retransmitted by a receiver.
43. **Transponder.** It is a device fitted on a satellite which receives the signal and retransmits it after amplification.

44. **Remote sensing.** It is the technique of obtaining information about an object, area or phenomenon, acquired by a sensor that is not in direct contact with the target of investigation. A satellite equipped with suitable sensors to acquire data is placed in an orbit around the earth at any height having a period of revolution. It takes photographs or collects any other information desired and transmits it back to an earth station. This is known as remote sensing.
45. **Line communication.** A communication system involving point-to-point contact between the transmitter and receiver is called line communication. This mode of communication requires use of a guided medium such as a parallel-wire line, twisted pair or coaxial cable.
46. **Parallel-wire line.** It consists of two conducting wires running parallel to each other inside a protective insulating PVC coating. This system is very lossy and suffers from electromagnetic interference from other sources.
47. **Twisted pair.** It consists of two insulated copper wires twisted around each other almost regularly. It can transmit both analog and digital signals and is less prone to electrical disturbances.
48. **Coaxial cable.** It consists of a hollow cylindrical conductor, surrounding a coaxial single inner conductor. The inner conductor is a copper wire surrounded by a low dielectric insulator and the outer conductor is a copper mesh covered with PVC. The outer conductor is normally grounded and thus provides an electrical shield to the signals carried by the central conductor. Coaxial cables are used for long distance high frequency transmission, local area computer networks for high-speed data transmission and as input / output channels for the computer systems.
49. **Optical communication.** It is a mode of communication in which a light beam carries information from one point to another by using a guided medium like optical fibre. This method has two main advantages :
 - (i) It can be used as a high bandwidth carrier.
 - (ii) It can carry large amount of information.
50. **Internet.** It is a global network of computers linked by high speed data lines and wireless systems. It allows communication and sharing of information between any two or more computers.
51. **Email : Electronic mail.** It is a message sent and received through a computer network.
52. **File Transfer.** An FTP (File Transfer Protocol) permits the transfer of files/software from one computer to another connected to the internet.
53. **WWW : World Wide Web.** It is a set of protocols that allows us to access any document on the internet.
54. **E-commerce.** It is the collection of tools and practices involving internet technologies that allow a company to create, maintain and optimise business relations with consumers and other businesses. It permits online banking and shopping.
55. **Chat.** It is the real time conversation among people with common interests through the typed messages on the net.
56. **FAX : Facsimile telegraphy.** The electronic reproduction of a document at a distant place is called Facsimile telegraphy or fax.

In order to send a document through fax, the following three functions are performed :
 - (i) At the sending end, the original document is read by an optical scanner.
 - (ii) The black and white dots of the document are encoded and compressed for transmission and reception via a telephone line.
 - (iii) At the receiving end, the received data is decompressed and processed by a printer to reproduce the original document.
57. **Mobile Telephony.** Instead of a single high-power transmitter for an entire service area, numerous lower power transmitters (*base stations*) are set up, each covering a fraction of that service area, called a *cell*. As a mobile receiver moves from one cell to another, the mobile user is handed over to the new cell's base station through a switching office called *mobile telephone switching office (MTSO)*.